CALCULUS II (MMT326J)

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Chapter 1

Methods of Integration Techniques

This unit focuses on mastering complex integration by decomposing rational functions into simpler partial fractions for easier evaluation. It further establishes the algebraic foundations required to handle irrational integrands and specific trigonometric forms involving reciprocals. Students will learn to transform difficult integrals into standard, solvable expressions using these systematic techniques.

1.1 Overview

Before proceeding with Unit I (Integration of Rational, Irrational, and Trigonometric functions), students are expected to be fluent with the following standard integration formulas. These serve as the building blocks for the advanced techniques covered in this semester. *Note: In all cases, C represents the constant of integration.*

1.1.1 Algebraic and Exponential Functions

These are essential for the final step of Partial Fraction decomposition.

- **Power Rule:**

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad (n \neq -1)$$

- **Logarithmic Form (Crucial for Linear Factors):**

$$\int \frac{1}{x} \, dx = \ln |x| + C$$

$$\int \frac{1}{ax+b} \, dx = \frac{1}{a} \ln |ax+b| + C$$

- **General Logarithmic Rule:**

$$\int \frac{f'(x)}{f(x)} \, dx = \ln |f(x)| + C$$

- **Exponential Functions:**

$$\int e^x \, dx = e^x + C$$

$$\int e^{ax} \, dx = \frac{1}{a} e^{ax} + C$$

$$\int a^x \, dx = \frac{a^x}{\ln a} + C, \quad (a > 0, a \neq 1)$$

1.1.2 Trigonometric Functions

Required for Unit I topics involving trigonometric denominators.

Basic Trigonometry:

1. $\int \sin x\, dx = -\cos x + C$
2. $\int \cos x\, dx = \sin x + C$
3. $\int \sec^2 x\, dx = \tan x + C$
4. $\int \text{cosec}^2 x\, dx = -\cot x + C$
5. $\int \sec x \tan x\, dx = \sec x + C$
6. $\int \text{cosec}\, x \cot x\, dx = -\text{cosec}\, x + C$

Advanced (Logarithmic):

7. $\int \tan x\, dx = \ln|\sec x| + C$
8. $\int \cot x\, dx = \ln|\sin x| + C$
9. $\int \sec x\, dx = \ln|\sec x + \tan x| + C$
10. $\int \text{cosec}\, x\, dx = \ln|\text{cosec}\, x - \cot x| + C$

1.1.3 Integrals Resulting in Inverse Trigonometry

These are **extremely important** for Unit I when dealing with partial fractions containing irreducible quadratic factors (e.g., $x^2 + a^2$).

1. **Inverse Tangent (Standard for Quadratic Denominators):**
$$\int \frac{1}{x^2 + a^2}\, dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$
2. **Inverse Sine:**
$$\int \frac{1}{\sqrt{a^2 - x^2}}\, dx = \sin^{-1}\left(\frac{x}{a}\right) + C$$
3. **Inverse Secant:**
$$\int \frac{1}{x\sqrt{x^2 - a^2}}\, dx = \frac{1}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$$

1.1.4 Standard Irrational Forms

Useful for the "Integration of Irrational functions" section of Unit I.

1. $\int \frac{1}{\sqrt{x^2 + a^2}}\, dx = \ln|x + \sqrt{x^2 + a^2}| + C$
2. $\int \frac{1}{\sqrt{x^2 - a^2}}\, dx = \ln|x + \sqrt{x^2 - a^2}| + C$
3. $\int \frac{1}{x^2 - a^2}\, dx = \frac{1}{2a} \ln\left|\frac{x - a}{x + a}\right| + C \quad (x > a)$
4. $\int \frac{1}{a^2 - x^2}\, dx = \frac{1}{2a} \ln\left|\frac{a + x}{a - x}\right| + C \quad (x < a)$

1.2 Integration using Partial Fractions

Partial fraction decomposition is an algebraic technique used to break down a complex rational function $\frac{P(x)}{Q(x)}$ into a sum of simpler rational expressions (partial fractions). Once decomposed, each term can be integrated using standard logarithmic, power, or inverse trigonometric formulas.

Condition: The rational function must be *proper*, meaning the degree of the numerator $P(x)$ must be strictly less than the degree of the denominator $Q(x)$. If the degree of $P(x) \geq Q(x)$, one must perform polynomial long division first.

The method of integration depends entirely on the nature of the factors of the denominator $Q(x)$. There are three standard cases generally encountered in this syllabus.

Case I: Denominator consists of non-repeated linear factors

If $Q(x)$ can be factored into distinct linear factors $(x - a_1)(x - a_2) \dots (x - a_n)$, the integral takes the form:

$$\int \frac{P(x)}{Q(x)}\, dx = \int \left(\frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \cdots + \frac{A_n}{x - a_n} \right) dx$$

which results in a sum of natural logarithms: $A_1 \ln|x - a_1| + \cdots + C$.

Example 1.2.1. Evaluate $\displaystyle\int \frac{1}{x^2 - 5x + 6}\, dx$.

Solution. First, factor the denominator: $x^2 - 5x + 6 = (x - 2)(x - 3)$. Perform partial fraction decomposition:

$$\frac{1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$$

Solving for constants (using the cover-up method or equating coefficients): $A = -1, B = 1$.

$$\text{Therefore: } \frac{1}{x^2 - 5x + 6} = \frac{-1}{x-2} + \frac{1}{x-3}$$

Now, integrate term by term:

$$\begin{aligned} I &= \int \left(\frac{-1}{x-2} + \frac{1}{x-3} \right) dx \\ &= -\ln|x-2| + \ln|x-3| + C \\ &= \ln\left|\frac{x-3}{x-2}\right| + C \end{aligned}$$

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Example 1.2.2. Evaluate $\displaystyle\int \frac{2x+1}{(x-1)(x+2)}\, dx$.

Solution. Decompose the integrand:

$$\frac{2x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

Multiplying by the denominator: $2x + 1 = A(x + 2) + B(x - 1)$.

- Let $x = 1 \implies 3 = 3A \implies A = 1$.
- Let $x = -2 \implies -3 = -3B \implies B = 1$.

$$\text{Therefore: } \frac{2x+1}{(x-1)(x+2)} = \frac{1}{x-1} + \frac{1}{x+2}$$

Now, integrate:

$$\begin{aligned} I &= \int \frac{1}{x-1}\, dx + \int \frac{1}{x+2}\, dx \\ &= \ln|x-1| + \ln|x+2| + C \\ &= \ln|(x-1)(x+2)| + C \end{aligned}$$

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Case II: Denominator consists of repeated linear factors

If $Q(x)$ contains a linear factor $(x-a)$ repeated k times (i.e., $(x-a)^k$), then the decomposition includes k terms for this factor. The integration involves both logarithmic forms and the power rule:

$$\int \frac{dx}{(x-a)^n} = \frac{(x-a)^{-n+1}}{-n+1} \quad \text{for } n \neq 1$$

Example 1.2.3. Evaluate $\displaystyle\int \frac{x}{(x-2)^2}\,dx$.

Solution. Decompose the integrand:

$$\frac{x}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2}$$

$x = A(x-2) + B$.

- Let $x = 2 \implies 2 = B$.
- Equate coefficients of x: $1 = A$.

$$\frac{x}{(x-2)^2} \quad \frac{1}{x-2} + \frac{2}{(x-2)^2}$$

Now, integrate:

$$\begin{aligned} I &= \int \frac{1}{x-2}\,dx + 2\int (x-2)^{-2}\,dx \\ &= \ln|x-2| + 2\left[\frac{(x-2)^{-1}}{-1}\right] + C \\ &= \ln|x-2| - \frac{2}{x-2} + C \end{aligned}$$

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Example 1.2.4. Evaluate $\displaystyle\int \frac{3x-1}{(x+1)(x-2)^2}\,dx$.

Solution. Decompose the integrand:

$$\frac{3x-1}{(x+1)(x-2)^2} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$$

Solving for constants yields $A = -4/9$, $B = 4/9$, $C = 5/3$.

$$\text{Therefore: } \frac{3x-1}{(x+1)(x-2)^2} \quad \frac{-4}{9(x+1)} + \frac{4}{9(x-2)} + \frac{5}{3(x-2)^2}$$

Now, integrate term by term:

$$\begin{aligned} I &= -\frac{4}{9}\int \frac{dx}{x+1} + \frac{4}{9}\int \frac{dx}{x-2} + \frac{5}{3}\int (x-2)^{-2}\,dx \\ &= -\frac{4}{9}\ln|x+1| + \frac{4}{9}\ln|x-2| + \frac{5}{3}\left(\frac{-1}{x-2}\right) + C \\ &= \frac{4}{9}\ln\left|\frac{x-2}{x+1}\right| - \frac{5}{3(x-2)} + C \end{aligned}$$

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Case III: Denominator contains irreducible quadratic factors

If $Q(x)$ contains a quadratic factor $ax^2 + bx + c$ that cannot be factored into real linear factors (i.e., discriminant $b^2 - 4ac < 0$), the numerator for this term must be linear ($Ax + B$). The integration usually results in a combination of a natural logarithm and an inverse tangent function.

Example 1.2.5. Evaluate $\int \frac{1}{(x+1)(x^2+1)}\,dx$.

Solution. Decompose the integrand:

$$\frac{1}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

$1 = A(x^2+1) + (Bx+C)(x+1)$. Solving yields $A = 1/2$, $B = -1/2$, $C = 1/2$.

$$\frac{1}{(x+1)(x^2+1)} = \frac{1}{2(x+1)} + \frac{-x+1}{2(x^2+1)}$$

Now, integrate. We split the quadratic term into two parts:

$$I = \frac{1}{2}\int \frac{dx}{x+1} - \frac{1}{2}\int \frac{x}{x^2+1}\,dx + \frac{1}{2}\int \frac{1}{x^2+1}\,dx$$

For the second integral, let $u = x^2 + 1 \implies du = 2x\,dx$:

$$\begin{aligned} I &= \frac{1}{2}\ln|x+1| - \frac{1}{2}\left(\frac{1}{2}\ln(x^2+1)\right) + \frac{1}{2}\tan^{-1}(x) + C \\ &= \frac{1}{2}\ln|x+1| - \frac{1}{4}\ln(x^2+1) + \frac{1}{2}\tan^{-1}(x) + C \end{aligned}$$

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Example 1.2.6. Evaluate $\int \frac{5x}{(x-1)(x^2+4)}\,dx$.

Solution. Decompose the integrand:

$$\frac{5x}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$$

$5x = A(x^2+4) + (Bx+C)(x-1)$. Solving yields $A = 1$, $B = -1$, $C = 4$.

$$\frac{5x}{(x-1)(x^2+4)} = \frac{1}{x-1} + \frac{-x+4}{x^2+4}$$

Now, integrate by splitting the quadratic term:

$$I = \int \frac{1}{x-1}\,dx - \int \frac{x}{x^2+4}\,dx + 4\int \frac{1}{x^2+4}\,dx$$

Using standard forms $\int \frac{f'(x)}{f(x)}dx$ and $\int \frac{1}{x^2+a^2}dx = \frac{1}{a}\tan^{-1}(\frac{x}{a})$:

$$\begin{aligned} I &= \ln|x-1| - \frac{1}{2}\ln(x^2+4) + 4\left(\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)\right) + C \\ &= \ln|x-1| - \frac{1}{2}\ln(x^2+4) + 2\tan^{-1}\left(\frac{x}{2}\right) + C \end{aligned}$$

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Example 1.2.7. Evaluate $\int \frac{x^3+x+2}{(x^2+1)^2}\,dx$.

Solution. Since the denominator has the repeated irreducible quadratic factor $(x^2+1)^2$, we set up the decomposition as:

$$\frac{x^3+x+2}{(x^2+1)^2} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2}$$

Multiply by $(x^2+1)^2$ to clear the denominators:

$$x^3+x+2 = (Ax+B)(x^2+1) + (Cx+D)$$

Expand the right side:

$$x^3+x+2 = Ax^3 + Ax + Bx^2 + B + Cx + D$$

Group coefficients of like powers of x:

$$x^3+x+2 = Ax^3 + Bx^2 + (A+C)x + (B+D)$$

Equate coefficients:

- x^3: $A = 1$
- x^2: $B = 0$
- x^1: $A + C = 1 \implies 1 + C = 1 \implies C = 0$
- Constant: $B + D = 2 \implies 0 + D = 2 \implies D = 2$

Thus, the decomposition is:

$$\frac{x^3+x+2}{(x^2+1)^2} = \frac{x}{x^2+1} + \frac{2}{(x^2+1)^2}$$

Now we integrate term by term:

$$I = \int \frac{x}{x^2+1}\,dx + \int \frac{2}{(x^2+1)^2}\,dx$$

Integral 1: Let $I_1 = \int \frac{x}{x^2+1}\,dx$.
Substitute $u = x^2+1$, then $du = 2x\,dx \implies \frac{1}{2}du = x\,dx$.

$$I_1 = \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln|u| = \frac{1}{2}\ln(x^2+1)$$

Integral 2: Let $I_2 = \int \frac{2}{(x^2+1)^2}\,dx$.
Use trigonometric substitution: Let $x = \tan\theta$, then $dx = \sec^2\theta\,d\theta$.
Note that $x^2+1 = \tan^2\theta + 1 = \sec^2\theta$.

$$\begin{aligned} I_2 &= 2\int \frac{\sec^2\theta}{(\sec^2\theta)^2}\,d\theta \\ &= 2\int \frac{1}{\sec^2\theta}\,d\theta \\ &= 2\int \cos^2\theta\,d\theta \end{aligned}$$

Use the identity $\cos^2\theta = \dfrac{1+\cos 2\theta}{2}$:

$$\begin{aligned} I_2 &= 2\int \frac{1+\cos 2\theta}{2}\,d\theta \\ &= \int (1+\cos 2\theta)\,d\theta \\ &= \theta + \frac{1}{2}\sin 2\theta \end{aligned}$$

We know $\theta = \tan^{-1} x$.
Also, $\sin 2\theta = 2\sin\theta\cos\theta$. From the triangle where $\tan\theta = x/1$, we have $\sin\theta = \dfrac{x}{\sqrt{x^2+1}}$ and $\cos\theta = \dfrac{1}{\sqrt{x^2+1}}$.

$$\frac{1}{2}\sin 2\theta = \sin\theta\cos\theta = \frac{x}{x^2+1}$$

So, $I_2 = \tan^{-1} x + \dfrac{x}{x^2+1}$.

Combine I_1 and I_2:

$$I = \frac{1}{2}\ln(x^2+1) + \tan^{-1} x + \frac{x}{x^2+1} + C$$

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1.3 Integration of irrational functions

Integration of irrational functions involves integrands containing fractional powers of the variable x or algebraic expressions. The general strategy for solving these integrals is **Rationalization**. We use specific substitutions to convert the irrational integrand into a rational function (a ratio of polynomials), which can then be integrated using standard methods or partial fractions.

1.3.1 Type I: Fractional Powers of x

Form: Integrands containing terms like $x^{1/n}, x^{1/m}$, etc.
Rule: Let k be the Least Common Multiple (LCM) of the denominators of the fractional powers $(n, m, \dots)$.

$$\text{Substitute } x = t^k \implies dx = kt^{k-1}\,dt$$

Example 1.3.1. Evaluate $\displaystyle\int \frac{1}{\sqrt{x}+\sqrt[3]{x}}\,dx$.

Solution. The powers of x are $1/2$ and $1/3$. The LCM of 2 and 3 is 6.
Substitution: Let $x = t^6$, then $dx = 6t^5\,dt$.

$$\begin{aligned} I &= \int \frac{6t^5}{(t^6)^{1/2}+(t^6)^{1/3}}\,dt \\ &= \int \frac{6t^5}{t^3+t^2}\,dt \\ &= 6\int \frac{t^5}{t^2(t+1)}\,dt = 6\int \frac{t^3}{t+1}\,dt \end{aligned}$$

Perform polynomial long division on $\dfrac{t^3}{t+1}$:

$$\frac{t^3}{t+1} = t^2 - t + 1 - \frac{1}{t+1}$$

Integrating term by term:

$$
\begin{aligned}
I &= 6\int \left(t^2 - t + 1 - \frac{1}{t+1}\right) dt \\
&= 6\left(\frac{t^3}{3} - \frac{t^2}{2} + t - \ln|t+1|\right) + C
\end{aligned}
$$

Substitute back $t = x^{1/6}$:

$$I = 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\ln|\sqrt[6]{x} + 1| + C$$

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Example 1.3.2. Evaluate $\int \frac{\sqrt{x}}{1+x}\,dx$.

Solution. The power is $1/2$. LCM is 2.
Substitution: Let $x = t^2$, then $dx = 2t\,dt$.

$$
\begin{aligned}
I &= \int \frac{t}{1+t^2}(2t)\,dt \\
&= 2\int \frac{t^2}{1+t^2}\,dt \\
&= 2\int \frac{(t^2+1)-1}{1+t^2}\,dt \\
&= 2\int \left(1 - \frac{1}{1+t^2}\right) dt
\end{aligned}
$$

Integrating standard forms:

$$I = 2\left(t - \tan^{-1}(t)\right) + C$$

Substitute back $t = \sqrt{x}$:

$$I = 2\sqrt{x} - 2\tan^{-1}(\sqrt{x}) + C$$

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1.3.2 Type II: Linear term outside, Linear term inside

Form: $\int \frac{dx}{(ax+b)\sqrt{cx+d}}$
Rule: Substitute the expression inside the radical equal to t^2.

$$\text{Substitute } cx + d = t^2$$

Example 1.3.3. Evaluate $\int \frac{dx}{(x+1)\sqrt{x+2}}$.

Solution. **Substitution:** Let $x+2 = t^2 \implies dx = 2t\,dt$. Also, $x = t^2 - 2$, so $(x+1) = (t^2-2)+1 = t^2-1$.
Substitute into the integral:

$$
\begin{aligned}
I &= \int \frac{2t\,dt}{(t^2-1)\sqrt{t^2}} \\
&= \int \frac{2t}{(t^2-1)t}\,dt \\
&= 2\int \frac{1}{t^2-1}\,dt
\end{aligned}
$$

Using the standard formula $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right|$:

$$I = 2 \cdot \frac{1}{2(1)}\ln\left|\frac{t-1}{t+1}\right| + C = \ln\left|\frac{t-1}{t+1}\right| + C$$

Substitute back $t = \sqrt{x+2}$:

$$I = \ln\left|\frac{\sqrt{x+2}-1}{\sqrt{x+2}+1}\right| + C$$

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Example 1.3.4. Evaluate $\int \frac{dx}{(2x-3)\sqrt{x+1}}$.

Solution. **Substitution:** Let $x+1 = t^2 \implies dx = 2t\,dt$. Also, $x = t^2 - 1$. Then denominator term $2x - 3 = 2(t^2-1) - 3 = 2t^2 - 5$.

Substitute:

$$\begin{aligned} I &= \int \frac{2t\,dt}{(2t^2-5)t} \\ &= 2\int \frac{dt}{2t^2-5} = \int \frac{dt}{t^2 - \frac{5}{2}} \\ &= \int \frac{dt}{t^2 - (\sqrt{5/2})^2} \end{aligned}$$

Using the standard formula:

$$I = \frac{1}{2(\sqrt{5/2})} \ln\left|\frac{t-\sqrt{5/2}}{t+\sqrt{5/2}}\right| + C$$

$$I = \frac{1}{\sqrt{10}} \ln\left|\frac{\sqrt{2}t-\sqrt{5}}{\sqrt{2}t+\sqrt{5}}\right| + C$$

Substitute back $t = \sqrt{x+1}$:

$$I = \frac{1}{\sqrt{10}} \ln\left|\frac{\sqrt{2(x+1)}-\sqrt{5}}{\sqrt{2(x+1)}+\sqrt{5}}\right| + C$$

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1.3.3 Type III: Linear term outside, Quadratic term inside

Form: $\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$

Rule: Substitute the linear expression outside the radical equal to $1/t$.

$$\text{Substitute } ax+b = \frac{1}{t}$$

Example 1.3.5. Evaluate $\int \frac{dx}{x\sqrt{x^2+4}}$.

Solution. **Substitution:** Let $x = \frac{1}{t} \implies dx = -\frac{1}{t^2}\,dt$.

Substitute into the integral:

$$\begin{aligned}
I &= \int \frac{-\frac{1}{t^2}\,dt}{\frac{1}{t}\sqrt{\left(\frac{1}{t}\right)^2+4}} \\
&= -\int \frac{\frac{1}{t^2}\,dt}{\frac{1}{t}\sqrt{\frac{1+4t^2}{t^2}}} \\
&= -\int \frac{\frac{1}{t^2}}{\frac{1}{t}\cdot\frac{1}{t}\sqrt{1+4t^2}}\,dt \\
&= -\int \frac{dt}{\sqrt{1+4t^2}}
\end{aligned}$$

This is a standard form. Factor out 4 inside the root: $\sqrt{4(1/4+t^2)} = 2\sqrt{t^2+(1/2)^2}$.

$$\begin{aligned}
I &= -\frac{1}{2}\int \frac{dt}{\sqrt{t^2+(1/2)^2}} \\
&= -\frac{1}{2}\ln\left|t+\sqrt{t^2+\frac{1}{4}}\right| + C
\end{aligned}$$

Substitute back $t = 1/x$:

$$I = -\frac{1}{2}\ln\left|\frac{1}{x}+\sqrt{\frac{1}{x^2}+\frac{1}{4}}\right| + C$$

Simplify:

$$I = -\frac{1}{2}\ln\left|\frac{2+\sqrt{4+x^2}}{2x}\right| + C$$

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Example 1.3.6. Evaluate $\displaystyle\int \frac{dx}{(x-1)\sqrt{x^2-1}}$.

Solution. **Substitution:** Let $x - 1 = \frac{1}{t} \implies x = 1 + \frac{1}{t}$. $dx = -\frac{1}{t^2}\,dt$.

Substitute into the quadratic term:

$$x^2 - 1 = \left(1+\frac{1}{t}\right)^2 - 1 = 1 + \frac{2}{t} + \frac{1}{t^2} - 1 = \frac{2t+1}{t^2}$$

Substitute into the integral:

$$\begin{aligned}
I &= \int \frac{-\frac{1}{t^2}\,dt}{\frac{1}{t}\sqrt{\frac{2t+1}{t^2}}} \\
&= -\int \frac{\frac{1}{t^2}}{\frac{1}{t^2}\sqrt{2t+1}}\,dt \\
&= -\int (2t+1)^{-1/2}\,dt
\end{aligned}$$

Integrate using power rule:

$$I = -\frac{(2t+1)^{1/2}}{1/2 \cdot 2} + C = -\sqrt{2t+1} + C$$

Substitute back $t = \dfrac{1}{x-1}$:

$$I = -\sqrt{\frac{2}{x-1} + 1} + C = -\sqrt{\frac{x+1}{x-1}} + C$$

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1.3.4 Type IV: Quadratic term outside, Linear term inside

Form: $\displaystyle\int \frac{dx}{(ax^2+b)\sqrt{cx+d}}$

Rule: Substitute the expression inside the radical equal to t^2.

$$\text{Substitute } cx + d = t^2$$

(This will convert the integrand into a form solvable by partial fractions).

Example 1.3.7. Evaluate $\displaystyle\int \frac{dx}{(x^2+1)\sqrt{x+1}}$.

Solution. **Substitution:** Let $x+1 = t^2 \implies dx = 2t\,dt$. Also, $x = t^2 - 1$, so $x^2 + 1 = (t^2-1)^2 + 1 = t^4 - 2t^2 + 2$.

Substitute:

$$I = \int \frac{2t\,dt}{(t^4 - 2t^2 + 2)(t)} = 2\int \frac{dt}{t^4 - 2t^2 + 2}$$

We have

$$\begin{aligned}
\int \frac{1}{t^4 - 2t^2 + 2}\,dt &= \frac{1}{2\sqrt{2}} \int \frac{(t^2+\sqrt{2}) - (t^2 - \sqrt{2})}{t^4 - 2t^2 + 2}\,dt \\
&= \frac{1}{2\sqrt{2}} \underbrace{\int \frac{1 + \frac{\sqrt{2}}{t^2}}{t^2 - 2 + \frac{2}{t^2}}\,dt}_{I_1} - \frac{1}{2\sqrt{2}} \underbrace{\int \frac{1 - \frac{\sqrt{2}}{t^2}}{t^2 - 2 + \frac{2}{t^2}}\,dt}_{I_2}
\end{aligned}$$

For I_1, let $u = t - \frac{\sqrt{2}}{t} \implies du = \left(1 + \frac{\sqrt{2}}{t^2}\right) dt$. The denominator transforms to $u^2 + (2\sqrt{2} - 2)$:

$$I_1 = \int \frac{du}{u^2 + (2\sqrt{2}-2)} = \frac{1}{\sqrt{2\sqrt{2}-2}} \arctan\left(\frac{t^2 - \sqrt{2}}{t\sqrt{2\sqrt{2}-2}}\right)$$

For I_2, let $v = t + \frac{\sqrt{2}}{t} \implies dv = \left(1 - \frac{\sqrt{2}}{t^2}\right) dt$. The denominator transforms to $v^2 - (2\sqrt{2} + 2)$:

$$I_2 = \int \frac{dv}{v^2 - (2\sqrt{2}+2)} = \frac{1}{2\sqrt{2\sqrt{2}+2}} \ln\left|\frac{t^2 - t\sqrt{2\sqrt{2}+2} + \sqrt{2}}{t^2 + t\sqrt{2\sqrt{2}+2} + \sqrt{2}}\right|$$

Substituting I_1 and I_2 back into $I = \frac{1}{2\sqrt{2}}(I_1 - I_2)$ and rationalizing the constant coefficients, we get the final answer:

$$\begin{aligned}
I &= \frac{\sqrt{\sqrt{2}+1}}{2} \arctan\left(\frac{t^2 - \sqrt{2}}{t\sqrt{2\sqrt{2}-2}}\right) - \frac{\sqrt{\sqrt{2}-1}}{4} \ln\left|\frac{t^2 - t\sqrt{2\sqrt{2}+2} + \sqrt{2}}{t^2 + t\sqrt{2\sqrt{2}+2} + \sqrt{2}}\right| + C \\
&= \frac{\sqrt{\sqrt{2}+1}}{2} \arctan\left(\frac{x + 1 - \sqrt{2}}{\sqrt{x+1}\sqrt{2\sqrt{2}-2}}\right) - \frac{\sqrt{\sqrt{2}-1}}{4} \ln\left|\frac{x + 1 - \sqrt{x+1}\sqrt{2\sqrt{2}+2} + \sqrt{2}}{x + 1 + \sqrt{x+1}\sqrt{2\sqrt{2}+2} + \sqrt{2}}\right| + C
\end{aligned}$$

■

Example 1.3.8. Evaluate $\int \frac{dx}{(x^2-4)\sqrt{x+1}}$.

Solution. **Substitution:** Let $x+1=t^2 \implies x=t^2-1, dx=2t\,dt$. $x^2-4=(t^2-1)^2-4=t^4-2t^2-3=(t^2-3)(t^2+1)$.

Substitute:

$$I=\int \frac{2t\,dt}{(t^2-3)(t^2+1)t}=2\int \frac{dt}{(t^2-3)(t^2+1)}$$

Use Partial Fractions (let $u=t^2$ for decomposition only):

$$\frac{1}{(u-3)(u+1)}=\frac{1}{4}\left(\frac{1}{u-3}-\frac{1}{u+1}\right)$$

So,

$$\begin{aligned} I &= 2\cdot\frac{1}{4}\int\left(\frac{1}{t^2-3}-\frac{1}{t^2+1}\right)dt \\ &= \frac{1}{2}\left[\frac{1}{2\sqrt{3}}\ln\left|\frac{t-\sqrt{3}}{t+\sqrt{3}}\right|-\tan^{-1}(t)\right]+C \end{aligned}$$

Substitute back $t=\sqrt{x+1}$. ■

1.4 The Universal Rule: Tangent Half-Angle Substitution

To integrate functions of the form $\int \frac{dx}{a+b\cos x}$ (or similarly involving $\sin x$), we use the **Tangent Half-Angle Substitution** (also known as the Weierstrass substitution). This method converts the trigonometric integral into a rational algebraic integral.

The Procedure:

1. **Substitute:** Let $t=\tan\left(\frac{x}{2}\right)$.

2. **Differentiate:**

$$x=2\tan^{-1}(t) \implies dx=\frac{2}{1+t^2}\,dt$$

3. **Replace Trigonometric Terms:** Using the half-angle identities:

$$\cos x=\frac{1-t^2}{1+t^2} \quad \text{and} \quad \sin x=\frac{2t}{1+t^2}$$

Substituting these into the integral yields:

$$I=\int \frac{1}{a+b\left(\frac{1-t^2}{1+t^2}\right)}\cdot\frac{2}{1+t^2}\,dt=\int \frac{2\,dt}{a(1+t^2)+b(1-t^2)}$$

Simplifying the denominator:

$$I=\int \frac{2\,dt}{(a+b)+(a-b)t^2}$$

The method of integration now depends on the signs of coefficients $(a+b)$ and $(a-b)$, leading to three distinct cases.

1.4.1 Case I: $a > b$ (Result involves Inverse Tangent)

If $a > b$, then both $(a+b)$ and $(a-b)$ are positive. The denominator takes the form of sum of squares $(A^2 + t^2)$.

Example 1.4.1. Evaluate $\int \frac{1}{5+4\cos x}\,dx$.

Solution. Here $a = 5, b = 4$ ($a > b$).

Let $t = \tan(x/2)$, $dx = \frac{2dt}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$.

$$\begin{aligned}
I &= \int \frac{1}{5+4\left(\frac{1-t^2}{1+t^2}\right)} \cdot \frac{2}{1+t^2}\,dt \\
&= \int \frac{2\,dt}{5(1+t^2)+4(1-t^2)} \\
&= \int \frac{2\,dt}{5+5t^2+4-4t^2} \\
&= \int \frac{2\,dt}{t^2+9}
\end{aligned}$$

This is the standard form $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}(\frac{x}{a})$.

$$I = 2\int \frac{dt}{t^2+3^2} = 2\left(\frac{1}{3}\tan^{-1}\left(\frac{t}{3}\right)\right) + C$$

Substitute $t = \tan(x/2)$:

$$\text{This gives:} \quad I = \frac{2}{3}\tan^{-1}\left(\frac{\tan(x/2)}{3}\right) + C$$

■

1.4.2 Case II: $b > a$ (Result involves Logarithms)

If $b > a$, then $(a-b)$ is negative. The denominator takes the form of difference of squares $(A^2 - t^2)$.

Example 1.4.2. Evaluate $\int \frac{1}{3+5\cos x}\,dx$.

Solution. Here $a = 3, b = 5$ ($b > a$).

Let $t = \tan(x/2)$.

$$\begin{aligned}
I &= \int \frac{2\,dt}{3(1+t^2)+5(1-t^2)} \\
&= \int \frac{2\,dt}{3+3t^2+5-5t^2} \\
&= \int \frac{2\,dt}{8-2t^2} \\
&= \int \frac{2\,dt}{2(4-t^2)} = \int \frac{dt}{4-t^2}
\end{aligned}$$

This is the standard form $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln|\frac{a+x}{a-x}|$.

$$I = \int \frac{dt}{2^2-t^2} = \frac{1}{2(2)}\ln\left|\frac{2+t}{2-t}\right| + C$$

$$\text{Using value of } x \text{ gives:} \qquad I = \frac{1}{4}\ln\left|\frac{2+\tan(x/2)}{2-\tan(x/2)}\right| + C$$

■

1.4.3 Case III: $a = b$ (Parabolic Case)

If $a = b$, the t^2 term in the denominator cancels out completely. While the t-substitution works, it is often faster to use trigonometric identities directly.

Example 1.4.3. Evaluate $\displaystyle\int \frac{1}{1+\cos x}\,dx$.

Solution. **Method A: Using t-substitution**

$$I = \int \frac{2\,dt}{1(1+t^2)+1(1-t^2)} = \int \frac{2\,dt}{2} = \int dt = t + C$$

Therefore: $\int \dfrac{1}{1+\cos x} dx = \tan(x/2) + C$.

Method B: Using Half-Angle Identity (Faster) Recall that $1+\cos x = 2\cos^2(x/2)$.

$$\begin{aligned} I &= \int \frac{dx}{2\cos^2(x/2)} \\ &= \frac{1}{2}\int \sec^2(x/2)\,dx \end{aligned}$$

Integration of $\sec^2 u$ is $\tan u$. Note the coefficient of x is $1/2$. Therefore,

$$I = \frac{1}{2}\cdot\frac{\tan(x/2)}{1/2} + C = \tan(x/2) + C$$

■

Exercise 1.4.4. Evaluate the following integrals:

1. $\int \frac{x}{(x-1)(x-2)(x-3)}\,dx$
2. $\int \frac{x+1}{x^2-5x+6}\,dx$
3. $\int \frac{2x-1}{(x+1)^2(x-2)}\,dx$
4. $\int \frac{x^2}{(x-1)^3}\,dx$
5. $\int \frac{x}{(x-1)(x^2+4)}\,dx$
6. $\int \frac{dx}{x^3-1}$
7. $\int \frac{dx}{(x^2+1)(x^2+2)}$
8. $\int \frac{x^2+1}{x(x^2+3)}\,dx$
9. $\int \frac{dx}{x\sqrt{x-1}}$
10. $\int \frac{dx}{(x-2)\sqrt{x+3}}$
11. $\int \frac{dx}{\sqrt{x}+\sqrt[3]{x}}$
12. $\int \frac{\sqrt{x}}{x+1}\,dx$
13. $\int \frac{dx}{(x+1)\sqrt{x^2+1}}$
14. $\int \frac{x}{(x-1)\sqrt{x+2}}\,dx$
15. $\int \frac{dx}{5+4\cos x}$
16. $\int \frac{dx}{3+5\sin x}$
17. $\int \frac{dx}{1+\sin x+\cos x}$
18. $\int \frac{dx}{4-5\cos x}$
19. $\int \frac{dx}{2\sin x+3\cos x}$
20. $\int \frac{dx}{5-3\cos x+4\sin x}$

Chapter 2

Reduction Formulae

This unit introduces reduction formulae, a powerful recursive method designed to integrate higher powers and products of trigonometric functions. By establishing relationships between integrals of different powers, students can systematically simplify and solve complex trigonometric problems. These formulas serve as essential computational tools that streamline the evaluation of otherwise lengthy integrals.

2.1 Introduction

Reduction formulae are recurrence relations that allow us to integrate higher powers of trigonometric functions by reducing the power n step-by-step until a basic integral is reached.

2.1.1 Reduction Formula for Sine

Formula 1 (Integral of $\sin^n x$)**.** If $I_n = \int \sin^n x\,dx$, where $n \geq 2$ is an integer, then:

$$I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$$

Derivation

Let $I_n = \displaystyle\int \sin^n x\,dx$. We write $\sin^n x = \sin^{n-1} x \cdot \sin x$.

$$I_n = \int \sin^{n-1} x \cdot \sin x\,dx$$

Apply **Integration by Parts**: $\int u\,dv = uv - \int v\,du$.

- Let $u = \sin^{n-1} x$. Then $du = (n-1)\sin^{n-2} x \cos x\,dx$.
- Let $dv = \sin x\,dx$. Then $v = -\cos x$.

Substituting these values:

$$I_n = \sin^{n-1} x(-\cos x) - \int (-\cos x) \cdot (n-1)\sin^{n-2} x \cos x\,dx$$
$$I_n = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x\,dx$$

Using the identity $\cos^2 x = 1 - \sin^2 x$:

$$I_n = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x(1 - \sin^2 x)\,dx$$
$$I_n = -\sin^{n-1} x \cos x + (n-1) \left[\int \sin^{n-2} x\,dx - \int \sin^n x\,dx \right]$$

Substitute back I_{n-2} and I_n:

$$I_n = -\sin^{n-1} x \cos x + (n-1)I_{n-2} - (n-1)I_n$$

Move the term $-(n-1)I_n$ to the left side:

$$\begin{aligned} I_n + (n-1)I_n &= -\sin^{n-1} x \cos x + (n-1)I_{n-2} \\ nI_n &= -\sin^{n-1} x \cos x + (n-1)I_{n-2} \end{aligned}$$

Dividing by n gives the final rule:

$$I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$$

Example 2.1.1. Evaluate $\int \sin^3 x \, dx$.

Solution. Here $n = 3$. Applying the reduction formula:

$$I_3 = -\frac{\sin^{3-1} x \cos x}{3} + \frac{3-1}{3} I_{3-2}$$

$$I_3 = -\frac{\sin^2 x \cos x}{3} + \frac{2}{3} \int \sin x \, dx$$

We know that $\int \sin x \, dx = -\cos x$. Substituting this back:

$$I_3 = -\frac{1}{3} \sin^2 x \cos x + \frac{2}{3}(-\cos x) + C$$

Therefore, $\int \sin^3 x \, dx = -\frac{1}{3} \sin^2 x \cos x - \frac{2}{3} \cos x + C$ ∎

Example 2.1.2. Evaluate $\int \sin^4 x \, dx$.

Solution. Here $n = 4$. Apply formula for $n = 4$.

$$I_4 = -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} I_2$$

Find I_2 using the formula for $n = 2$.

$$I_2 = -\frac{\sin x \cos x}{2} + \frac{1}{2} I_0$$

Here, $I_0 = \int \sin^0 x \, dx = \int 1 \, dx = x$. So, $I_2 = -\frac{1}{2} \sin x \cos x + \frac{1}{2} x$.

Substitute I_2 back into the equation for I_4.

$$\begin{aligned} I_4 &= -\frac{\sin^3 x \cos x}{4} + \frac{3}{4} \left(-\frac{1}{2} \sin x \cos x + \frac{1}{2} x \right) \\ &= -\frac{1}{4} \sin^3 x \cos x - \frac{3}{8} \sin x \cos x + \frac{3}{8} x + C \end{aligned}$$

Therefore, $\int \sin^4 x \, dx = \frac{3}{8} x - \frac{3}{8} \sin x \cos x - \frac{1}{4} \sin^3 x \cos x + C$ ∎

2.1.2 Reduction Formula for Cosine

Formula 2 (Integral of $\cos^n x$)**.** If $J_n = \int \cos^n x\,dx$, where $n \geq 2$ is an integer, then:

$$J_n = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} J_{n-2}$$

The derivation process is identical to the sine derivation. 1. Split $\cos^n x$ into $\cos^{n-1} x \cdot \cos x$. 2. Use Integration by Parts with $u = \cos^{n-1} x$ and $dv = \cos x\,dx$. 3. Use the identity $\sin^2 x = 1 - \cos^2 x$ in the intermediate step.

Solution. Here $n = 3$. Using the cosine rule:

$$J_3 = \frac{\cos^2 x \sin x}{3} + \frac{2}{3} J_1$$

Where $J_1 = \int \cos x\,dx = \sin x$.

$$J_3 = \frac{\cos^2 x \sin x}{3} + \frac{2}{3} \sin x + C$$

■

2.1.3 Wallis's Formula (Definite Integrals)

For definite integrals from 0 to $\frac{\pi}{2}$, the reduction formula simplifies into a direct multiplication rule.

Formula 3. Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x\,dx = \int_0^{\frac{\pi}{2}} \cos^n x\,dx$.

- **Case 1 (n is Even):**

$$I_n = \frac{n-1}{n} \times \frac{n-3}{n-2} \times \cdots \times \frac{1}{2} \times \frac{\pi}{2}$$

- **Case 2 (n is Odd):**

$$I_n = \frac{n-1}{n} \times \frac{n-3}{n-2} \times \cdots \times \frac{2}{3} \times 1$$

By repeatedly applying the recursive relation $I_n = \frac{n-1}{n} I_{n-2}$, we can expand I_n down to the base cases I_1 or I_0.

Base Cases

- **If n is Even (ends at I_0):**

$$I_0 = \int_0^{\frac{\pi}{2}} \sin^0 x\,dx = \int_0^{\frac{\pi}{2}} 1\,dx = [x]_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

- **If n is Odd (ends at I_1):**

$$I_1 = \int_0^{\frac{\pi}{2}} \sin^1 x\,dx = [-\cos x]_0^{\frac{\pi}{2}} = -(\cos \frac{\pi}{2} - \cos 0) = -(0-1) = 1$$

Theorem 2.1.3 (Wallis's Formula)**.** ***Case 1: When n is Even***

$$\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \int_0^{\frac{\pi}{2}} \cos^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\dots\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

Case 2: When n is Odd

$$\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \int_0^{\frac{\pi}{2}} \cos^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\dots\cdot\frac{2}{3}\cdot 1$$

Example 2.1.4. Evaluate $\int_0^{\frac{\pi}{2}} \sin^5 x\,dx$.

Solution. Here $n = 5$ (Odd). We start with $\frac{n-1}{n}$ and subtract 2 from numerator and denominator until we reach 1.

$$I_5 = \frac{5-1}{5}\times\frac{5-3}{5-2}\times 1$$
$$I_5 = \frac{4}{5}\times\frac{2}{3}\times 1$$
$$I_5 = \frac{8}{15}$$

■

Example 2.1.5. Evaluate $\int_0^{\frac{\pi}{2}} \sin^6 x\,dx$.

Solution. Here $n = 6$ (Even). Since n is even, we multiply by the factor $\frac{\pi}{2}$ at the end.

$$I_6 = \frac{6-1}{6}\times\frac{6-3}{6-2}\times\frac{6-5}{6-4}\times\frac{\pi}{2}$$
$$I_6 = \frac{5}{6}\times\frac{3}{4}\times\frac{1}{2}\times\frac{\pi}{2}$$
$$I_6 = \frac{15\pi}{96} = \frac{5\pi}{32}$$

■

Example 2.1.6. Evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x\,dx$.

Solution. $n = 6$ is even.

$$J_6 = \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{15\pi}{96} = \frac{5\pi}{32}$$

■

2.2 Reduction Formula for $\int \tan^n x\,dx$

Let $I_n = \int \tan^n x\,dx$, where n is an integer and $n > 1$.

Derivation

We split the power of $\tan x$ into $\tan^{n-2} x$ and $\tan^2 x$:

$$I_n = \int \tan^{n-2} x \cdot \tan^2 x \, dx$$

Using the trigonometric identity $\tan^2 x = \sec^2 x - 1$:

$$I_n = \int \tan^{n-2} x(\sec^2 x - 1) \, dx$$
$$I_n = \int \tan^{n-2} x \sec^2 x \, dx - \int \tan^{n-2} x \, dx$$

The second integral is simply I_{n-2}. For the first integral $\int \tan^{n-2} x \sec^2 x \, dx$, we use substitution:

- Let $u = \tan x$, then $du = \sec^2 x \, dx$.
- The integral becomes $\int u^{n-2} \, du = \dfrac{u^{n-1}}{n-1} = \dfrac{\tan^{n-1} x}{n-1}$.

Combining these results gives the reduction formula:

$$\boxed{I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}}$$

Example 2.2.1. Evaluate $\int \tan^4 x \, dx$.

Solution. We use the reduction formula: $I_n = \dfrac{\tan^{n-1} x}{n-1} - I_{n-2}$.

Here $n = 4$.

$$I_4 = \frac{\tan^3 x}{3} - I_2$$

Now we find I_2 (where $n = 2$):

$$I_2 = \frac{\tan^1 x}{1} - I_0$$

We know that $I_0 = \int \tan^0 x \, dx = \int 1 \, dx = x$. Therefore:

$$I_2 = \tan x - x$$

Substituting I_2 back into the equation for I_4:

$$I_4 = \frac{1}{3} \tan^3 x - (\tan x - x) + C$$

Therefor, $\int \tan^4 x \, dx = \dfrac{1}{3} \tan^3 x - \tan x + x + C$. ■

2.3 Reduction Formula for $\int \cot^n x \, dx$

Let $I_n = \int \cot^n x \, dx$, where n is an integer and $n > 1$.

Derivation

Similar to the tangent case, we split the power into $\cot^{n-2} x$ and $\cot^2 x$:

$$I_n = \int \cot^{n-2} x \cdot \cot^2 x \, dx$$

Using the trigonometric identity $\cot^2 x = \operatorname{cosec}^2 x - 1$:

$$I_n = \int \cot^{n-2} x(\operatorname{cosec}^2 x - 1)\,dx$$
$$I_n = \int \cot^{n-2} x \operatorname{cosec}^2 x\,dx - \int \cot^{n-2} x\,dx$$

The second integral is I_{n-2}. For the first integral, let $u = \cot x$, then $du = -\operatorname{cosec}^2 x\,dx$ (so $-du = \operatorname{cosec}^2 x\,dx$).

$$\int \cot^{n-2} x \operatorname{cosec}^2 x\,dx = \int u^{n-2}(-du) = -\frac{u^{n-1}}{n-1} = -\frac{\cot^{n-1} x}{n-1}$$

Combining these results gives the reduction formula:

$$\boxed{I_n = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}}$$

Example 2.3.1. Evaluate $\int \cot^3 x\,dx$.

Solution. We use the reduction formula: $I_n = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}$.

Here $n = 3$.

$$I_3 = -\frac{\cot^2 x}{2} - I_1$$

Now we evaluate I_1:

$$I_1 = \int \cot x\,dx = \int \frac{\cos x}{\sin x}\,dx = \ln|\sin x|$$

Substituting I_1 back into the equation for I_3:

$$I_3 = -\frac{1}{2}\cot^2 x - \ln|\sin x| + C$$

Therefore, $\int \cot^3 x\,dx = -\frac{1}{2}\cot^2 x - \ln|\sin x| + C$. ■

2.4 Reduction Formula for $\int \sec^n x\,dx$

Let $I_n = \int \sec^n x\,dx$, where $n > 1$.

Derivation

We split the term $\sec^n x$ into $\sec^{n-2} x \cdot \sec^2 x$ to apply **Integration by Parts**.

$$I_n = \int \sec^{n-2} x \cdot \sec^2 x\,dx$$

Let $u = \sec^{n-2} x$ and $dv = \sec^2 x\,dx$.

- $du = (n-2)\sec^{n-3} x \cdot (\sec x \tan x)\,dx = (n-2)\sec^{n-2} x \tan x\,dx$
- $v = \int \sec^2 x\,dx = \tan x$

Applying integration by parts ($uv - \int v\,du$):

$$I_n = \sec^{n-2} x \tan x - \int \tan x \cdot (n-2)\sec^{n-2} x \tan x\,dx$$
$$I_n = \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x \tan^2 x\,dx$$

Using the identity $\tan^2 x = \sec^2 x - 1$:

$$
\begin{aligned}
I_n &= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1)\, dx \\
I_n &= \sec^{n-2} x \tan x - (n-2) \left[\int \sec^n x\, dx - \int \sec^{n-2} x\, dx \right] \\
I_n &= \sec^{n-2} x \tan x - (n-2) I_n + (n-2) I_{n-2}
\end{aligned}
$$

Now, bring the term with I_n to the left side:

$$
\begin{aligned}
I_n + (n-2) I_n &= \sec^{n-2} x \tan x + (n-2) I_{n-2} \\
I_n (1 + n - 2) &= \sec^{n-2} x \tan x + (n-2) I_{n-2} \\
(n-1) I_n &= \sec^{n-2} x \tan x + (n-2) I_{n-2}
\end{aligned}
$$

Dividing by $(n-1)$, we get the final formula:

$$
\boxed{I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}}
$$

Example 2.4.1. Evaluate $\int \sec^3 x\, dx$.

Solution. We use the reduction formula: $I_n = \dfrac{\sec^{n-2} x \tan x}{n-1} + \dfrac{n-2}{n-1} I_{n-2}$.

Here $n = 3$.

$$
I_3 = \frac{\sec^{3-2} x \tan x}{3-1} + \frac{3-2}{3-1} I_{3-2}
$$

$$
I_3 = \frac{\sec x \tan x}{2} + \frac{1}{2} I_1
$$

Now we evaluate I_1:

$$
I_1 = \int \sec x\, dx = \ln |\sec x + \tan x|
$$

Substituting I_1 back into the equation for I_3:

$$
I_3 = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C
$$

Therefore, $\int \sec^3 x\, dx = \dfrac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C$. ■

2.5 Reduction Formula for $\int \operatorname{cosec}^n x\, dx$

Let $I_n = \int \operatorname{cosec}^n x\, dx$, where $n > 1$.

Derivation

Split $\operatorname{cosec}^n x$ into $\operatorname{cosec}^{n-2} x \cdot \operatorname{cosec}^2 x$ and use **Integration by Parts**.

$$
I_n = \int \operatorname{cosec}^{n-2} x \cdot \operatorname{cosec}^2 x\, dx
$$

Let $u = \operatorname{cosec}^{n-2} x$ and $dv = \operatorname{cosec}^2 x\, dx$.

- $du = (n-2) \operatorname{cosec}^{n-3} x (-\operatorname{cosec} x \cot x)\, dx = -(n-2) \operatorname{cosec}^{n-2} x \cot x\, dx$
- $v = \int \operatorname{cosec}^2 x\, dx = -\cot x$

Applying integration by parts:

$$I_n = \operatorname{cosec}^{n-2} x(-\cot x) - \int (-\cot x)[-(n-2)\operatorname{cosec}^{n-2} x \cot x]\,dx$$
$$I_n = -\operatorname{cosec}^{n-2} x \cot x - (n-2)\int \operatorname{cosec}^{n-2} x \cot^2 x\,dx$$

Using the identity $\cot^2 x = \operatorname{cosec}^2 x - 1$:

$$I_n = -\operatorname{cosec}^{n-2} x \cot x - (n-2)\int \operatorname{cosec}^{n-2} x(\operatorname{cosec}^2 x - 1)\,dx$$
$$I_n = -\operatorname{cosec}^{n-2} x \cot x - (n-2)[I_n - I_{n-2}]$$
$$I_n = -\operatorname{cosec}^{n-2} x \cot x - (n-2)I_n + (n-2)I_{n-2}$$

Move $(n-2)I_n$ to the left side:

$$I_n + (n-2)I_n = -\operatorname{cosec}^{n-2} x \cot x + (n-2)I_{n-2}$$
$$(n-1)I_n = -\operatorname{cosec}^{n-2} x \cot x + (n-2)I_{n-2}$$

Dividing by $(n-1)$, we get the final formula:

$$\boxed{I_n = \frac{-\operatorname{cosec}^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1}I_{n-2}}$$

Example 2.5.1. Evaluate $\int \operatorname{cosec}^4 x\,dx$.

Solution. We use the reduction formula: $I_n = \frac{-\operatorname{cosec}^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1}I_{n-2}$.

Here $n = 4$.

$$I_4 = \frac{-\operatorname{cosec}^2 x \cot x}{3} + \frac{2}{3}I_2$$

Now we evaluate I_2:

$$I_2 = \int \operatorname{cosec}^2 x\,dx = -\cot x$$

Substituting I_2 back into the equation for I_4:

$$I_4 = -\frac{1}{3}\operatorname{cosec}^2 x \cot x + \frac{2}{3}(-\cot x) + C$$

Therefore, $\int \operatorname{cosec}^4 x\,dx = -\frac{1}{3}\operatorname{cosec}^2 x \cot x - \frac{2}{3}\cot x + C$. ■

Summary Table

Function	**Reduction Formula**
$\int \sin^n x\,dx$	$-\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}I_{n-2}$
$\int \cos^n x\,dx$	$\frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n}I_{n-2}$
$\int \tan^n x\,dx$	$\frac{\tan^{n-1} x}{n-1} - I_{n-2}$
$\int \cot^n x\,dx$	$-\frac{\cot^{n-1} x}{n-1} - I_{n-2}$
$\int \sec^n x\,dx$	$\frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1}I_{n-2}$
$\int \operatorname{cosec}^n x\,dx$	$\frac{-\operatorname{cosec}^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1}I_{n-2}$

Exercise 2.5.2. Evaluate the following integrals using appropriate reduction formulae.

1. $\int \sin^5 x \, dx$
2. $\int \cos^6 x \, dx$
3. $\int \tan^5 x \, dx$
4. $\int \cot^4 x \, dx$
5. $\int \sec^3 x \, dx$
6. $\int \operatorname{cosec}^5 x \, dx$
7. $\int_0^{\frac{\pi}{2}} \sin^8 x \, dx$
8. $\int_0^{\frac{\pi}{2}} \cos^9 x \, dx$
9. $\int_0^{\frac{\pi}{4}} \tan^4 x \, dx$
10. $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^3 x \, dx$
11. $\int \sec^6 x \, dx$
12. $\int \operatorname{cosec}^4 x \, dx$
13. $\int \sin^4 x \, dx$
14. $\int_0^{\frac{\pi}{4}} \sec^4 x \, dx$
15. If $I_n = \int \tan^n x \, dx$, evaluate $I_7 + I_5$.

Answers to Exercises 2.5.2

1. $-\frac{1}{5}\sin^4 x \cos x - \frac{4}{15}\sin^2 x \cos x - \frac{8}{15}\cos x + C$
2. $\frac{1}{6}\cos^5 x \sin x + \frac{5}{24}\cos^3 x \sin x + \frac{5}{16}\cos x \sin x + \frac{5}{16}x + C$
3. $\frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + \ln|\sec x| + C$
4. $-\frac{1}{3}\cot^3 x + \cot x + x + C$
5. $\frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$
6. $-\frac{1}{4}\operatorname{cosec}^3 x \cot x - \frac{3}{8}\operatorname{cosec} x \cot x + \frac{3}{8}\ln|\operatorname{cosec} x - \cot x| + C$
7. $\frac{35\pi}{256}$
8. $\frac{128}{315}$
9. $3 - \frac{\pi}{4}$ (Simplifies to $\frac{12-\pi}{4}$) – *Wait, check calculation:* $\tan^3/3 - \tan + x$. *At* $\pi/4$: $1/3 - 1 + \pi/4 = \pi/4 - 2/3$
10. $\frac{1}{2}(1 - \ln 2)$
11. $\frac{1}{5}\sec^4 x \tan x + \frac{4}{15}\sec^2 x \tan x + \frac{8}{15}\tan x + C$
12. $-\frac{1}{3}\operatorname{cosec}^2 x \cot x - \frac{2}{3}\cot x + C$
13. $-\frac{1}{4}\sin^3 x \cos x - \frac{3}{8}\sin x \cos x + \frac{3}{8}x + C$
14. $\frac{4}{3}$
15. $\frac{1}{6}\tan^6 x + C$

2.6 Reduction Formula for $\int \sin^m x \cos^n x \, dx$

Let the integral be denoted by $I_{m,n}$:

$$I_{m,n} = \int \sin^m x \cos^n x \, dx$$

To derive a reduction formula connecting $I_{m,n}$ with $I_{m-2,n}$, we employ the method of integration by parts. We begin by splitting the sine term, writing $\sin^m x$ as $\sin^{m-1} x \cdot \sin x$. Thus, the integrand becomes $\sin^{m-1} x(\cos^n x \sin x)$.

Let $u = \sin^{m-1} x$ and $dv = \cos^n x \sin x \, dx$. Differentiating u with respect to x yields $du = (m - 1)\sin^{m-2} x \cos x \, dx$. To find v, we integrate dv:

$$v = \int \cos^n x \sin x \, dx = -\frac{\cos^{n+1} x}{n+1}$$

Applying the integration by parts formula $\int u\,dv = uv - \int v\,du$, we obtain:

$$I_{m,n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} \int \sin^{m-2} x \cos^{n+2} x\,dx$$

We now manipulate the remaining integral to reintroduce $I_{m,n}$. By using the trigonometric identity $\cos^2 x = 1 - \sin^2 x$, we rewrite $\cos^{n+2} x$ as $\cos^n x(1 - \sin^2 x)$. Substituting this into the equation gives:

$$I_{m,n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} \int \sin^{m-2} x \cos^n x(1 - \sin^2 x)\,dx$$

Distributing the terms within the integral leads to:

$$I_{m,n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} \left[\int \sin^{m-2} x \cos^n x\,dx - \int \sin^m x \cos^n x\,dx\right]$$

Recognizing that $\int \sin^{m-2} x \cos^n x\,dx = I_{m-2,n}$ and $\int \sin^m x \cos^n x\,dx = I_{m,n}$, the equation becomes:

$$I_{m,n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} I_{m-2,n} - \frac{m-1}{n+1} I_{m,n}$$

To isolate $I_{m,n}$, we add $\dfrac{m-1}{n+1} I_{m,n}$ to both sides:

$$I_{m,n}\left(1 + \frac{m-1}{n+1}\right) = -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} I_{m-2,n}$$

Simplifying the coefficient on the left side:

$$I_{m,n}\left(\frac{n+1+m-1}{n+1}\right) = I_{m,n}\left(\frac{m+n}{n+1}\right)$$

Thus:

$$\frac{m+n}{n+1} I_{m,n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{n+1} + \frac{m-1}{n+1} I_{m-2,n}$$

Multiplying the entire equation by $\dfrac{n+1}{m+n}$ yields the final reduction formula:

$$\boxed{I_{m,n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{m+n} + \frac{m-1}{m+n} I_{m-2,n}}$$

Example 2.6.1. Evaluate $\displaystyle\int \sin^3 x \cos^2 x\,dx$.

Solution. Here, $m = 3$ and $n = 2$. We apply the reduction formula to reduce the power of sine.

$$I_{3,2} = -\frac{\sin^{3-1} x \cos^{2+1} x}{3+2} + \frac{3-1}{3+2} I_{3-2,2}$$

$$I_{3,2} = -\frac{\sin^2 x \cos^3 x}{5} + \frac{2}{5} I_{1,2}$$

Now we need to evaluate $I_{1,2} = \int \sin^1 x \cos^2 x\,dx$. Using simple substitution (let $u = \cos x, du = -\sin x\,dx$):

$$I_{1,2} = \int \cos^2 x \sin x\,dx = -\frac{\cos^3 x}{3}$$

Substituting this back into our expression for $I_{3,2}$:

$$I_{3,2} = -\frac{\sin^2 x \cos^3 x}{5} + \frac{2}{5}\left(-\frac{\cos^3 x}{3}\right) + C$$

Therefore:

$$I = -\frac{1}{5} \sin^2 x \cos^3 x - \frac{2}{15} \cos^3 x + C$$

■

Example 2.6.2. Evaluate $\int \sin^2 x \cos^4 x \, dx$.

Solution. Here, $m = 2$ and $n = 4$.

$$I_{2,4} = -\frac{\sin^{2-1} x \cos^{4+1} x}{2+4} + \frac{2-1}{2+4} I_{0,4}$$

$$I_{2,4} = -\frac{\sin x \cos^5 x}{6} + \frac{1}{6} I_{0,4}$$

Now we evaluate $I_{0,4} = \int \sin^0 x \cos^4 x \, dx = \int \cos^4 x \, dx$. We use the standard reduction formula for $\cos^n x$: $\int \cos^n x = \dfrac{\cos^{n-1} x \sin x}{n} + \dfrac{n-1}{n} I_{n-2}$.

$$I_{0,4} = \frac{\cos^3 x \sin x}{4} + \frac{3}{4} \int \cos^2 x \, dx$$

For $\int \cos^2 x \, dx$, we use the identity $\cos^2 x = \dfrac{1 + \cos 2x}{2}$:

$$\int \cos^2 x \, dx = \frac{x}{2} + \frac{\sin 2x}{4} = \frac{x}{2} + \frac{2 \sin x \cos x}{4} = \frac{x}{2} + \frac{\sin x \cos x}{2}$$

Substituting back into $I_{0,4}$:

$$I_{0,4} = \frac{\cos^3 x \sin x}{4} + \frac{3}{4} \left(\frac{x}{2} + \frac{\sin x \cos x}{2} \right)$$

Finally, substituting $I_{0,4}$ back into $I_{2,4}$:

$$I_{2,4} = -\frac{\sin x \cos^5 x}{6} + \frac{1}{6} \left[\frac{\cos^3 x \sin x}{4} + \frac{3}{8} x + \frac{3}{8} \sin x \cos x \right] + C$$

Therefore:

$$I = -\frac{1}{6} \sin x \cos^5 x + \frac{1}{24} \sin x \cos^3 x + \frac{1}{16} x + \frac{1}{16} \sin x \cos x + C$$

■

2.7 Reduction Formula for $\int x^m \cos nx \, dx$

Let the integral be denoted by J_m:

$$J_m = \int x^m \cos nx \, dx$$

Since the integrand contains a power of x multiplied by a trigonometric function, we must apply integration by parts twice to reduce the power of x from m to $m-2$ while returning to the cosine function.

First Integration by Parts: Let $u = x^m$ and $dv = \cos nx \, dx$. Then $du = m x^{m-1} \, dx$ and $v = \dfrac{1}{n} \sin nx$. Applying the integration by parts formula:

$$J_m = \frac{x^m \sin nx}{n} - \frac{m}{n} \int x^{m-1} \sin nx \, dx$$

Second Integration by Parts: We focus on the remaining integral $\int x^{m-1} \sin nx \, dx$. Let $u = x^{m-1}$ and $dv = \sin nx \, dx$. Then $du = (m-1) x^{m-2} \, dx$ and $v = -\dfrac{1}{n} \cos nx$. Applying the formula again:

$$\int x^{m-1} \sin nx \, dx = -\frac{x^{m-1} \cos nx}{n} + \frac{m-1}{n} \int x^{m-2} \cos nx \, dx$$

Substituting this result back into our expression for J_m:

$$J_m = \frac{x^m \sin nx}{n} - \frac{m}{n}\left[-\frac{x^{m-1}\cos nx}{n} + \frac{m-1}{n}\int x^{m-2}\cos nx\,dx\right]$$

Recognizing that $\int x^{m-2}\cos nx\,dx = J_{m-2}$, we expand and simplify the expression:

$$J_m = \frac{x^m \sin nx}{n} + \frac{mx^{m-1}\cos nx}{n^2} - \frac{m(m-1)}{n^2}J_{m-2}$$

This yields the final reduction formula:

$$\boxed{\int x^m \cos nx\,dx = \frac{x^m \sin nx}{n} + \frac{mx^{m-1}\cos nx}{n^2} - \frac{m(m-1)}{n^2}\int x^{m-2}\cos nx\,dx}$$

Example 2.7.1. Evaluate $\int x^2 \cos x\,dx$.

Solution. Here $m = 2$ and $n = 1$. Applying the reduction formula:

$$J_2 = \frac{x^2 \sin(1x)}{1} + \frac{2x^{2-1}\cos(1x)}{1^2} - \frac{2(2-1)}{1^2}J_{2-2}$$

$$J_2 = x^2 \sin x + 2x\cos x - 2J_0$$

We must evaluate J_0 (where $m = 0$):

$$J_0 = \int x^0 \cos x\,dx = \int \cos x\,dx = \sin x$$

Substituting J_0 back:

$$J_2 = x^2 \sin x + 2x\cos x - 2(\sin x) + C$$

Therefore:

$$I = x^2 \sin x + 2x\cos x - 2\sin x + C$$

■

Example 2.7.2. Evaluate $\int x^2 \cos 3x\,dx$.

Solution. Here $m = 2$ and $n = 3$. Applying the formula with careful attention to the n^2 terms:

$$J_2 = \frac{x^2 \sin 3x}{3} + \frac{2x^{2-1}\cos 3x}{3^2} - \frac{2(2-1)}{3^2}J_0$$

$$J_2 = \frac{x^2 \sin 3x}{3} + \frac{2x\cos 3x}{9} - \frac{2}{9}J_0$$

Evaluating J_0 (where $m = 0, n = 3$):

$$J_0 = \int x^0 \cos 3x\,dx = \int \cos 3x\,dx = \frac{\sin 3x}{3}$$

Substituting J_0 back into J_2:

$$J_2 = \frac{x^2 \sin 3x}{3} + \frac{2x\cos 3x}{9} - \frac{2}{9}\left(\frac{\sin 3x}{3}\right) + C$$

Therefore:

$$I = \frac{x^2 \sin 3x}{3} + \frac{2x\cos 3x}{9} - \frac{2\sin 3x}{27} + C$$

■

2.8 Reduction Formula for $\int \sin^m x \sin nx \, dx$

Let the integral be denoted by $I_{m,n}$:

$$I_{m,n} = \int \sin^m x \sin nx \, dx$$

We employ the method of **Integration by Parts**. Let:

$$u = \sin^m x \qquad dv = \sin nx \, dx$$
$$du = m \sin^{m-1} x \cos x \, dx \qquad v = -\frac{\cos nx}{n}$$

Applying the formula $\int u \, dv = uv - \int v \, du$:

$$I_{m,n} = -\frac{\sin^m x \cos nx}{n} - \int \left(-\frac{\cos nx}{n}\right) (m \sin^{m-1} x \cos x) \, dx$$

Simplifying the expression:

$$I_{m,n} = -\frac{\sin^m x \cos nx}{n} + \frac{m}{n} \int \sin^{m-1} x (\cos x \cos nx) \, dx$$

We now use the trigonometric addition identity to handle the $\cos x \cos nx$ term. Recall that:

$$\cos(nx - x) = \cos nx \cos x + \sin nx \sin x$$

Rearranging this gives:

$$\cos x \cos nx = \cos((n-1)x) - \sin x \sin nx$$

Substituting this into the integral:

$$I_{m,n} = -\frac{\sin^m x \cos nx}{n} + \frac{m}{n} \int \sin^{m-1} x \left[\cos((n-1)x) - \sin x \sin nx\right] \, dx$$

Distributing the integral:

$$I_{m,n} = -\frac{\sin^m x \cos nx}{n} + \frac{m}{n} \int \sin^{m-1} x \cos((n-1)x) \, dx - \frac{m}{n} \int \sin^m x \sin nx \, dx$$

We recognize the last term as the original integral $I_{m,n}$. Thus:

$$I_{m,n} = -\frac{\sin^m x \cos nx}{n} + \frac{m}{n} \int \sin^{m-1} x \cos((n-1)x) \, dx - \frac{m}{n} I_{m,n}$$

Now, we collect the $I_{m,n}$ terms on the left side:

$$I_{m,n} + \frac{m}{n} I_{m,n} = -\frac{\sin^m x \cos nx}{n} + \frac{m}{n} \int \sin^{m-1} x \cos((n-1)x) \, dx$$
$$I_{m,n} \left(\frac{n+m}{n}\right) = -\frac{\sin^m x \cos nx}{n} + \frac{m}{n} \int \sin^{m-1} x \cos((n-1)x) \, dx$$

Multiplying the entire equation by $\dfrac{n}{m+n}$ gives the final reduction formula:

$$\boxed{\int \sin^m x \sin nx \, dx = -\frac{\sin^m x \cos nx}{m+n} + \frac{m}{m+n} \int \sin^{m-1} x \cos((n-1)x) \, dx}$$

Example 2.8.1. Evaluate $\int \sin x \sin 3x \, dx$.

Solution. Here, $m = 1$ and $n = 3$. Substitute these values into the reduction formula:

$$I = -\frac{\sin^1 x \cos 3x}{1+3} + \frac{1}{1+3}\int \sin^{1-1} x \cos((3-1)x)\,dx$$

Simplifying the powers and terms:

$$I = -\frac{\sin x \cos 3x}{4} + \frac{1}{4}\int \sin^0 x \cos 2x\,dx$$

Since $\sin^0 x = 1$:

$$I = -\frac{\sin x \cos 3x}{4} + \frac{1}{4}\int \cos 2x\,dx$$

Now, integrate $\cos 2x$:

$$\int \cos 2x\,dx = \frac{\sin 2x}{2}$$

Substituting this back gives the final result:

$$I = -\frac{\sin x \cos 3x}{4} + \frac{1}{4}\left(\frac{\sin 2x}{2}\right) + C$$

$$\boxed{I = -\frac{1}{4}\sin x \cos 3x + \frac{1}{8}\sin 2x + C}$$

■

Example 2.8.2. Evaluate $\int \sin^2 x \sin 2x\,dx$.

Solution. Here, $m = 2$ and $n = 2$. Applying the formula:

$$I = -\frac{\sin^2 x \cos 2x}{2+2} + \frac{2}{2+2}\int \sin^{2-1} x \cos((2-1)x)\,dx$$

$$I = -\frac{\sin^2 x \cos 2x}{4} + \frac{2}{4}\int \sin x \cos x\,dx$$

We now evaluate the remaining integral $\int \sin x \cos x\,dx$. Using substitution ($u = \sin x, du = \cos x\,dx$):

$$\int \sin x \cos x\,dx = \frac{\sin^2 x}{2}$$

Substituting this back into the expression for I:

$$I = -\frac{\sin^2 x \cos 2x}{4} + \frac{1}{2}\left(\frac{\sin^2 x}{2}\right) + C$$

$$\boxed{I = -\frac{1}{4}\sin^2 x \cos 2x + \frac{1}{4}\sin^2 x + C}$$

■

2.9 Reduction Formula for $\int \cos^m x \cos nx\,dx$

Let the integral be denoted by $K_{m,n}$:

$$K_{m,n} = \int \cos^m x \cos nx\,dx$$

We again use **Integration by Parts**. Let:

$$u = \cos^m x \qquad dv = \cos nx\,dx$$

$$du = -m\cos^{m-1} x \sin x\,dx \qquad v = \frac{\sin nx}{n}$$

Applying the formula $\int u\,dv = uv - \int v\,du$:

$$K_{m,n} = \frac{\cos^m x \sin nx}{n} - \int \left(\frac{\sin nx}{n}\right)(-m\cos^{m-1} x \sin x)\,dx$$

Simplifying the expression:

$$K_{m,n} = \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x(\sin x \sin nx)\,dx$$

We use the trigonometric identity involving $\sin x \sin nx$. Recall that:

$$\cos(nx - x) = \cos nx \cos x + \sin nx \sin x$$

Rearranging this gives:

$$\sin x \sin nx = \cos((n-1)x) - \cos x \cos nx$$

Substituting this into the integral:

$$K_{m,n} = \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x\left[\cos((n-1)x) - \cos x \cos nx\right]\,dx$$

Distributing the integral:

$$K_{m,n} = \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x \cos((n-1)x)\,dx - \frac{m}{n}\int \cos^m x \cos nx\,dx$$

We recognize the last term as the original integral $K_{m,n}$. Thus:

$$K_{m,n} = \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x \cos((n-1)x)\,dx - \frac{m}{n}K_{m,n}$$

Collecting the $K_{m,n}$ terms on the left side:

$$K_{m,n}\left(1 + \frac{m}{n}\right) = \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x \cos((n-1)x)\,dx$$

$$K_{m,n}\left(\frac{n+m}{n}\right) = \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x \cos((n-1)x)\,dx$$

Multiplying by $\dfrac{n}{m+n}$ yields the final reduction formula:

$$\boxed{\int \cos^m x \cos nx\,dx = \frac{\cos^m x \sin nx}{m+n} + \frac{m}{m+n}\int \cos^{m-1} x \cos((n-1)x)\,dx}$$

Example 2.9.1. Evaluate $\int \cos x \cos 2x\,dx$.

Solution. Here, $m = 1$ and $n = 2$. Substitute these values into the reduction formula:

$$I = \frac{\cos^1 x \sin 2x}{1+2} + \frac{1}{1+2}\int \cos^{1-1} x \cos((2-1)x)\,dx$$

Simplifying:

$$I = \frac{\cos x \sin 2x}{3} + \frac{1}{3}\int \cos^0 x \cos x\,dx$$

Since $\cos^0 x = 1$:

$$I = \frac{\cos x \sin 2x}{3} + \frac{1}{3}\int \cos x\,dx$$

The integral of $\cos x$ is $\sin x$.

$$\boxed{I = \frac{1}{3}\cos x \sin 2x + \frac{1}{3}\sin x + C}$$

■

Example 2.9.2. Evaluate $\int \cos^2 x \cos 2x \, dx$.

Solution. Here, $m = 2$ and $n = 2$. Applying the formula:

$$I = \frac{\cos^2 x \sin 2x}{2+2} + \frac{2}{2+2} \int \cos^{2-1} x \cos((2-1)x) \, dx$$

$$I = \frac{\cos^2 x \sin 2x}{4} + \frac{2}{4} \int \cos x \cos x \, dx$$

$$I = \frac{\cos^2 x \sin 2x}{4} + \frac{1}{2} \int \cos^2 x \, dx$$

Now we solve $\int \cos^2 x \, dx$ using the identity $\cos^2 x = \dfrac{1 + \cos 2x}{2}$:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos 2x}{2} \, dx = \frac{x}{2} + \frac{\sin 2x}{4}$$

Substituting this back into the main equation:

$$I = \frac{\cos^2 x \sin 2x}{4} + \frac{1}{2} \left(\frac{x}{2} + \frac{\sin 2x}{4} \right) + C$$

$$\boxed{I = \frac{1}{4} \cos^2 x \sin 2x + \frac{x}{4} + \frac{1}{8} \sin 2x + C}$$

■

Apply the appropriate reduction formulae to evaluate the following integrals. For problems involving powers of x, apply the formula repeatedly until the power of x becomes zero. For trigonometric powers, apply the formula until the integral becomes a standard known form (like $\int \sin x \, dx$ or $\int \cos nx \, dx$).

Exercise 2.9.3. *Use the formula:* $I_{m,n} = -\dfrac{\sin^{m-1} x \cos^{n+1} x}{m+n} + \dfrac{m-1}{m+n} I_{m-2,n}$

1.1 $\int \sin^3 x \cos^4 x \, dx$

1.2 $\int \sin^5 x \cos^2 x \, dx$

1.3 $\int \sin^4 x \cos^2 x \, dx$

Use the formula: $J_m = \dfrac{x^m \sin nx}{n} + \dfrac{m x^{m-1} \cos nx}{n^2} - \dfrac{m(m-1)}{n^2} J_{m-2}$

2.1 $\int x^2 \cos 3x \, dx$

2.2 $\int x^3 \cos 2x \, dx$

2.3 $\int x^2 \cos 5x \, dx$

Use the formula: $I_{m,n} = -\dfrac{\sin^m x \cos nx}{m+n} + \dfrac{m}{m+n} \int \sin^{m-1} x \cos((n-1)x) \, dx$

3.1 $\int \sin^3 x \sin 3x \, dx$

3.2 $\int \sin^4 x \sin 2x \, dx$

3.3 $\int \sin^2 x \sin 4x \, dx$

Use the formula: $K_{m,n} = \dfrac{\cos^m x \sin nx}{m+n} + \dfrac{m}{m+n} \int \cos^{m-1} x \cos((n-1)x) \, dx$

4.1 $\int \cos^3 x \cos 3x \, dx$

4.2 $\int \cos^2 x \cos 4x \, dx$

4.3 $\int \cos^4 x \cos 2x \, dx$

Answers to Exercises 2.9.3

1.1 $I_{3,4} = -\dfrac{\sin^2 x \cos^5 x}{7} + \dfrac{2}{7} I_{1,4}$

1.2 $I_{5,2} = -\dfrac{\sin^4 x \cos^3 x}{7} + \dfrac{4}{7} I_{3,2}$

1.3 $I_{4,2} = -\dfrac{\sin^3 x \cos^3 x}{6} + \dfrac{3}{6} I_{2,2}$

2.1 $\dfrac{x^2 \sin 3x}{3} + \dfrac{2x \cos 3x}{9} - \dfrac{2}{9} \int \cos 3x \, dx$

2.2 $\dfrac{x^3 \sin 2x}{2} + \dfrac{3x^2 \cos 2x}{4} - \dfrac{6}{4} \int x \cos 2x \, dx$

2.3 $\dfrac{x^2 \sin 5x}{5} + \dfrac{2x \cos 5x}{25} - \dfrac{2}{25} \int \cos 5x \, dx$

3.1 $-\dfrac{\sin^3 x \cos 3x}{6} + \dfrac{3}{6} \int \sin^2 x \cos 2x \, dx$

3.2 $-\dfrac{\sin^4 x \cos 2x}{6} + \dfrac{4}{6} \int \sin^3 x \cos x \, dx$

3.3 $-\dfrac{\sin^2 x \cos 4x}{6} + \dfrac{2}{6} \int \sin x \cos 3x \, dx$

4.1 $\dfrac{\cos^3 x \sin 3x}{6} + \dfrac{3}{6} \int \cos^2 x \cos 2x \, dx$

4.2 $\dfrac{\cos^2 x \sin 4x}{6} + \dfrac{2}{6} \int \cos x \cos 3x \, dx$

4.3 $\dfrac{\cos^4 x \sin 2x}{6} + \dfrac{4}{6} \int \cos^3 x \cos x \, dx$

Chapter 3

Definite Integrals and Rectification

This unit bridges the gap between indefinite integration and practical calculation through the rigorous study of definite integrals and the Fundamental Theorem of Calculus. It explores essential properties to simplify evaluations and introduces the concept of the integral as a limit of a sum. Furthermore, it covers rectification, enabling students to calculate the precise arc length of plane curves.

3.1 Properties of Definite Integrals

The following properties simplify the evaluation of definite integrals. Let $f(x)$ be a continuous function defined on the interval $[a, b]$, and let $F(x)$ be the antiderivative of $f(x)$, i.e., $F'(x) = f(x)$. By the Fundamental Theorem of Calculus:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Property $\mathbf{P_0}$ (Change of Variable)**.** The value of a definite integral does not depend on the dummy variable of integration.

$$\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$$

Proof. Let $F(x)$ be the antiderivative of $f(x)$.

$$\text{L.H.S.} = \int_a^b f(x)\,dx = F(b) - F(a)$$

$$\text{R.H.S.} = \int_a^b f(t)\,dt = F(b) - F(a)$$

Since L.H.S. = R.H.S., the property holds. ■

Property $\mathbf{P_1}$ (Interchange of Limits)**.** Interchanging the limits of integration changes the sign of the integral.

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

Proof. Using the Fundamental Theorem of Calculus:

$$\int_a^b f(x)\,dx = F(b) - F(a) = -[F(a) - F(b)]$$

Also,

$$\int_b^a f(x)\,dx = F(a) - F(b)$$

Therefore,

$$\int_a^b f(x)\,dx = -\left(\int_b^a f(x)\,dx\right)$$

■

Property P_2 (Splitting the Interval)**.** For any point c such that $a < c < b$:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Proof. Using the Fundamental Theorem of Calculus on the right-hand side (R.H.S.):

$$\begin{aligned}
\text{R.H.S.} &= \int_a^c f(x)\,dx + \int_c^b f(x)\,dx \\
&= [F(c) - F(a)] + [F(b) - F(c)] \\
&= F(b) - F(a) \\
&= \int_a^b f(x)\,dx \quad \text{(L.H.S.)}
\end{aligned}$$

■

Property P_3 (Sum of Limits Substitution)**.**

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

Proof. Consider the integral on the right: $I = \int_a^b f(a+b-x)\,dx$.

Let $t = a + b - x$. Then $dt = -dx \implies dx = -dt$.
Change Limits:

- When $x = a$, $t = a + b - a = b$.
- When $x = b$, $t = a + b - b = a$.

Substituting these:

$$I = \int_b^a f(t)(-dt) = -\int_b^a f(t)\,dt$$

Using Property 1 (swapping limits removes the negative sign):

$$I = \int_a^b f(t)\,dt$$

Using Property 0 (changing dummy variable back to x):

$$I = \int_a^b f(x)\,dx$$

■

Property P_4 (Reflection about $a/2$)**.**

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

Proof. This is a special case of Property 3 where the lower limit is 0 and the upper limit is a. Replace a with 0 and b with a in Property 3:

$$\int_0^a f(x)\,dx = \int_0^a f(0+a-x)\,dx = \int_0^a f(a-x)\,dx$$

∎

Property P_5 (Double Limit Property)**.**

$$\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(2a-x)\,dx$$

Proof. Using Property 2, split the interval $[0, 2a]$ at a:

$$I = \int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_a^{2a} f(x)\,dx$$

Consider the second integral: $I_2 = \int_a^{2a} f(x)\,dx$.
Let $t = 2a - x$. Then $dt = -dx$.
Change Limits:

- When $x = a$, $t = a$.
- When $x = 2a$, $t = 0$.

$$I_2 = \int_a^0 f(2a-t)(-dt) = -\int_a^0 f(2a-t)\,dt = \int_0^a f(2a-t)\,dt$$

Using Property 0, write t as x: $I_2 = \int_0^a f(2a-x)\,dx$.
Adding I_1 and I_2:

$$I = \int_0^a f(x)\,dx + \int_0^a f(2a-x)\,dx$$

∎

Property P_6 (Symmetric Interval)**.**

$$\int_{-a}^a f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & \text{if } f(x) \text{ is Even, i.e., } f(-x) = f(x) \\ 0 & \text{if } f(x) \text{ is Odd, i.e., } f(-x) = -f(x) \end{cases}$$

Proof. Split the integral at $x = 0$:

$$I = \int_{-a}^a f(x)\,dx = \int_{-a}^0 f(x)\,dx + \int_0^a f(x)\,dx$$

Consider the first integral $I_1 = \int\limits_{-a}^{0} f(x)\,dx$.

Let $t = -x \implies dx = -dt$. Limits change from a to 0.

$$I_1 = \int\limits_{a}^{0} f(-t)(-dt) = \int\limits_{0}^{a} f(-t)\,dt = \int\limits_{0}^{a} f(-x)\,dx$$

Thus, the total integral is:

$$I = \int\limits_{0}^{a} f(-x)\,dx + \int\limits_{0}^{a} f(x)\,dx = \int\limits_{0}^{a} [f(-x) + f(x)]\,dx$$

Case 1 (Even): $f(-x) = f(x) \implies I = \int\limits_{0}^{a} 2f(x)\,dx = 2\int\limits_{0}^{a} f(x)\,dx$.

Case 2 (Odd): $f(-x) = -f(x) \implies I = \int\limits_{0}^{a} [-f(x) + f(x)]\,dx = 0$. ■

Example 3.1.1. Evaluate $\int\limits_{0}^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx$.

Solution. Let the given integral be I.

$$I = \int\limits_{0}^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx \tag{3.1}$$

Using the property $\int\limits_{0}^{a} f(x)\,dx = \int\limits_{0}^{a} f(a-x)\,dx$: Here, $a = \frac{\pi}{2}$, so we replace x with $(\frac{\pi}{2} - x)$. Recall that $\sin(\frac{\pi}{2} - x) = \cos x$ and $\cos(\frac{\pi}{2} - x) = \sin x$.

$$I = \int\limits_{0}^{\frac{\pi}{2}} \frac{\sqrt{\sin(\frac{\pi}{2} - x)}}{\sqrt{\sin(\frac{\pi}{2} - x)} + \sqrt{\cos(\frac{\pi}{2} - x)}}\,dx$$

$$I = \int\limits_{0}^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}}\,dx \tag{3.2}$$

Adding equations (5.2) and (3.2):

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx + \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx$$

$$2I = \int_0^{\frac{\pi}{2}} \left(\frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} \right) dx$$

$$2I = \int_0^{\frac{\pi}{2}} 1\,dx$$

$$2I = [x]_0^{\pi/2} = \frac{\pi}{2} - 0$$

$$I = \frac{\pi}{4}$$

■

Example 3.1.2. Evaluate $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^3 + x\cos x + \tan^5 x + 1)\,dx.$

Solution. Let $I = \displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^3 + x\cos x + \tan^5 x + 1)\,dx$. We use the property for symmetric intervals $[-a, a]$:

$$\int_{-a}^{a} f(x)\,dx = 0 \quad \text{if } f(x) \text{ is Odd (i.e., } f(-x) = -f(x))$$

We split the integral into four parts:

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^3\,dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x\cos x\,dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tan^5 x\,dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1\,dx$$

Analyze each term:

1. Let $f_1(x) = x^3$. Since $(-x)^3 = -x^3$, this is **Odd**. Integral is 0.
2. Let $f_2(x) = x\cos x$. Since $(-x)\cos(-x) = -x\cos x$, this is **Odd**. Integral is 0.
3. Let $f_3(x) = \tan^5 x$. Since $\tan^5(-x) = (-\tan x)^5 = -\tan^5 x$, this is **Odd**. Integral is 0.
4. Let $f_4(x) = 1$. This is an **Even** function (constant).

Therefore, the complex terms vanish instantly:

$$I = 0 + 0 + 0 + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1\,dx = [x]_{-\pi/2}^{\pi/2} = \frac{\pi}{2} - \left(-\frac{\pi}{2}\right) = \pi$$

■

Example 3.1.3. Evaluate $\int\limits_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\, dx$.

Solution. Let $I = \int\limits_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\, dx$. — (1)

Using the property $\int\limits_0^{a} f(x)\, dx = \int\limits_0^{a} f(a-x)\, dx$: Replace x with $(\pi - x)$. Note that $\sin(\pi - x) = \sin x$ and $\cos(\pi - x) = -\cos x$, so $\cos^2(\pi - x) = \cos^2 x$.

$$I = \int\limits_0^{\pi} \frac{(\pi - x)\sin x}{1+\cos^2 x}\, dx \quad \text{— (2)}$$

Adding (1) and (2):

$$2I = \int\limits_0^{\pi} \frac{(x + \pi - x)\sin x}{1+\cos^2 x}\, dx$$
$$2I = \pi \int\limits_0^{\pi} \frac{\sin x}{1+\cos^2 x}\, dx$$

Now we perform substitution. Let $\cos x = t$, then $-\sin x\, dx = dt \implies \sin x\, dx = -dt$. Limits: When $x = 0, t = 1$. When $x = \pi, t = -1$.

$$2I = \pi \int\limits_1^{-1} \frac{-dt}{1+t^2} = \pi \int\limits_{-1}^{1} \frac{dt}{1+t^2}$$
$$2I = \pi[\tan^{-1} t]_{-1}^{1}$$
$$2I = \pi[\tan^{-1}(1) - \tan^{-1}(-1)]$$
$$2I = \pi\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = \pi\left[\frac{\pi}{2}\right] = \frac{\pi^2}{2}$$

$$\therefore I = \frac{\pi^2}{4}$$

■

Example 3.1.4. Evaluate $\int\limits_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x}\, dx$.

Solution. Let $I = \int\limits_{-\pi/2}^{\pi/2} \frac{\cos x}{1+e^x}\, dx$. — (1)

We use the property $\int\limits_a^{b} f(x)\, dx = \int\limits_a^{b} f(a+b-x)\, dx$. Here, $a = -\frac{\pi}{2}$ and $b = \frac{\pi}{2}$, so $a + b = 0$. We replace x with $-x$.

$$I = \int\limits_{-\pi/2}^{\pi/2} \frac{\cos(-x)}{1+e^{-x}}\, dx$$

Since $\cos(-x) = \cos x$ and $e^{-x} = \dfrac{1}{e^x}$:

$$I = \int\limits_{-\pi/2}^{\pi/2} \frac{\cos x}{1 + \dfrac{1}{e^x}}\,dx = \int\limits_{-\pi/2}^{\pi/2} \frac{e^x \cos x}{e^x + 1}\,dx \quad \text{— (2)}$$

Adding equations (1) and (2):

$$2I = \int\limits_{-\pi/2}^{\pi/2} \frac{\cos x}{1 + e^x}\,dx + \int\limits_{-\pi/2}^{\pi/2} \frac{e^x \cos x}{1 + e^x}\,dx$$

$$2I = \int\limits_{-\pi/2}^{\pi/2} \frac{(1 + e^x)\cos x}{1 + e^x}\,dx$$

$$2I = \int\limits_{-\pi/2}^{\pi/2} \cos x\,dx$$

Since $\cos x$ is an even function:

$$2I = 2\int\limits_{0}^{\pi/2} \cos x\,dx$$

$$I = [\sin x]_0^{\pi/2} = 1 - 0 = 1$$

■

Example 3.1.5. Evaluate $\displaystyle\int\limits_0^{\pi} \ln(\sin x)\,dx$.

Solution. Let $I = \displaystyle\int\limits_0^{\pi} \ln(\sin x)\,dx$. Using the symmetry property $\sin(\pi - x) = \sin x$, we know that $\displaystyle\int\limits_0^{2a} f(x)dx = 2\int\limits_0^{a} f(x)dx$ if $f(2a - x) = f(x)$.

$$I = 2\int\limits_0^{\pi/2} \ln(\sin x)\,dx \quad \text{— (1)}$$

Now let $I_1 = \displaystyle\int\limits_0^{\pi/2} \ln(\sin x)\,dx$. Apply $x \to \dfrac{\pi}{2} - x$:

$$I_1 = \int\limits_0^{\pi/2} \ln(\cos x)\,dx$$

Adding the two forms of I_1:

$$2I_1 = \int\limits_0^{\pi/2} (\ln(\sin x) + \ln(\cos x))\,dx = \int\limits_0^{\pi/2} \ln(\sin x \cos x)\,dx$$

$$2I_1 = \int\limits_0^{\pi/2} \ln\left(\frac{\sin 2x}{2}\right)\,dx = \int\limits_0^{\pi/2} \ln(\sin 2x)\,dx - \int\limits_0^{\pi/2} \ln 2\,dx$$

In the first integral, let $2x = t \implies dx = dt/2$. Limits $0 \to \pi$.

$$2I_1 = \frac{1}{2}\int_0^{\pi} \ln(\sin t)\, dt - [x \ln 2]_0^{\pi/2}$$

Notice $\int_0^{\pi} \ln(\sin t)\, dt$ is exactly our original I.

$$2I_1 = \frac{1}{2}I - \frac{\pi}{2}\ln 2$$

From (1), we know $I = 2I_1$, so substitute $I_1 = I/2$:

$$2(I/2) = \frac{1}{2}I - \frac{\pi}{2}\ln 2 \implies I = \frac{1}{2}I - \frac{\pi}{2}\ln 2$$

$$\frac{1}{2}I = -\frac{\pi}{2}\ln 2 \implies I = -\pi \ln 2$$

■

Example 3.1.6. Evaluate $\int_{-1}^{2} |2x - 1|\, dx$.

Solution. We use the **Splitting Property**. The critical point is where $2x - 1 = 0 \implies x = 1/2$.

- For $x < 1/2$, $|2x - 1| = -(2x - 1) = 1 - 2x$.
- For $x > 1/2$, $|2x - 1| = 2x - 1$.

$$I = \int_{-1}^{1/2} (1 - 2x)\, dx + \int_{1/2}^{2} (2x - 1)\, dx$$

Integrating:

$$\begin{aligned} I &= \left[x - x^2\right]_{-1}^{1/2} + \left[x^2 - x\right]_{1/2}^{2} \\ &= \left[(\frac{1}{2} - \frac{1}{4}) - (-1 - 1)\right] + \left[(4 - 2) - (\frac{1}{4} - \frac{1}{2})\right] \\ &= \left[\frac{1}{4} - (-2)\right] + \left[2 - (-\frac{1}{4})\right] \\ &= \frac{9}{4} + \frac{9}{4} = \frac{18}{4} = \frac{9}{2} \end{aligned}$$

■

Example 3.1.7. Evaluate $\int_0^{100\pi} |\sin x|\, dx$.

Solution. We use the **Periodicity Property**. The function $f(x) = |\sin x|$ is periodic with period $T = \pi$.

Property: $\int_0^{nT} f(x)\, dx = n\int_0^{T} f(x)\, dx$. Here, $n = 100$ and $T = \pi$.

$$I = 100\int_0^{\pi} |\sin x|\, dx$$

In the interval $[0, \pi]$, $\sin x$ is positive, so $|\sin x| = \sin x$.

$$
\begin{aligned}
I &= 100 \int_0^{\pi} \sin x \, dx \\
&= 100[-\cos x]_0^{\pi} \\
&= 100[-(-1) - (-1)] \\
&= 100[1 + 1] = 200
\end{aligned}
$$

■

Example 3.1.8. Evaluate $\int_0^{1.5} [x^2] \, dx$, where $[\cdot]$ denotes the greatest integer function.

Solution. We must split the integral at points where x^2 becomes an integer. The range of x is 0 to 1.5. Consequently, x^2 ranges from 0 to 2.25. The integer values for x^2 occur at:

- $x^2 = 1 \implies x = 1$
- $x^2 = 2 \implies x = \sqrt{2} \approx 1.414$

We split the interval $[0, 1.5]$ into $[0, 1]$, $[1, \sqrt{2}]$, and $[\sqrt{2}, 1.5]$.

1. For $0 \leq x < 1$, $0 \leq x^2 < 1 \implies [x^2] = 0$.
2. For $1 \leq x < \sqrt{2}$, $1 \leq x^2 < 2 \implies [x^2] = 1$.
3. For $\sqrt{2} \leq x < 1.5$, $2 \leq x^2 < 2.25 \implies [x^2] = 2$.

The integral becomes:

$$
\begin{aligned}
I &= \int_0^1 0 \, dx + \int_1^{\sqrt{2}} 1 \, dx + \int_{\sqrt{2}}^{1.5} 2 \, dx \\
&= 0 + [x]_1^{\sqrt{2}} + 2[x]_{\sqrt{2}}^{1.5} \\
&= (\sqrt{2} - 1) + 2(1.5 - \sqrt{2}) \\
&= \sqrt{2} - 1 + 3 - 2\sqrt{2} \\
&= 2 - \sqrt{2}
\end{aligned}
$$

■

Example 3.1.9. Evaluate $\int_0^1 \frac{\ln(1+x)}{1+x^2} \, dx$.

Solution. Let $I = \int_0^1 \frac{\ln(1+x)}{1+x^2} \, dx$.

Let $x = \tan\theta \implies dx = \sec^2\theta \, d\theta$. Limits: When $x = 0, \theta = 0$. When $x = 1, \theta = \frac{\pi}{4}$.

$$
I = \int_0^{\frac{\pi}{4}} \frac{\ln(1+\tan\theta)}{1+\tan^2\theta} (\sec^2\theta \, d\theta)
$$

Since $1 + \tan^2\theta = \sec^2\theta$, terms cancel out:

$$I = \int_0^{\frac{\pi}{4}} \ln(1 + \tan\theta)\, d\theta \quad \text{— (1)}$$

Apply Property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$**.** Replace θ with $(\frac{\pi}{4} - \theta)$:

$$I = \int_0^{\frac{\pi}{4}} \ln\left(1 + \tan\left(\frac{\pi}{4} - \theta\right)\right)\, d\theta$$

Using the identity $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$:

$$\tan\left(\frac{\pi}{4} - \theta\right) = \frac{1 - \tan\theta}{1 + \tan\theta}$$

Substitute this back:

$$I = \int_0^{\frac{\pi}{4}} \ln\left(1 + \frac{1 - \tan\theta}{1 + \tan\theta}\right)\, d\theta$$
$$I = \int_0^{\frac{\pi}{4}} \ln\left(\frac{1 + \tan\theta + 1 - \tan\theta}{1 + \tan\theta}\right)\, d\theta$$
$$I = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan\theta}\right)\, d\theta$$
$$I = \int_0^{\frac{\pi}{4}} \ln 2\, d\theta - \int_0^{\frac{\pi}{4}} \ln(1 + \tan\theta)\, d\theta$$

Notice the second integral is the original I from equation (1).

$$I = [\theta \ln 2]_0^{\pi/4} - I$$
$$2I = \frac{\pi}{4}\ln 2$$
$$I = \frac{\pi}{8}\ln 2$$

■

Example 3.1.10. Evaluate $\displaystyle\int_0^{\infty} \frac{\ln x}{1 + x^2}\, dx.$

Solution. Let $I = \displaystyle\int_0^{\infty} \frac{\ln x}{1 + x^2}\, dx.$

Let $x = \frac{1}{t} \implies dx = -\frac{1}{t^2}\,dt$. Limits: When $x \to 0, t \to \infty$. When $x \to \infty, t \to 0$.

$$I = \int_{\infty}^{0} \frac{\ln(1/t)}{1+(1/t)^2} \left(-\frac{1}{t^2}\right) dt$$

$$I = \int_{\infty}^{0} \frac{-\ln t}{\frac{t^2+1}{t^2}} \left(-\frac{1}{t^2}\right) dt$$

$$I = \int_{\infty}^{0} \frac{\ln t}{1+t^2}\,dt$$

Using the property $\int_b^a f(x)dx = -\int_a^b f(x)dx$:

$$I = -\int_0^{\infty} \frac{\ln t}{1+t^2}\,dt$$

Since the variable of integration is a dummy variable, $\int_0^{\infty} \frac{\ln t}{1+t^2}\,dt = I$.

$$I = -I \implies 2I = 0 \implies I = 0$$

■

Example 3.1.11. Evaluate $\int_0^{\pi} \frac{x}{1+\sin x}\,dx$.

Solution. Let $I = \int_0^{\pi} \frac{x}{1+\sin x}\,dx$. — (1)

Apply property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$:

$$I = \int_0^{\pi} \frac{\pi - x}{1+\sin(\pi - x)}\,dx = \int_0^{\pi} \frac{\pi - x}{1+\sin x}\,dx \quad \text{— (2)}$$

Adding (1) and (2) eliminates x from the numerator:

$$2I = \int_0^{\pi} \frac{x+\pi-x}{1+\sin x}\,dx = \pi \int_0^{\pi} \frac{1}{1+\sin x}\,dx$$

Multiply numerator and denominator by the conjugate $(1-\sin x)$:

$$2I = \pi \int_0^{\pi} \frac{1-\sin x}{1-\sin^2 x}\,dx$$

$$2I = \pi \int_0^{\pi} \frac{1-\sin x}{\cos^2 x}\,dx$$

$$2I = \pi \left[\int_0^{\pi} \sec^2 x\,dx - \int_0^{\pi} \sec x \tan x\,dx\right]$$

Note: $\sec x$ *and* $\tan x$ *are discontinuous at* $\pi/2$*. However, we can evaluate limits or integrate directly if we note the symmetry or split limits. Alternatively, evaluate using standard results carefully.* Let's integrate directly:

$$2I = \pi[\tan x - \sec x]_0^\pi$$

Upper limit (π): $\tan\pi - \sec\pi = 0 - (-1) = 1$. Lower limit (0): $\tan 0 - \sec 0 = 0 - 1 = -1$.

$$2I = \pi[1 - (-1)] = 2\pi \implies I = \pi$$

■

Example 3.1.12. Evaluate $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x}\,dx$.

Solution. Let $I = \displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{\sin x + \cos x}\,dx$. — (1)

Apply the King's Property ($x \to \frac{\pi}{2} - x$):

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos^2 x}{\cos x + \sin x}\,dx \quad \text{— (2)}$$

Adding (1) and (2):

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^2 x + \cos^2 x}{\sin x + \cos x}\,dx = \int_0^{\frac{\pi}{2}} \frac{1}{\sin x + \cos x}\,dx$$

To integrate $\dfrac{1}{\sin x + \cos x}$, divide and multiply by $\sqrt{2}$:

$$2I = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{1}{\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{2}}\cos x}\,dx$$

$$2I = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{1}{\sin(x + \frac{\pi}{4})}\,dx = \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \text{cosec}(x + \frac{\pi}{4})\,dx$$

Using $\int \text{cosec}\, u\, du = \ln|\text{cosec}\, u - \cot u| = \ln|\tan(u/2)|$:

$$2I = \frac{1}{\sqrt{2}} \left[\ln\left|\tan\left(\frac{x}{2} + \frac{\pi}{8}\right)\right|\right]_0^{\pi/2}$$
$$2I = \frac{1}{\sqrt{2}} \left[\ln(\tan(\frac{\pi}{4} + \frac{\pi}{8})) - \ln(\tan(\frac{\pi}{8}))\right]$$
$$2I = \frac{1}{\sqrt{2}} \ln\left(\frac{\cot(\pi/8)}{\tan(\pi/8)}\right) \quad \text{(Since } \tan(\frac{3\pi}{8}) = \cot\frac{\pi}{8}\text{)}$$
$$2I = \frac{1}{\sqrt{2}} \ln(\cot^2(\pi/8)) = \frac{2}{\sqrt{2}} \ln(\cot(\pi/8))$$

Since $\cot(\pi/8) = \sqrt{2} + 1$:

$$I = \frac{1}{\sqrt{2}} \ln(\sqrt{2} + 1)$$

■

Example 3.1.13. Evaluate $\displaystyle\int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan^3 x} dx.$

Solution. Let $I = \displaystyle\int_0^{\frac{\pi}{2}} \frac{1}{1 + \frac{\sin^3 x}{\cos^3 x}} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx.$ — (1)

Apply King's Property ($x \to \frac{\pi}{2} - x$):

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx \quad \text{— (2)}$$

Adding (1) and (2):

$$2I = \int_0^{\frac{\pi}{2}} \frac{\cos^3 x + \sin^3 x}{\cos^3 x + \sin^3 x} dx = \int_0^{\frac{\pi}{2}} 1\, dx$$

$$2I = [x]_0^{\pi/2} = \frac{\pi}{2} \implies I = \frac{\pi}{4}$$

Note: This result holds for any power n in $\displaystyle\int_0^{\pi/2} \frac{dx}{1 + \tan^n x}$. ■

3.2 Fundamental Concepts

The definite integral is historically and conceptually rooted in the problem of finding the area of a region bounded by a curve. Let $f : [a, b] \to \mathbb{R}$ be a continuous function. To define the integral $\int_a^b f(x)dx$, we employ the method of exhaustion by partitioning the interval $[a, b]$ into infinitesimally small sub-intervals.

Definition 3.2.1 (The Riemann Sum)**.** Let the interval $[a, b]$ be divided into n equal sub-intervals, each of length h. The length h is defined by the relation:

$$nh = b - a \quad \iff \quad h = \frac{b - a}{n} \tag{3.3}$$

The partition points are given by $x_r = a + rh$ for $r = 0, 1, \ldots, n$.

The definite integral is defined as the limit of the sum of the areas of rectangles formed on these sub-intervals as $n \to \infty$ (implying $h \to 0$):

$$\int_a^b f(x)dx = \lim_{n\to\infty} h \sum_{r=1}^{n} f(a + rh) \tag{3.4}$$

Remark 3.2.2. The definition above utilizes the *right-end points* of the sub-intervals (r running from 1 to n). One may equivalently use the *left-end points* (r running from 0 to $n - 1$). For continuous functions, both limits converge to the same value.

3.3 Auxiliary Summation Identities

To evaluate integrals using this definition, we frequently utilize the following algebraic identities for the summation of the first n natural numbers and geometric progressions.

Lemma 3.3.1 (Power Sums). *For any $n \in \mathbb{N}$:*

(i) $\displaystyle\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$

(ii) $\displaystyle\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$

(iii) $\displaystyle\sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2$

Lemma 3.3.2 (Geometric Series). *For a geometric progression with common ratio $q \neq 1$:*

$$\sum_{r=1}^{n} aq^{r-1} = a + aq + \cdots + aq^{n-1} = \frac{a(q^n - 1)}{q-1} \tag{3.5}$$

We now demonstrate the application of the definition to evaluate definite integrals.

Example 3.3.3. Evaluate the integral $\int_0^2 (2x+1)dx$ from first principles.

Solution. We identify the limits of integration as $a = 0$ and $b = 2$. Let the interval $[0, 2]$ be divided into n equal sub-intervals of width h. Thus, we have:

$$h = \frac{2-0}{n} = \frac{2}{n} \implies nh = 2.$$

The function is $f(x) = 2x + 1$. We evaluate the function at the partition points $a + rh = rh$:

$$f(rh) = 2(rh) + 1.$$

Applying the definition of the integral as the limit of a sum:

$$\begin{aligned} I &= \lim_{n\to\infty} h \sum_{r=1}^{n} f(rh) \\ &= \lim_{n\to\infty} h \sum_{r=1}^{n} (2rh + 1) \\ &= \lim_{n\to\infty} h \left[2h \sum_{r=1}^{n} r + \sum_{r=1}^{n} 1\right]. \end{aligned}$$

Using the standard summation identities $\sum r = \frac{n(n+1)}{2}$ and $\sum 1 = n$, we obtain:

$$\begin{aligned} I &= \lim_{n\to\infty} h \left[2h \frac{n(n+1)}{2} + n\right] \\ &= \lim_{n\to\infty} \left[h^2 n(n+1) + nh\right]. \end{aligned}$$

To evaluate the limit, we substitute $h = 2/n$ (noting that $nh = 2$):

$$\begin{aligned} I &= \lim_{n\to\infty} \left[\left(\frac{2}{n}\right)^2 n(n+1) + 2\right] \\ &= \lim_{n\to\infty} \left[\frac{4}{n^2}(n^2 + n) + 2\right] \\ &= \lim_{n\to\infty} \left[4\left(1 + \frac{1}{n}\right) + 2\right]. \end{aligned}$$

As $n \to \infty$, the term $1/n$ vanishes. Thus:

$$I = 4(1) + 2 = 6.$$

■

Example 3.3.4. Evaluate the integral $\int_1^3 x^2 dx$ as a limit of a sum.

Solution. Here, $a = 1$, $b = 3$, and $f(x) = x^2$. The width of the sub-interval is $h = \frac{3-1}{n} = \frac{2}{n}$, implying $nh = 2$. The general term at the partition point is:

$$f(a + rh) = f(1 + rh) = (1 + rh)^2 = 1 + 2rh + r^2h^2.$$

By the definition of the integral:

$$\begin{aligned} I &= \lim_{n\to\infty} h \sum_{r=1}^{n} \left(1 + 2rh + r^2h^2\right) \\ &= \lim_{n\to\infty} h \left[\sum_{r=1}^{n} 1 + 2h \sum_{r=1}^{n} r + h^2 \sum_{r=1}^{n} r^2\right]. \end{aligned}$$

Substituting the summation formulas:

$$\begin{aligned} I &= \lim_{n\to\infty} h \left[n + 2h\frac{n(n+1)}{2} + h^2\frac{n(n+1)(2n+1)}{6}\right] \\ &= \lim_{n\to\infty} \left[nh + h^2n(n+1) + \frac{h^3n(n+1)(2n+1)}{6}\right]. \end{aligned}$$

We now group terms to form factors of nh (where $nh = 2$):

$$\begin{aligned} I &= \lim_{n\to\infty} \left[nh + (nh)(nh+h) + \frac{(nh)(nh+h)(2nh+h)}{6}\right] \\ &= \lim_{n\to\infty} \left[2 + 2(2+h) + \frac{2(2+h)(4+h)}{6}\right]. \end{aligned}$$

Taking the limit as $n \to \infty$ corresponds to $h \to 0$:

$$\begin{aligned} I &= 2 + 2(2+0) + \frac{2(2)(4)}{6} \\ &= 2 + 4 + \frac{16}{6} \\ &= 6 + \frac{8}{3} = \frac{26}{3}. \end{aligned}$$

■

Example 3.3.5. Evaluate $\int_0^1 e^x dx$.

Solution. We have $a = 0$, $b = 1$, and $f(x) = e^x$. The step size is $h = \frac{1}{n}$, so $nh = 1$. The term for summation is $f(rh) = e^{rh} = (e^h)^r$.

$$\begin{aligned} I &= \lim_{n\to\infty} h \sum_{r=1}^{n} e^{rh} \\ &= \lim_{n\to\infty} h \left(e^h + e^{2h} + e^{3h} + \cdots + e^{nh}\right). \end{aligned}$$

The expression in the parenthesis is a geometric progression with first term e^h, common ratio e^h, and n terms.

$$\begin{aligned} I &= \lim_{n\to\infty} h \cdot \frac{e^h((e^h)^n - 1)}{e^h - 1} \\ &= \lim_{n\to\infty} h \cdot \frac{e^h(e^{nh} - 1)}{e^h - 1}. \end{aligned}$$

Substituting $nh = 1$:

$$I = \lim_{n\to\infty} \frac{h}{e^h - 1} \cdot e^h(e^1 - 1).$$

Since $n \to \infty$ is equivalent to $h \to 0$, we utilize the standard limit $\lim_{h\to 0} \frac{e^h-1}{h} = 1$, which implies $\lim_{h\to 0} \frac{h}{e^h-1} = 1$. Therefore:

$$\begin{aligned} I &= 1 \cdot e^0 \cdot (e - 1) \\ &= e - 1. \end{aligned}$$

■

Example 3.3.6. Evaluate the integral $\int_1^3 (2x^2 + 5)dx$ as the limit of a sum.

Solution. Here $a = 1$, $b = 3$, and $f(x) = 2x^2 + 5$. The length of the sub-interval is determined by $h = \frac{3-1}{n} = \frac{2}{n}$, which implies $nh = 2$.

We evaluate the function at the general partition point $x_r = 1 + rh$:

$$\begin{aligned} f(1 + rh) &= 2(1 + rh)^2 + 5 \\ &= 2(1 + 2rh + r^2h^2) + 5 \\ &= 2 + 4rh + 2r^2h^2 + 5 \\ &= 7 + 4rh + 2r^2h^2. \end{aligned}$$

Applying the definition of the definite integral:

$$\begin{aligned} I &= \lim_{n\to\infty} h \sum_{r=1}^{n} \left(7 + 4rh + 2r^2h^2\right) \\ &= \lim_{n\to\infty} h \left[\sum_{r=1}^{n} 7 + 4h \sum_{r=1}^{n} r + 2h^2 \sum_{r=1}^{n} r^2\right]. \end{aligned}$$

Substituting the standard summation identities:

$$\begin{aligned} I &= \lim_{n\to\infty} h \left[7n + 4h\frac{n(n+1)}{2} + 2h^2\frac{n(n+1)(2n+1)}{6}\right] \\ &= \lim_{n\to\infty} \left[7nh + 2h^2n(n+1) + \frac{h^3n(n+1)(2n+1)}{3}\right]. \end{aligned}$$

We organize the terms to highlight the factors of nh (where $nh = 2$):

$$\begin{aligned} I &= \lim_{n\to\infty} \left[7(nh) + 2(nh)(nh + h) + \frac{(nh)(nh + h)(2nh + h)}{3}\right] \\ &= \lim_{n\to\infty} \left[7(2) + 2(2)(2 + h) + \frac{2(2 + h)(4 + h)}{3}\right]. \end{aligned}$$

As $n \to \infty$, $h \to 0$. Thus:

$$\begin{aligned} I &= 14 + 4(2) + \frac{2(2)(4)}{3} \\ &= 14 + 8 + \frac{16}{3} \\ &= 22 + \frac{16}{3} = \frac{66+16}{3} = \frac{82}{3}. \end{aligned}$$

■

Example 3.3.7. Evaluate $\int_0^1 e^{2x} dx$ using the method of summation.

Solution. We have $a = 0$, $b = 1$, and $f(x) = e^{2x}$. The interval width is $h = \frac{1-0}{n} = \frac{1}{n}$, so $nh = 1$. The general term is $f(rh) = e^{2(rh)} = (e^{2h})^r$. This forms a geometric progression with common ratio $R = e^{2h}$.

Using the integral definition:

$$\begin{aligned} I &= \lim_{n\to\infty} h \sum_{r=1}^{n} (e^{2h})^r \\ &= \lim_{n\to\infty} h \left(e^{2h} + e^{4h} + \cdots + e^{2nh} \right). \end{aligned}$$

Applying the sum of a geometric series formula $S_n = \dfrac{\text{first term} \cdot (R^n - 1)}{R - 1}$:

$$I = \lim_{n\to\infty} h \cdot \frac{e^{2h}((e^{2h})^n - 1)}{e^{2h} - 1}.$$

Simplifying the exponent $(e^{2h})^n = e^{2nh}$. Since $nh = 1$, we have $e^{2nh} = e^2$:

$$I = \lim_{n\to\infty} h \cdot \frac{e^{2h}(e^2 - 1)}{e^{2h} - 1}.$$

We rearrange the limit to utilize the standard form $\lim_{x\to 0} \frac{x}{e^x - 1} = 1$:

$$I = (e^2 - 1) \cdot \lim_{h\to 0} e^{2h} \cdot \frac{h}{e^{2h} - 1}.$$

To match the denominator, we multiply and divide by 2:

$$I = (e^2 - 1) \cdot \lim_{h\to 0} e^{2h} \cdot \frac{1}{2} \left(\frac{2h}{e^{2h} - 1} \right).$$

Evaluating the limits as $h \to 0$:

$$I = (e^2 - 1) \cdot 1 \cdot \frac{1}{2} \cdot 1 = \frac{e^2 - 1}{2}.$$

■

Example 3.3.8. Evaluate $\int_a^b x dx$ for general constants a and b ($b > a$).

Solution. Let $f(x) = x$. The step size is $h = \frac{b-a}{n}$, implying $nh = b - a$. The partition points are $x_r = a + rh$. Thus, $f(x_r) = a + rh$.

By definition:

$$\begin{aligned} I &= \lim_{n\to\infty} h\sum_{r=1}^{n}(a+rh) \\ &= \lim_{n\to\infty} h\left[\sum_{r=1}^{n} a + h\sum_{r=1}^{n} r\right] \\ &= \lim_{n\to\infty} h\left[na + h\frac{n(n+1)}{2}\right]. \end{aligned}$$

Distributing h and grouping terms with nh:

$$I = \lim_{n\to\infty}\left[(nh)a + \frac{(nh)(nh+h)}{2}\right].$$

Substituting $nh = b - a$ and letting $h \to 0$:

$$\begin{aligned} I &= (b-a)a + \frac{(b-a)(b-a+0)}{2} \\ &= ab - a^2 + \frac{(b-a)^2}{2} \\ &= \frac{2ab - 2a^2 + (b^2 - 2ab + a^2)}{2} \\ &= \frac{b^2 - a^2}{2}. \end{aligned}$$

This confirms the standard result $\left[\frac{x^2}{2}\right]_a^b = \frac{b^2}{2} - \frac{a^2}{2}$. ■

Example 3.3.9. Evaluate the limit:

$$I = \lim_{n\to\infty}\frac{1}{n}\left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \cdots + \sqrt{\frac{n}{n}}\right)$$

Solution. We express the given series in summation notation:

$$I = \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\sqrt{\frac{r}{n}}.$$

This summation represents a Riemann sum for the function $f(x) = \sqrt{x}$ over the interval $[0, 1]$. We identify the conversion to the definite integral:

$$\frac{r}{n} \to x, \quad \frac{1}{n} \to dx, \quad \lim_{n\to\infty}\sum_{r=1}^{n} \to \int_0^1.$$

Thus, the limit becomes:

$$\begin{aligned} I &= \int_0^1 \sqrt{x}dx \\ &= \int_0^1 x^{1/2}dx \\ &= \left[\frac{x^{3/2}}{3/2}\right]_0^1 \\ &= \frac{2}{3}\left[1^{3/2} - 0^{3/2}\right] \\ &= \frac{2}{3}. \end{aligned}$$

■

Example 3.3.10. Evaluate the limit L:

$$L = \lim_{n\to\infty} \left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\dots\left(1+\frac{n}{n}\right)\right]^{\frac{1}{n}}$$

Solution. To handle the product, we take the natural logarithm of the limit. Since the logarithm is a continuous function, we can move the limit operator outside:

$$\ln L = \lim_{n\to\infty} \ln\left(\prod_{r=1}^{n}\left(1+\frac{r}{n}\right)^{\frac{1}{n}}\right).$$

Using the properties of logarithms ($\ln(a^b) = b\ln a$ and $\ln(ab) = \ln a + \ln b$), we convert the product into a sum:

$$\ln L = \lim_{n\to\infty} \frac{1}{n}\sum_{r=1}^{n} \ln\left(1+\frac{r}{n}\right).$$

This is a Riemann sum for the function $f(x) = \ln(1+x)$ over the interval $[0,1]$:

$$\ln L = \int_0^1 \ln(1+x)dx.$$

To evaluate this integral, we use the substitution $t = 1+x$, hence $dx = dt$. The limits change from $[0,1]$ to $[1,2]$:

$$\ln L = \int_1^2 \ln(t)dt.$$

Using the standard antiderivative $\int \ln t dt = t\ln t - t$:

$$\begin{aligned}\ln L &= [t\ln t - t]_1^2 \\ &= (2\ln 2 - 2) - (1\ln 1 - 1) \\ &= 2\ln 2 - 2 - 0 + 1 \\ &= 2\ln 2 - 1.\end{aligned}$$

We can rewrite the result using logarithm properties:

$$\ln L = \ln(2^2) - \ln e = \ln 4 - \ln e = \ln\left(\frac{4}{e}\right).$$

Exponentiating both sides to solve for L:

$$L = \frac{4}{e}.$$

■

Example 3.3.11. Evaluate the limit of the sum:

$$L = \lim_{n\to\infty}\left(\frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \dots + \frac{n}{n^2+n^2}\right)$$

Solution. We write the series in summation notation:

$$L = \lim_{n\to\infty}\sum_{r=1}^{n}\frac{n}{n^2+r^2}.$$

To transform this into a Riemann sum, we must create factors of $\frac{r}{n}$ and an explicit $\frac{1}{n}$. We divide the numerator and denominator of the general term by n^2:

$$\begin{aligned}L &= \lim_{n\to\infty}\sum_{r=1}^{n}\frac{n/n^2}{(n^2+r^2)/n^2} \\ &= \lim_{n\to\infty}\sum_{r=1}^{n}\frac{1/n}{1+(r/n)^2} \\ &= \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\frac{1}{1+(r/n)^2}.\end{aligned}$$

Recognizing this as the limit of a sum for $f(x) = \frac{1}{1+x^2}$ over $[0, 1]$:

$$\begin{aligned} L &= \int_0^1 \frac{1}{1+x^2} dx \\ &= \left[\tan^{-1} x\right]_0^1 \\ &= \tan^{-1}(1) - \tan^{-1}(0) \\ &= \frac{\pi}{4}. \end{aligned}$$

■

Example 3.3.12 (Limit of a Root of Factorials)**.** Evaluate the limit:

$$L = \lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n}$$

Solution. We first rewrite the expression to unify the n terms under the radical. Note that $n = \sqrt[n]{n^n}$.

$$L = \lim_{n\to\infty} \left(\frac{n!}{n^n}\right)^{\frac{1}{n}} = \lim_{n\to\infty} \left(\frac{1 \cdot 2 \cdot 3 \dots n}{n \cdot n \cdot n \dots n}\right)^{\frac{1}{n}}.$$

This can be written as a product:

$$L = \lim_{n\to\infty} \left[\prod_{r=1}^{n} \left(\frac{r}{n}\right)\right]^{\frac{1}{n}}.$$

Taking the natural logarithm of both sides to convert the product into a sum:

$$\begin{aligned} \ln L &= \lim_{n\to\infty} \ln \left(\prod_{r=1}^{n} \frac{r}{n}\right)^{\frac{1}{n}} \\ &= \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} \ln\left(\frac{r}{n}\right). \end{aligned}$$

This corresponds to the integral of $\ln x$ over the interval $[0, 1]$.

$$\ln L = \int_0^1 \ln x dx.$$

We evaluate this improper integral using integration by parts ($\int \ln x dx = x \ln x - x$):

$$\begin{aligned} \ln L &= [x \ln x - x]_0^1 \\ &= (1 \ln 1 - 1) - \lim_{t\to 0^+} (t \ln t - t). \end{aligned}$$

Recalling that $\lim_{t\to 0^+} t \ln t = 0$, we have:

$$\ln L = (0 - 1) - (0) = -1.$$

Exponentiating to find L:

$$L = e^{-1} = \frac{1}{e}.$$

■

Example 3.3.13. Evaluate the limit:

$$L = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{\sqrt{n^2 - r^2}}$$

Solution. We begin by manipulating the general term to extract the form $f(r/n) \cdot \frac{1}{n}$. Factor n^2 out of the square root in the denominator:

$$\begin{aligned} L &= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{\sqrt{n^2(1 - r^2/n^2)}} \\ &= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n\sqrt{1 - (r/n)^2}} \\ &= \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} \frac{1}{\sqrt{1 - (r/n)^2}}. \end{aligned}$$

This is the Riemann sum for the function $f(x) = \frac{1}{\sqrt{1-x^2}}$ over the interval $[0, 1]$.

$$L = \int_0^1 \frac{1}{\sqrt{1 - x^2}} dx.$$

This is a standard integral resulting in the inverse sine function:

$$\begin{aligned} L &= \left[\sin^{-1} x\right]_0^1 \\ &= \sin^{-1}(1) - \sin^{-1}(0) \\ &= \frac{\pi}{2} - 0 \\ &= \frac{\pi}{2}. \end{aligned}$$

■

3.4 Differentiation under the integral sign

Differentiation under the integral sign, famously associated with Richard Feynman (often called "Feynman's Trick"), is a powerful technique for evaluating definite integrals that are difficult to solve using standard methods. The core idea is to introduce a parameter into the integral, differentiate with respect to that parameter to simplify the integrand, and then integrate the result back.

Leibniz Rule (Constant Limits)

If the function $f(x, t)$ and its partial derivative $\frac{\partial f}{\partial t}$ are continuous in a region containing the rectangle $[a, b] \times [c, d]$, then for $t \in (c, d)$:

$$\frac{d}{dt} \int_a^b f(x, t)\, dx = \int_a^b \frac{\partial}{\partial t} f(x, t)\, dx \tag{3.6}$$

Theorem 3.4.1 (General Leibniz Integral Rule). *Let $f(x, \alpha)$ and $\frac{\partial f}{\partial \alpha}$ be continuous functions. Let $a(\alpha)$ and $b(\alpha)$ be differentiable functions of α. If*

$$I(\alpha) = \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha) dx, \tag{3.7}$$

then the derivative of I with respect to α is:

$$\frac{dI}{d\alpha} = \int_{a(\alpha)}^{b(\alpha)} \frac{\partial f}{\partial \alpha} dx + f(b(\alpha), \alpha) \cdot \frac{db}{d\alpha} - f(a(\alpha), \alpha) \cdot \frac{da}{d\alpha}. \tag{3.8}$$

Example 3.4.2. Evaluate the integral:

$$I = \int_0^1 \frac{x^7 - 1}{\ln x}\, dx$$

Proof. We define a more general function $I(\alpha)$ such that:

$$I(\alpha) = \int_0^1 \frac{x^\alpha - 1}{\ln x}\, dx$$

Our goal is to find $I(7)$, and we note that $I(0) = \int_0^1 0\, dx = 0$. Differentiating with respect to α:

$$\frac{dI}{d\alpha} = \int_0^1 \frac{\partial}{\partial\alpha}\left(\frac{x^\alpha - 1}{\ln x}\right) dx$$

Since $\frac{\partial}{\partial\alpha}(x^\alpha) = x^\alpha \ln x$, we have:

$$\frac{dI}{d\alpha} = \int_0^1 \frac{x^\alpha \ln x}{\ln x}\, dx = \int_0^1 x^\alpha\, dx$$

Evaluating the integral:

$$\frac{dI}{d\alpha} = \left[\frac{x^{\alpha+1}}{\alpha+1}\right]_0^1 = \frac{1}{\alpha+1}$$

Now, we integrate $\frac{dI}{d\alpha}$ with respect to α from 0 to 7:

$$I(7) - I(0) = \int_0^7 \frac{1}{\alpha+1}\, d\alpha$$

$$I(7) = [\ln(\alpha+1)]_0^7 = \ln 8 - \ln 1 = \ln 8$$

■

Example 3.4.3. Evaluate the integral:

$$J(a) = \int_0^\infty e^{-x^2} \cos(2ax)\, dx$$

Proof. Differentiate with respect to a:

$$J'(a) = \int_0^\infty -2xe^{-x^2} \sin(2ax)\, dx$$

Apply Integration by Parts letting $u = \sin(2ax)$ and $dv = -2xe^{-x^2} dx$:

- $du = 2a\cos(2ax)dx$
- $v = e^{-x^2}$

$$J'(a) = \underbrace{\left[e^{-x^2}\sin(2ax)\right]_0^\infty}_{0} - \int_0^\infty 2ae^{-x^2}\cos(2ax)\, dx$$

This yields the differential equation:

$$J'(a) = -2aJ(a) \implies \frac{dJ}{J} = -2a\, da$$

Integrating both sides:

$$\ln J(a) = -a^2 + C \implies J(a) = Ce^{-a^2}$$

When $a = 0$, $J(0) = \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$. Thus $C = \frac{\sqrt{\pi}}{2}$. **Hence,** $J(a) = \frac{\sqrt{\pi}}{2}e^{-a^2}$. ■

Example 3.4.4. Evaluate the Frullani-type integral:

$$I = \int_0^\infty \frac{e^{-ax} - e^{-bx}}{x}\, dx \quad (a, b > 0)$$

Solution. Define a function $I(a) = \int_0^\infty \frac{e^{-ax}-e^{-bx}}{x}\,dx$. Note that $I(b) = 0$. Differentiating with respect to a:

$$\frac{dI}{da} = \int_0^\infty \frac{\partial}{\partial a}\left(\frac{e^{-ax} - e^{-bx}}{x}\right) dx = \int_0^\infty \frac{-xe^{-ax}}{x}\,dx = -\int_0^\infty e^{-ax}\,dx$$

Evaluating the integral:

$$\frac{dI}{da} = \left[\frac{e^{-ax}}{a}\right]_0^\infty = -\frac{1}{a}$$

Now, integrate $I'(a)$ with respect to a:

$$I(a) = -\ln(a) + C$$

Using the condition $I(b) = 0$:

$$0 = -\ln(b) + C \implies C = \ln(b)$$

Thus, $I(a) = \ln(b) - \ln(a) = \ln\left(\frac{b}{a}\right)$. ■

Example 3.4.5. Evaluate the integral:

$$I(a) = \int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)}\,dx$$

Solution. Differentiate with respect to a:

$$I'(a) = \int_0^\infty \frac{x}{x(1+a^2x^2)(1+x^2)}\,dx = \int_0^\infty \frac{1}{(1+a^2x^2)(1+x^2)}\,dx$$

Using partial fraction decomposition:

$$\frac{1}{(1+a^2x^2)(1+x^2)} = \frac{1}{1-a^2}\left[\frac{1}{1+x^2} - \frac{a^2}{1+a^2x^2}\right]$$

Then

$$I'(a) = \int_0^\infty \frac{1}{1-a^2}\left[\frac{1}{1+x^2} - \frac{a^2}{1+a^2x^2}\right] dx$$

Integrating RHS gives:

$$I'(a) = \frac{1}{1-a^2}\left[\tan^{-1}(x) - a\tan^{-1}(ax)\right]_0^\infty = \frac{1}{1-a^2}\left(\frac{\pi}{2} - \frac{a\pi}{2}\right)$$

$$I'(a) = \frac{\pi}{2}\frac{1-a}{(1-a)(1+a)} = \frac{\pi}{2(1+a)}$$

Integrating $I'(a)$ back:

$$I(a) = \frac{\pi}{2}\ln(1+a) + C$$

Since $I(0) = 0$, we have $C = 0$. The final result is $I(a) = \frac{\pi}{2}\ln(1+a)$. ■

Example 3.4.6. Evaluate the "Glasser's Master Theorem" related integral:

$$I(a) = \int_0^\infty e^{-\left(x^2 + \frac{a^2}{x^2}\right)}\,dx \quad (a > 0)$$

Solution. Differentiate with respect to a:

$$I'(a) = \int_0^\infty -\frac{2a}{x^2}e^{-(x^2+a^2/x^2)}\,dx$$

Substitute $u = a/x$, then $du = -a/x^2 dx$, which means $dx = -a/u^2 du$. As $x \to 0, u \to \infty$ and $x \to \infty, u \to 0$:

$$I'(a) = -2\int_0^\infty e^{-(a^2/u^2+u^2)}\,du = -2I(a)$$

We have a differential equation $I'(a) = -2I(a)$. The solution is:

$$I(a) = Ce^{-2a}$$

When $a = 0$, $I(0) = \displaystyle\int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}$. Thus, $I(a) = \frac{\sqrt{\pi}}{2}e^{-2a}$. ■

Example 3.4.7. Evaluate:

$$I(a) = \int_0^\infty \frac{\ln(1+a^2x^2)}{1+x^2}\,dx$$

Solution. Differentiate with respect to a:

$$I'(a) = \int_0^\infty \frac{2ax^2}{(1+a^2x^2)(1+x^2)}\,dx$$

Using partial fractions on the integrand $\dfrac{x^2}{(1+a^2x^2)(1+x^2)}$:

$$I'(a) = \frac{2a}{a^2-1}\int_0^\infty \left(\frac{a^2x^2+1-1}{1+a^2x^2} - \frac{x^2+1-1}{1+x^2}\right)dx = \cdots = \frac{\pi}{a+1}$$

Integrating $I'(a) = \frac{\pi}{a+1}$:

$$I(a) = \pi\ln(a+1) + C$$

Since $I(0) = \displaystyle\int_0^\infty \frac{\ln(1)}{1+x^2}dx = 0$, we find $C = 0$. The result is $I(a) = \pi\ln(a+1)$. ■

Example 3.4.8. Evaluate the integral:

$$I(a,b) = \int_0^\infty \frac{\cos(ax)-\cos(bx)}{x^2}\,dx$$

Solution. Differentiate with respect to a:

$$\frac{\partial I}{\partial a} = \int_0^\infty \frac{-x\sin(ax)}{x^2}\,dx = -\int_0^\infty \frac{\sin(ax)}{x}\,dx$$

Using the Dirichlet Integral result ($\displaystyle\int_0^\infty \frac{\sin(ax)}{x}dx = \frac{\pi}{2}\text{sgn}(a)$), for $a > 0$:

$$\frac{\partial I}{\partial a} = -\frac{\pi}{2}$$

Integrating with respect to a:

$$I(a,b) = -\frac{\pi}{2}a + f(b)$$

Since $I(b,b) = 0$, we have $0 = -\frac{\pi}{2}b + f(b)$, so $f(b) = \frac{\pi}{2}b$. Thus, $I(a,b) = \frac{\pi}{2}(b-a)$. ■

Example 3.4.9. Evaluate:

$$I(a) = \int_0^\pi \ln(1-2a\cos x+a^2)\,dx$$

Solution. We define the integral as a function of the parameter a:

$$I(a) = \int_0^\pi \ln(a^2 - 2a\cos x + 1)\,dx$$

First, we observe the initial condition at $a = 0$:

$$I(0) = \int_0^\pi \ln(1)\,dx = 0$$

Differentiating $I(a)$ with respect to a using Leibniz's Integral Rule:

$$I'(a) = \int_0^\pi \frac{\partial}{\partial a}\ln(a^2 - 2a\cos x + 1)\,dx = \int_0^\pi \frac{2a - 2\cos x}{a^2 - 2a\cos x + 1}\,dx$$

To evaluate this, we manipulate the numerator to relate it to the denominator:

$$I'(a) = \frac{1}{a}\int_0^\pi \frac{2a^2 - 2a\cos x}{a^2 - 2a\cos x + 1}\,dx$$

$$I'(a) = \frac{1}{a}\int_0^\pi \frac{(a^2 - 2a\cos x + 1) + (a^2 - 1)}{a^2 - 2a\cos x + 1}\,dx$$

Splitting the integral into two parts:

$$I'(a) = \frac{1}{a}\left[\int_0^\pi 1\,dx + (a^2 - 1)\int_0^\pi \frac{1}{(a^2+1) - 2a\cos x}\,dx\right]$$

$$I'(a) = \frac{1}{a}\left[\pi + (a^2 - 1)\int_0^\pi \frac{dx}{(a^2+1) - 2a\cos x}\right]$$

We utilize the standard integral formula:

$$\int_0^\pi \frac{dx}{A + B\cos x} = \frac{\pi}{\sqrt{A^2 - B^2}} \quad \text{for } A^2 > B^2$$

Setting $A = a^2 + 1$ and $B = -2a$, we find:

$$A^2 - B^2 = (a^2+1)^2 - (-2a)^2 = a^4 + 2a^2 + 1 - 4a^2 = (a^2 - 1)^2$$

Thus, $\sqrt{A^2 - B^2} = |a^2 - 1|$. We now evaluate $I'(a)$ by cases:

Case 1: $|a| < 1$
In this region, $a^2 - 1$ is negative, so $|a^2 - 1| = 1 - a^2$. Substituting this into our expression for $I'(a)$:

$$I'(a) = \frac{1}{a}\left[\pi + (a^2 - 1)\frac{\pi}{1 - a^2}\right] = \frac{1}{a}[\pi - \pi] = 0$$

Since $I'(a) = 0$ for all $|a| < 1$ and $I(0) = 0$, it follows that:

$$I(a) = 0 \quad \text{for } |a| \le 1$$

Case 2: $|a| > 1$
In this region, $a^2 - 1$ is positive, so $|a^2 - 1| = a^2 - 1$. Substituting this into our expression for $I'(a)$:

$$I'(a) = \frac{1}{a}\left[\pi + (a^2 - 1)\frac{\pi}{a^2 - 1}\right] = \frac{1}{a}[\pi + \pi] = \frac{2\pi}{a}$$

Integrating $I'(a) = \frac{2\pi}{a}$ with respect to a:

$$I(a) = \int \frac{2\pi}{a}\,da = 2\pi\ln|a| + C$$

Because the original integral $I(a)$ is a continuous function of a, we use the boundary value from Case 1 where $I(1) = 0$:

$$0 = 2\pi \ln(1) + C \implies C = 0$$

Thus, $I(a) = 2\pi \ln |a|$ for $|a| > 1$.

Thus, we conclude that:

$$\int_0^\pi \ln(1 - 2a\cos x + a^2)\,dx = \begin{cases} 0 & |a| \le 1 \\ 2\pi \ln |a| & |a| > 1 \end{cases}$$

■

Example 3.4.10. Evaluate the integral:

$$I(a) = \int_0^\infty \frac{1 - \cos(ax)}{x^2}\,dx$$

Solution. We note that $I(0) = 0$. Differentiating with respect to the parameter a:

$$I'(a) = \int_0^\infty \frac{\partial}{\partial a}\left(\frac{1 - \cos(ax)}{x^2}\right) dx = \int_0^\infty \frac{x\sin(ax)}{x^2}\,dx = \int_0^\infty \frac{\sin(ax)}{x}\,dx$$

This is the Dirichlet integral. For $a > 0$, we substitute $u = ax$, which yields:

$$I'(a) = \int_0^\infty \frac{\sin u}{u}\,du = \frac{\pi}{2}$$

Now, we integrate $I'(a)$ with respect to a to return to $I(a)$:

$$I(a) = \int \frac{\pi}{2}\,da = \frac{\pi}{2}a + C$$

Using the condition $I(0) = 0$, we find $C = 0$. Thus, for $a \ge 0$:

$$I(a) = \frac{\pi a}{2}$$

■

Example 3.4.11. Evaluate the integral:

$$I(a) = \int_0^\pi \frac{\ln(1 + a\cos x)}{\cos x}\,dx, \quad |a| < 1$$

Solution. Differentiating with respect to a:

$$I'(a) = \int_0^\pi \frac{\partial}{\partial a}\left(\frac{\ln(1 + a\cos x)}{\cos x}\right) dx = \int_0^\pi \frac{1}{1 + a\cos x}\,dx$$

Using the standard result $\int_0^\pi \frac{dx}{1+a\cos x} = \frac{\pi}{\sqrt{1-a^2}}$ for $|a| < 1$:

$$I'(a) = \frac{\pi}{\sqrt{1 - a^2}}$$

Integrating with respect to a:

$$I(a) = \int \frac{\pi}{\sqrt{1 - a^2}}\,da = \pi \arcsin(a) + C$$

Since $I(0) = \int_0^\pi \frac{\ln(1)}{\cos x}\,dx = 0$, and $\arcsin(0) = 0$, it follows that $C = 0$. The final result is:

$$I(a) = \pi \arcsin(a)$$

■

Example 3.4.12. Evaluate the integral ($a, b > 0$):

$$I(a,b) = \int_0^{\pi/2} \ln(a^2\cos^2 x + b^2 \sin^2 x)\,dx$$

Solution. Differentiate with respect to a:

$$\frac{\partial I}{\partial a} = \int_0^{\pi/2} \frac{2a\cos^2 x}{a^2\cos^2 x + b^2\sin^2 x}\,dx = \int_0^{\pi/2} \frac{2a}{a^2 + b^2\tan^2 x}\,dx$$

Let $u = \tan x$, then $du = (1+u^2)dx$:

$$\frac{\partial I}{\partial a} = \int_0^{\infty} \frac{2a}{(a^2+b^2u^2)(1+u^2)}\,du$$

Using partial fractions:

$$\frac{2a}{(a^2+b^2u^2)(1+u^2)} = \frac{2a}{b^2-a^2}\left[\frac{b^2}{b^2u^2+a^2} - \frac{1}{u^2+1}\right]$$

Integrating:

$$\frac{\partial I}{\partial a} = \frac{2a}{b^2-a^2}\left[\frac{b}{a}\tan^{-1}\left(\frac{bu}{a}\right) - \tan^{-1}(u)\right]_0^{\infty} = \frac{2a}{b^2-a^2}\left(\frac{b\pi}{2a} - \frac{\pi}{2}\right) = \frac{\pi}{a+b}$$

Integrating with respect to a:

$$I(a,b) = \pi\ln(a+b) + C(b)$$

When $a = b$, $I(b,b) = \int_0^{\pi/2}\ln(b^2)dx = \pi\ln b$. Substituting $a = b$ into our formula: $\pi \ln b = \pi\ln(2b) + C(b) \implies C(b) = \pi\ln(1/2)$. Thus:

$$I(a,b) = \pi\ln\left(\frac{a+b}{2}\right)$$

■

Example 3.4.13. Evaluate:

$$I(a) = \int_0^{\infty} e^{-x}\frac{\sin^2(ax)}{x}\,dx$$

Solution. Note $I(0) = 0$. Differentiating with respect to a:

$$I'(a) = \int_0^{\infty} e^{-x}\frac{2x\sin(ax)\cos(ax)}{x}\,dx = \int_0^{\infty} e^{-x}\sin(2ax)\,dx$$

This is the Laplace transform of $\sin(2ax)$ evaluated at $s = 1$:

$$I'(a) = \mathcal{L}\{\sin(2ax)\}_{s=1} = \frac{2a}{1^2+(2a)^2} = \frac{2a}{1+4a^2}$$

Integrating with respect to a:

$$I(a) = \int \frac{2a}{1+4a^2}\,da = \frac{1}{4}\ln(1+4a^2) + C$$

Since $I(0) = 0$, $C = 0$. The result is:

$$I(a) = \frac{1}{4}\ln(1+4a^2)$$

■

Example 3.4.14. Find the derivative $F'(x)$ of the function with variable limits:

$$F(x) = \int_x^{x^2} \frac{\sin(xt)}{t}\, dt$$

Solution. Using the General Leibniz Rule:

$$F'(x) = f(x^2, x) \cdot \frac{d}{dx}(x^2) - f(x, x) \cdot \frac{d}{dx}(x) + \int_x^{x^2} \frac{\partial}{\partial x}\left(\frac{\sin(xt)}{t}\right) dt$$

Substituting $f(t, x) = \frac{\sin(xt)}{t}$:

1. $f(x^2, x) \cdot 2x = \frac{\sin(x^3)}{x^2} \cdot 2x = \frac{2\sin(x^3)}{x}$
2. $f(x, x) \cdot 1 = \frac{\sin(x^2)}{x}$
3. $\int_x^{x^2} \cos(xt)\, dt = \left[\frac{\sin(xt)}{x}\right]_x^{x^2} = \frac{\sin(x^3) - \sin(x^2)}{x}$

Summing the terms:

$$F'(x) = \frac{2\sin(x^3)}{x} - \frac{\sin(x^2)}{x} + \frac{\sin(x^3) - \sin(x^2)}{x} = \frac{3\sin(x^3) - 2\sin(x^2)}{x}$$

■

3.5 The Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus (FTC) serves as the bridge between the two principal branches of calculus: differential calculus (the study of rates of change) and integral calculus (the study of accumulation and area). Independently developed by Isaac Newton and Gottfried Wilhelm Leibniz, this theorem establishes that differentiation and integration are, in essence, inverse operations.

The theorem is typically presented in two parts. The first part asserts that the accumulation of a function is an antiderivative of that function. The second part provides a powerful method for evaluating definite integrals using antiderivatives.

3.5.1 The First Fundamental Theorem of Calculus

This part of the theorem deals with the derivative of a function defined by an integral. It formally establishes the existence of antiderivatives for continuous functions.

Theorem 3.5.1 (FTC Part I)**.** *Let f be a continuous real-valued function defined on a closed interval $[a, b]$. Let F be the function defined, for all $x \in [a, b]$, by*

$$F(x) = \int_a^x f(t)dt. \tag{3.9}$$

Then F is uniformly continuous on $[a, b]$ and differentiable on the open interval (a, b), and its derivative is given by

$$F'(x) = \frac{d}{dx}\int_a^x f(t)dt = f(x). \tag{3.10}$$

Proof. We apply the definition of the derivative to the function $F(x)$. For a fixed $x \in (a, b)$ and a small increment $h \neq 0$ such that $x + h \in [a, b]$, the difference quotient is:

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\left(\int_a^{x+h} f(t)dt - \int_a^x f(t)dt\right).$$

Using the additive property of definite integrals, $\int_a^{x+h} = \int_a^x + \int_x^{x+h}$, the expression simplifies to:

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}\int_x^{x+h} f(t)dt.$$

According to the **Mean Value Theorem for Integrals**, since f is continuous on $[x, x+h]$ (assuming $h > 0$ for simpler notation, though the logic holds for $h < 0$), there exists a number c_h in the interval $[x, x+h]$ such that:

$$\int_x^{x+h} f(t)dt = f(c_h) \cdot h.$$

Substituting this back into our difference quotient:

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h}(f(c_h) \cdot h) = f(c_h).$$

Now, we take the limit as $h \to 0$. As $h \to 0$, the interval $[x, x+h]$ shrinks to the single point x. Consequently, c_h is forced to approach x. Since f is continuous at x:

$$\lim_{h \to 0} f(c_h) = f(x).$$

Therefore:

$$F'(x) = \lim_{h \to 0} \frac{F(x+h) - F(x)}{h} = f(x).$$

This completes the proof. ■

3.5.2 The Second Fundamental Theorem of Calculus

While the first part relates the derivative to the integral, the second part (often called the Evaluation Theorem) provides the practical method for calculating definite integrals without using Riemann sums.

Theorem 3.5.2 (FTC Part II)**.** *Let f be a continuous function on the closed interval $[a, b]$ and let G be any antiderivative of f on $[a, b]$, meaning that $G'(x) = f(x)$. Then:*

$$\int_a^b f(x)dx = G(b) - G(a). \tag{3.11}$$

This is often denoted as $[G(x)]_a^b$.

Proof. Let $F(x) = \int_a^x f(t)dt$. By the First Fundamental Theorem of Calculus, we know that F is an antiderivative of f; that is, $F'(x) = f(x)$.

We are given that G is also an antiderivative of f. A standard corollary of the Mean Value Theorem states that if two functions have the same derivative on an interval, they differ only by a constant. Thus, there exists a constant C such that:

$$F(x) = G(x) + C \quad \text{for all } x \in [a, b].$$

We evaluate this relationship at $x = a$:

$$F(a) = \int_a^a f(t)dt = 0.$$

Thus,

$$0 = G(a) + C \implies C = -G(a).$$

Substituting C back into the equation yields $F(x) = G(x) - G(a)$. Finally, we evaluate this relationship at $x = b$:

$$F(b) = G(b) - G(a)$$
$$\int_a^b f(t)dt = G(b) - G(a).$$

This completes the proof. ■

These examples involve finding the derivative of functions defined by integrals.

Example 3.5.3. Find the derivative of $g(x) = \int_1^x \sqrt{1+t^2}dt$.

Solution. Since $f(t) = \sqrt{1+t^2}$ is continuous, we apply FTC Part I directly. The theorem states that if the upper limit is x and the lower limit is a constant, the derivative of the integral is simply the integrand evaluated at x.

$$g'(x) = \frac{d}{dx}\int_1^x \sqrt{1+t^2}dt = \sqrt{1+x^2}.$$

■

Example 3.5.4 (Chain Rule Application)**.** Calculate $\frac{dy}{dx}$ if $y = \int_0^{x^4} \cos(t^2)dt$.

Solution. Here, the upper limit of integration is a function of x, specifically $u(x) = x^4$. We must apply the Chain Rule in conjunction with FTC Part I. Let $y(u) = \int_0^u \cos(t^2)dt$. Then:

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}.$$

By FTC Part I, $\frac{dy}{du} = \cos(u^2)$. We also know $\frac{du}{dx} = \frac{d}{dx}(x^4) = 4x^3$. Substituting these back:

$$\begin{aligned}\frac{dy}{dx} &= \cos((x^4)^2)\cdot(4x^3)\\ &= 4x^3\cos(x^8).\end{aligned}$$

■

These examples involve evaluating definite integrals.

Example 3.5.5. Evaluate $\int_1^2 (4x^3 - 6x)dx$.

Solution. We first identify an antiderivative $G(x)$ for the integrand $f(x) = 4x^3 - 6x$. Using the power rule for integration:

$$G(x) = 4\left(\frac{x^4}{4}\right) - 6\left(\frac{x^2}{2}\right) = x^4 - 3x^2.$$

Note that we omit the constant of integration C as it cancels out during subtraction. By FTC Part II:

$$\begin{aligned}\int_1^2 (4x^3 - 6x)dx &= [x^4 - 3x^2]_1^2\\ &= G(2) - G(1)\\ &= \left(2^4 - 3(2)^2\right) - \left(1^4 - 3(1)^2\right)\\ &= (16-12) - (1-3)\\ &= 4 - (-2)\\ &= 6.\end{aligned}$$

■

Example 3.5.6. Evaluate $\int_0^{\pi/2} \sin(2x)dx$.

Solution. We require an antiderivative for $f(x) = \sin(2x)$. Recall that $\frac{d}{dx}(-\cos(2x)) = 2\sin(2x)$. Therefore, an antiderivative is $G(x) = -\frac{1}{2}\cos(2x)$. Applying FTC Part II:

$$\begin{aligned}\int_0^{\pi/2} \sin(2x)dx &= \left[-\frac{1}{2}\cos(2x)\right]_0^{\pi/2} \\ &= -\frac{1}{2}\left(\cos\left(2\cdot\frac{\pi}{2}\right) - \cos(2\cdot 0)\right) \\ &= -\frac{1}{2}\left(\cos(\pi) - \cos(0)\right) \\ &= -\frac{1}{2}(-1-1) \\ &= -\frac{1}{2}(-2) \\ &= 1.\end{aligned}$$

■

3.6 Rectification

Rectification is the process of determining the length of an arc of a plane curve. To successfully evaluate the arc length, one must often visualize the curve to determine the appropriate limits of integration and exploit symmetries. Therefore, we first discuss the general rules for curve tracing before establishing the formulas for rectification.

3.6.1 Rules for Curve Tracing

To trace a curve given by a Cartesian equation $f(x, y) = 0$, we generally investigate the following five characteristics:

1. **Symmetry:**
   - *About the x-axis:* If the equation contains only **even powers of** y, the curve is symmetric about the x-axis.
   - *About the y-axis:* If the equation contains only **even powers of** x, the curve is symmetric about the y-axis.
   - *About the line $y = x$:* If the equation remains unchanged when x and y are interchanged.
   - *About the Origin:* If the equation remains unchanged when (x, y) is replaced by $(-x, -y)$. This often occurs if all terms have odd degrees or all have even degrees.
2. **Origin and Tangents at the Origin:**
   - Does the curve pass through the origin? Check if $(0, 0)$ satisfies the equation (usually indicated by the absence of a constant term).
   - If it passes through the origin, the equations of the tangents at the origin are found by equating the **lowest degree terms** to zero.
3. **Intersections with Axes:**
   - *x-intercepts:* Set $y = 0$ and solve for x.
   - *y-intercepts:* Set $x = 0$ and solve for y.
4. **Asymptotes:**
   - *Parallel to x-axis:* Equate the coefficient of the highest power of x to zero.
   - *Parallel to y-axis:* Equate the coefficient of the highest power of y to zero.
5. **Region of Existence:** Determine the values of x for which y becomes imaginary. The curve does not exist in these regions. Similarly, check for values of y where x becomes imaginary.

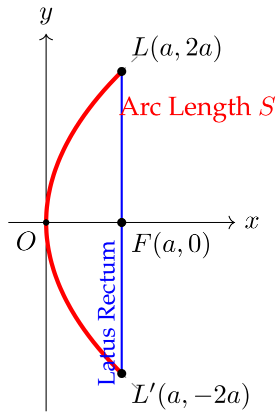


Figure 3.1: Area bounded by the parabola $y^2 = 4ax$ and its latus rectum

3.6.2 Formulas for Rectification

Let s denote the arc length of a curve. The differential of the arc length, ds, depends on the coordinate system used.

3.6.3 Cartesian Coordinates

For a curve defined by $y = f(x)$ with continuous derivative $f'(x)$ on $[a, b]$, the length of the arc is:

$$s = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \tag{3.12}$$

Alternatively, if $x = g(y)$ on $[c, d]$:

$$s = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \tag{3.13}$$

3.6.4 Parametric Equations

If the curve is defined by $x = \phi(t)$ and $y = \psi(t)$ for $t \in [\alpha, \beta]$:

$$s = \int_\alpha^\beta \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \tag{3.14}$$

3.6.5 Polar Coordinates

For a curve defined by $r = f(\theta)$ for $\theta \in [\alpha, \beta]$:

$$s = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta \tag{3.15}$$

Example 3.6.1 (Cartesian Form)**.** Find the length of the arc of the parabola $y^2 = 4ax$ cut off by the latus rectum.

Solution. **1. Curve Tracing:**

- *Symmetry:* Powers of y are even, so it is symmetric about the x-axis.
- *Origin:* Passes through $(0, 0)$. Tangent is $x = 0$ (y-axis).
- *Region:* x must be ≥ 0.

The latus rectum is the line perpendicular to the axis passing through the focus $(a, 0)$. Thus, we integrate from $x = 0$ (vertex) to $x = a$ (focus) and multiply by 2 due to symmetry.

2. Calculation: Equation: $y^2 = 4ax \implies y = 2\sqrt{a}\sqrt{x}$. Differentiation:

$$\frac{dy}{dx} = 2\sqrt{a} \cdot \frac{1}{2\sqrt{x}} = \sqrt{\frac{a}{x}}.$$

The formula for arc length is:

$$\begin{aligned} s &= 2\int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= 2\int_0^a \sqrt{1 + \frac{a}{x}} dx = 2\int_0^a \sqrt{\frac{x+a}{x}} dx. \end{aligned}$$

To evaluate, put $x = a\tan^2\theta$. Then $dx = 2a\tan\theta\sec^2\theta d\theta$. Limits: When $x = 0, \theta = 0$. When $x = a, \tan^2\theta = 1 \implies \theta = \pi/4$.

$$\begin{aligned} s &= 2\int_0^{\pi/4} \sqrt{\frac{a(\tan^2\theta + 1)}{a\tan^2\theta}} \cdot 2a\tan\theta\sec^2\theta d\theta \\ &= 4a\int_0^{\pi/4} \frac{\sec\theta}{\tan\theta} \cdot \tan\theta\sec^2\theta d\theta \\ &= 4a\int_0^{\pi/4} \sec^3\theta d\theta. \end{aligned}$$

Using the reduction formula or standard integral $\int \sec^3 x dx = \frac{1}{2}(\sec x\tan x + \ln|\sec x + \tan x|)$:

$$\begin{aligned} s &= 4a\left[\frac{1}{2}(\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|)\right]_0^{\pi/4} \\ &= 2a\left[(\sqrt{2}\cdot 1 + \ln(\sqrt{2}+1)) - (1\cdot 0 + \ln(1))\right] \\ &= 2a\left[\sqrt{2} + \ln(\sqrt{2}+1)\right]. \end{aligned}$$

■

Example 3.6.2 (Parametric Form)**.** Find the total length of the curve of the Cycloid:

$$x = a(\theta - \sin\theta), \quad y = a(1 - \cos\theta).$$

Solution. **1. Curve Tracing:** Since $y = a(1 - \cos\theta)$, $y \geq 0$ for all θ. The curve lies above the x-axis. $y = 0$ when $\cos\theta = 1 \implies \theta = 0, 2\pi$. One complete arch of the cycloid is generated as θ varies from 0 to 2π.

2. Calculation: Find derivatives with respect to the parameter θ:

$$\frac{dx}{d\theta} = a(1 - \cos\theta), \quad \frac{dy}{d\theta} = a\sin\theta.$$

Calculate the sum of squares:

$$\begin{aligned} \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= a^2(1-\cos\theta)^2 + a^2\sin^2\theta \\ &= a^2(1 - 2\cos\theta + \cos^2\theta + \sin^2\theta) \\ &= a^2(2 - 2\cos\theta) = 2a^2(1-\cos\theta). \end{aligned}$$

Using the half-angle identity $1 - \cos\theta = 2\sin^2(\theta/2)$:

$$\text{Integrand} = \sqrt{4a^2\sin^2(\theta/2)} = 2a\sin(\theta/2).$$

Now, integrate from 0 to 2π:

$$\begin{aligned} s &= \int_0^{2\pi} 2a\sin(\theta/2)d\theta \\ &= 2a\left[-2\cos(\theta/2)\right]_0^{2\pi} \\ &= -4a\left[\cos(\pi) - \cos(0)\right] \\ &= -4a\left[-1-1\right] \\ &= 8a. \end{aligned}$$

The length of one arch of a cycloid is $8a$. ■

Example 3.6.3 (Polar Form)**.** Find the perimeter of the Cardioid $r = a(1+\cos\theta)$.

Solution. **1. Curve Tracing:**

- *Symmetry:* Since $\cos(-\theta) = \cos\theta$, the curve is symmetric about the initial line (the polar axis).
- *Limits:* The curve starts at $\theta = 0$ ($r = 2a$) and closes at the pole when $\theta = \pi$ ($r = 0$).

Due to symmetry, we calculate the length of the upper half (θ from 0 to π) and multiply by 2.

2. Calculation:

$$r = a(1+\cos\theta) \implies \frac{dr}{d\theta} = -a\sin\theta.$$

Calculate the term inside the radical:

$$\begin{aligned} r^2 + \left(\frac{dr}{d\theta}\right)^2 &= a^2(1+\cos\theta)^2 + (-a\sin\theta)^2 \\ &= a^2(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta) \\ &= a^2(2+2\cos\theta) \\ &= 2a^2(1+\cos\theta) \\ &= 2a^2(2\cos^2(\theta/2)) = 4a^2\cos^2(\theta/2). \end{aligned}$$

The arc length is:

$$\begin{aligned} s &= 2\int_0^{\pi} \sqrt{4a^2\cos^2(\theta/2)}d\theta \\ &= 2\int_0^{\pi} 2a\cos(\theta/2)d\theta \quad (\cos(\theta/2) > 0 \text{ in } [0,\pi]) \\ &= 4a\left[2\sin(\theta/2)\right]_0^{\pi} \\ &= 8a(\sin(\pi/2) - \sin(0)) \\ &= 8a(1-0) \\ &= 8a. \end{aligned}$$

■

Example 3.6.4 (The Catenary)**.** Find the length of the arc of the catenary $y = c\cosh(x/c)$ measured from the vertex $(0, c)$ to any point (x, y).

Solution. The curve $y = c\cosh(x/c)$ is symmetric about the y-axis because $\cosh(-x/c) = \cosh(x/c)$. The vertex is the lowest point where $x = 0$. We integrate from 0 to an arbitrary point x.

First, we compute the derivative:

$$\frac{dy}{dx} = c\sinh(x/c)\cdot\frac{1}{c} = \sinh(x/c).$$

We form the integrand for the Cartesian arc length formula:

$$\sqrt{1+\left(\frac{dy}{dx}\right)^2} = \sqrt{1+\sinh^2(x/c)}.$$

Utilizing the fundamental hyperbolic identity $\cosh^2 t - \sinh^2 t = 1$, this simplifies to:

$$\sqrt{\cosh^2(x/c)} = \cosh(x/c).$$

Therefore, the arc length s is:

$$\begin{aligned} s &= \int_0^x \cosh(t/c)dt \\ &= [c\sinh(t/c)]_0^x \\ &= c(\sinh(x/c) - \sinh(0)). \end{aligned}$$

Since $\sinh(0) = 0$, the arc length is $s = c\sinh(x/c)$. ■

Example 3.6.5 (Trigonometric Cartesian Form)**.** Find the length of the arc of the curve $y = \ln(\sec x)$ from $x = 0$ to $x = \frac{\pi}{3}$.

Solution. This curve passes through the origin since $\ln(\sec 0) = \ln(1) = 0$. The derivative is found using the chain rule:

$$\frac{dy}{dx} = \frac{1}{\sec x} \cdot (\sec x \tan x) = \tan x.$$

The expression under the radical becomes:

$$1+\left(\frac{dy}{dx}\right)^2 = 1+\tan^2 x = \sec^2 x.$$

Thus, the differential of the arc length is $ds = \sqrt{\sec^2 x}dx = \sec x dx$. We evaluate the integral with the given limits:

$$\begin{aligned} s &= \int_0^{\pi/3} \sec x dx \\ &= [\ln|\sec x + \tan x|]_0^{\pi/3}. \end{aligned}$$

Substituting the upper limit ($\sec(\pi/3) = 2, \tan(\pi/3) = \sqrt{3}$) and the lower limit ($\sec 0 = 1, \tan 0 = 0$):

$$\begin{aligned} s &= \ln(2+\sqrt{3}) - \ln(1+0) \\ &= \ln(2+\sqrt{3}). \end{aligned}$$

■

Example 3.6.6 (The Astroid)**.** Find the total length of the curve $x^{2/3} + y^{2/3} = a^{2/3}$.

Solution. This curve is known as the Astroid (or a hypocycloid of four cusps). It is generally more convenient to use the standard parametric equations:

$$x = a\cos^3 t, \quad y = a\sin^3 t, \quad t \in [0, 2\pi].$$

Curve Tracing: The curve is symmetric about both axes and the origin (replacing x with $-x$ or y with $-y$ leaves the Cartesian equation unchanged). Therefore, the total length is 4 times the length of the arc in the first quadrant, where t varies from 0 to $\pi/2$.

Differentiating with respect to t:

$$\frac{dx}{dt} = -3a\cos^2 t\sin t, \quad \frac{dy}{dt} = 3a\sin^2 t\cos t.$$

We calculate the sum of squares:

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2 &= 9a^2\cos^4 t\sin^2 t+9a^2\sin^4 t\cos^2 t\\ &= 9a^2\sin^2 t\cos^2 t(\cos^2 t+\sin^2 t)\\ &= 9a^2\sin^2 t\cos^2 t.\end{aligned}$$

Taking the square root (considering the first quadrant where sine and cosine are positive):

$$\frac{ds}{dt}=3a\sin t\cos t.$$

The total length L is:

$$\begin{aligned}L &= 4\int_0^{\pi/2} 3a\sin t\cos t dt\\ &= 6a\int_0^{\pi/2}\sin(2t)dt.\end{aligned}$$

Integrating:

$$\begin{aligned}L &= 6a\left[-\frac{\cos(2t)}{2}\right]_0^{\pi/2}\\ &= -3a(\cos\pi-\cos 0)\\ &= -3a(-1-1)\\ &= 6a.\end{aligned}$$

■

Example 3.6.7 (Involute of a Circle)**.** Find the length of the curve given by $x=a(\cos t+t\sin t)$ and $y=a(\sin t-t\cos t)$ from $t=0$ to $t=2\pi$.

Solution. We differentiate the parametric equations with respect to t. For x:

$$\frac{dx}{dt}=a(-\sin t+\sin t+t\cos t)=at\cos t.$$

For y:

$$\frac{dy}{dt}=a(\cos t-\cos t+t\sin t)=at\sin t.$$

Observe the simplification resulting from the cancellation of terms. We now compute the differential of arc length:

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2 &= (at\cos t)^2+(at\sin t)^2\\ &= a^2t^2(\cos^2 t+\sin^2 t)\\ &= a^2t^2.\end{aligned}$$

Taking the square root, we get $\sqrt{\dots}=at$ (assuming $t\geq 0$). The arc length is:

$$\begin{aligned}s &= \int_0^{2\pi} atdt\\ &= a\left[\frac{t^2}{2}\right]_0^{2\pi}\\ &= \frac{a}{2}(4\pi^2-0)\\ &= 2a\pi^2.\end{aligned}$$

■

Example 3.6.8 (The Logarithmic Spiral)**.** Find the length of the arc of the equiangular spiral $r = ae^{\theta\cot\alpha}$ from the pole to the point where the radius vector is r_1.

Solution. The curve spirals outward from the pole (origin). At the pole, $r \to 0$, which corresponds to $\theta \to -\infty$. Let us integrate from an arbitrary angle θ_0 to θ_1, and then take the limit. First, compute the derivative with respect to θ:

$$\frac{dr}{d\theta} = ae^{\theta\cot\alpha} \cdot \cot\alpha = r\cot\alpha.$$

Using the polar rectification formula:

$$\begin{aligned} ds &= \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\ &= \sqrt{r^2 + r^2\cot^2\alpha d\theta} \\ &= r\sqrt{1 + \cot^2\alpha d\theta} \\ &= r\csc\alpha d\theta \quad (\text{assuming } \alpha \in (0, \pi)). \end{aligned}$$

Substitute $r = ae^{\theta\cot\alpha}$ back into the expression:

$$s = \int_{\theta_0}^{\theta_1} ae^{\theta\cot\alpha}\csc\alpha d\theta.$$

Since $\csc\alpha$ is constant:

$$\begin{aligned} s &= a\csc\alpha\left[\frac{e^{\theta\cot\alpha}}{\cot\alpha}\right]_{\theta_0}^{\theta_1} \\ &= a\frac{\csc\alpha}{\cot\alpha}(e^{\theta_1\cot\alpha} - e^{\theta_0\cot\alpha}) \\ &= a\sec\alpha(e^{\theta_1\cot\alpha} - e^{\theta_0\cot\alpha}). \end{aligned}$$

To find the length measured *from the pole*, we let $\theta_0 \to -\infty$. Provided $\cot\alpha > 0$, the term $e^{\theta_0\cot\alpha} \to 0$. The term $ae^{\theta_1\cot\alpha}$ corresponds to the radius vector r_1 at the upper limit. Thus, the length is:

$$s = r_1\sec\alpha.$$

■

Exercise 3.6.9. Solve the following exercises:

1. **E**valuate the integral:
$$\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}dx$$

2. **E**valuate using the property $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$:
$$\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x} + \sqrt{\tan x}}dx$$

3. **E**valuate the integral by splitting the limits:
$$\int_{-1}^{2} |x^3 - x|dx$$

4. **E**valuate the following, where $[x]$ denotes the greatest integer less than or equal to x:
$$\int_0^2 [x^2]dx$$

5. **E**valuate as a limit of a sum:
$$\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}\right)$$
6. **E**valuate the limit:
$$\lim_{n\to\infty}\frac{1}{n}\left(\sqrt{\frac{1}{n}}+\sqrt{\frac{2}{n}}+\cdots+\sqrt{\frac{n}{n}}\right)$$
7. **E**valuate the limit L:
$$L=\lim_{n\to\infty}\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\cdots\left(1+\frac{n}{n}\right)\right]^{\frac{1}{n}}$$
8. **I**f $F(x)=\int_x^{x^2}\ln(t)dt$, find $F'(x)$ for $x>1$.
9. **E**valuate the following limit using L'Hôpital's Rule and the Fundamental Theorem:
$$\lim_{x\to 0}\frac{\int_0^{x^2}\sin(\sqrt{t})dt}{x^3}$$
10. **F**ind the points of local maxima and minima for the function defined by:
$$f(x)=\int_0^x(t-1)(t-2)dt$$
11. **U**sing differentiation with respect to the parameter a, show that for $b>0$:
$$\int_0^\infty\frac{e^{-ax}-e^{-bx}}{x}dx=\ln\left(\frac{b}{a}\right)$$
12. **E**valuate the integral by differentiating with respect to α:
$$\int_0^1\frac{x^\alpha-x^\beta}{\ln x}dx,\quad(\alpha,\beta>-1)$$
13. **E**valuate for $a\geq 0$:
$$I(a)=\int_0^\infty\frac{1-\cos(ax)}{x^2}dx$$
(Hint: Compute $\frac{d^2I}{da^2}$ or $\frac{dI}{da}$)
14. **E**valuate via differentiation under the integral sign: $\displaystyle\int_0^1\frac{\tan^{-1}(ax)}{x\sqrt{1-x^2}}dx$
15. **F**ind the length of the curve $y=\ln(1-x^2)$ from $x=0$ to $x=\frac{1}{2}$.
16. **F**ind the length of one arch of the cycloid given by:
$$x=r(t-\sin t),\quad y=r(1-\cos t)$$
(One arch corresponds to $t\in[0,2\pi]$).
17. **F**ind the total length (perimeter) of the cardioid $r=a(1-\cos\theta)$.
18. **T**race the curve $3ay^2=x(x-a)^2$ and find the length of the loop.
19. **F**ind the length of the arc of the Tractrix defined by:
$$x=a\left(\cos t+\ln\tan\frac{t}{2}\right),\quad y=a\sin t$$
measured from $t=\pi/2$ to any point t.
20. **F**ind the length of the arc of the Spiral of Archimedes $r=a\theta$ from the pole ($\theta=0$) to the point where $\theta=1$.

Chapter 4

Applications of Integration

Introduction

This unit applies the tools of calculus to geometric problems, focusing on Quadrature to determine areas bounded by Cartesian and polar curves. It extends these concepts to three-dimensional space by calculating the volumes and surface areas of solids generated by revolution. Students will learn to translate physical shapes into integral expressions for precise geometric measurement. In classical mathematics, **Quadrature** refers to the process of constructing a square having the same area as a given plane figure. In the context of modern calculus, the term is synonymous with the determination of the area of a region bounded by curves and lines using the definite integral. The fundamental principle relies on the summation of elementary rectangular strips. As the width of these strips approaches zero, the limit of their sum yields the exact area.

4.1 Area Bounded by Cartesian Curves

We consider the region bounded by a function, a coordinate axis, and two fixed perpendicular lines (ordinates or abscissae).

4.1.1 Area Bounded by the x-axis

Theorem 4.1.1. *Let $f(x)$ be a continuous, non-negative function defined on the interval $[a, b]$. The area A of the region bounded by the curve $y = f(x)$, the x-axis, and the ordinates $x = a$ and $x = b$ is given by:*

$$A = \int_a^b y dx = \int_a^b f(x) dx. \tag{4.1}$$

Proof. Let $A(x)$ denote the area function from the fixed point a to a variable point x. Consider a small increment Δx. The change in area, ΔA, corresponds to the strip between x and $x + \Delta x$. This area is approximately a rectangle of height $f(x)$ and width Δx:

$$\Delta A \approx f(x) \cdot \Delta x$$

Dividing by Δx and taking the limit as $\Delta x \to 0$:

$$\lim_{\Delta x \to 0} \frac{\Delta A}{\Delta x} = f(x) \implies \frac{dA}{dx} = f(x)$$

By the Fundamental Theorem of Calculus, the total change in area from a to b is:

$$A(b) - A(a) = \int_a^b f(x)\, dx$$

Since $A(a) = 0$ (area at the starting point is zero) and $A(b) = A$ (total area), we have:

$$A = \int_a^b f(x)\, dx$$

■

Remark 4.1.2 (Sign Convention). If the curve $y = f(x)$ lies below the x-axis (i.e., $f(x) < 0$) for $x \in [a, b]$, the definite integral $\int_a^b f(x)dx$ will yield a negative value. Since area represents a physical magnitude and must be non-negative, the area is defined as:

$$A = \left| \int_a^b f(x)dx \right| = \int_a^b -f(x)dx.$$

If the curve crosses the x-axis at a point $c \in (a, b)$, the total area is the sum of the absolute values of the integrals over the sub-intervals:

$$A = \int_a^c |f(x)|dx + \int_c^b |f(x)|dx.$$

4.1.2 Area Bounded by the y-axis

By interchanging the roles of the independent and dependent variables, we derive the formula for area measured against the vertical axis.

Theorem 4.1.3. *Let $x = g(y)$ be a continuous function defined on the interval $[c, d]$. The area A of the region bounded by the curve $x = g(y)$, the y-axis, and the abscissae $y = c$ and $y = d$ is given by:*

$$A = \int_c^d x dy = \int_c^d g(y)dy. \tag{4.2}$$

4.2 Area of Curves in Parametric Form

When the curve is defined parametrically by $x = \phi(t)$ and $y = \psi(t)$, the area can be found by substitution. Assuming x varies from a to b as the parameter t varies from t_1 to t_2:

$$A = \int_a^b y dx = \int_{t_1}^{t_2} \psi(t) \cdot \phi'(t)dt.$$

Care must be taken regarding the direction of integration. If $\phi(t)$ is decreasing on the interval, the limits of integration may need to be swapped to ensure a positive area, or the absolute value should be taken.

Example 4.2.1 (Standard Parabola)**.** Find the area of the region bounded by the parabola $y^2 = 4ax$ and its latus rectum.

Solution. The equation of the parabola is $y^2 = 4ax$. The latus rectum is the vertical line passing through the focus $(a, 0)$, given by $x = a$. Due to the even power of y, the curve is symmetric about the x-axis. The total area is therefore twice the area of the region in the first quadrant bounded by $x = 0$ and $x = a$.

In the first quadrant, $y = \sqrt{4ax} = 2\sqrt{a}x^{1/2}$.

$$\begin{aligned}
A &= 2\int_0^a y dx \\
&= 2\int_0^a 2\sqrt{a}x^{1/2}dx \\
&= 4\sqrt{a}\left[\frac{x^{3/2}}{3/2}\right]_0^a \\
&= 4\sqrt{a} \cdot \frac{2}{3}\left(a^{3/2} - 0\right) \\
&= \frac{8\sqrt{a}}{3} \cdot a\sqrt{a} \\
&= \frac{8a^2}{3}.
\end{aligned}$$

■

Example 4.2.2 (Area of an Ellipse)**.** Find the total area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution. We utilize the parametric equations of the ellipse:

$$x = a\cos t, \quad y = b\sin t.$$

The ellipse is symmetric about both axes. We calculate the area of the quadrant in the first quadrant ($x > 0, y > 0$) and multiply by 4. In the first quadrant, x ranges from 0 to a. In terms of the parameter t:

- At $x = 0$, $a\cos t = 0 \implies t = \pi/2$.
- At $x = a$, $a\cos t = a \implies t = 0$.

The differential is $dx = -a\sin t dt$.

$$\begin{aligned}
\text{Area} &= 4\left|\int_{x=0}^{x=a} y dx\right| \\
&= 4\left|\int_{\pi/2}^{0} (b\sin t)(-a\sin t)dt\right| \\
&= 4ab\left|\int_{\pi/2}^{0} -\sin^2 t dt\right| \\
&= 4ab\int_{0}^{\pi/2} \sin^2 t dt.
\end{aligned}$$

Using the reduction formula or Wallis' formula $\displaystyle\int_0^{\pi/2} \sin^2 t dt = \frac{1}{2}\cdot\frac{\pi}{2}$:

$$A = 4ab\left(\frac{\pi}{4}\right) = \pi ab.$$

■

Example 4.2.3 (Integration with respect to y)**.** Find the area of the region bounded by the curve $xy = c^2$, the y-axis, and the lines $y = c$ and $y = 2c$ ($c > 0$).

Solution. The curve $xy = c^2$ is a rectangular hyperbola. We are evaluating the area bounded by the y-axis, so we use the formula $A = \int x dy$. From the equation of the curve, we express x in terms of y:

$$x = \frac{c^2}{y}.$$

The limits of integration are the ordinates $y = c$ and $y = 2c$.

$$\begin{aligned}
A &= \int_c^{2c} \frac{c^2}{y} dy \\
&= c^2\left[\ln|y|\right]_c^{2c} \\
&= c^2(\ln(2c) - \ln(c)) \\
&= c^2\ln\left(\frac{2c}{c}\right) \\
&= c^2\ln 2.
\end{aligned}$$

■

Example 4.2.4 (Area of a Loop)**.** Find the area of the loop of the curve $ay^2 = x^2(a - x)$.

Solution. We first trace the curve to determine the limits.

1. **Symmetry:** Since the power of y is even, the curve is symmetric about the x-axis.

2. **Limits:** For y to be real, $y^2 \geq 0$, which implies $x^2(a-x) \geq 0$. Since $x^2 \geq 0$, we require $a - x \geq 0 \implies x \leq a$. The curve passes through the origin $(0,0)$ and meets the x-axis again at $x = a$. Thus, a loop is formed between $x = 0$ and $x = a$.

We integrate $y = \frac{x}{\sqrt{a}}\sqrt{a-x}$ from 0 to a and multiply by 2 (for the upper and lower halves).

$$\begin{aligned}
A &= 2\int_0^a y dx \\
&= 2\int_0^a \frac{x}{\sqrt{a}}\sqrt{a-x}dx \\
&= \frac{2}{\sqrt{a}}\int_0^a x\sqrt{a-x}dx.
\end{aligned}$$

Using the property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$:

$$\begin{aligned}
A &= \frac{2}{\sqrt{a}}\int_0^a (a-x)\sqrt{a-(a-x)}dx \\
&= \frac{2}{\sqrt{a}}\int_0^a (a-x)\sqrt{x}dx \\
&= \frac{2}{\sqrt{a}}\int_0^a (ax^{1/2} - x^{3/2})dx \\
&= \frac{2}{\sqrt{a}}\left[a\frac{x^{3/2}}{3/2} - \frac{x^{5/2}}{5/2}\right]_0^a \\
&= \frac{2}{\sqrt{a}}\left[\frac{2a}{3}(a^{3/2}) - \frac{2}{5}(a^{5/2})\right] \\
&= \frac{2}{\sqrt{a}} \cdot a^{5/2}\left(\frac{2}{3} - \frac{2}{5}\right) \\
&= 2a^2\left(\frac{10-6}{15}\right) = 2a^2\left(\frac{4}{15}\right) \\
&= \frac{8a^2}{15}.
\end{aligned}$$

■

Example 4.2.5. Find the area of the region bounded by the curve $y = x^2 + 2$, the x-axis, and the lines $x = 1$ and $x = 3$.

Solution. Here $f(x) = x^2 + 2$, which is positive on the interval $[1, 3]$.

$$\begin{aligned}
A &= \int_1^3 (x^2+2)\,dx \\
A &= \left[\frac{x^3}{3} + 2x\right]_1^3 \\
A &= \left(\frac{3^3}{3} + 2(3)\right) - \left(\frac{1^3}{3} + 2(1)\right) \\
A &= (9+6) - \left(\frac{1}{3} + 2\right) \\
A &= 15 - \frac{7}{3} = \frac{45-7}{3} = \frac{38}{3} \text{ sq. units.}
\end{aligned}$$

■

Example 4.2.6. Find the area bounded by one arch of the sine curve $y = \sin x$ and the x-axis.

Solution. One arch of the sine curve extends from $x = 0$ to $x = \pi$. In this interval, $\sin x \geq 0$.

$$A = \int_0^\pi \sin x \, dx$$
$$A = [-\cos x]_0^\pi$$
$$A = -\cos(\pi) - (-\cos(0))$$
$$A = -(-1) - (-1)$$
$$A = 1 + 1 = 2 \text{ sq. units.}$$

■

Example 4.2.7. Find the area of the region bounded by the parabola $x = y^2 - 4y$, the y-axis, and the lines $y = 4$ and $y = 5$.

Solution. The region lies between the curve and the y-axis. Note that for $y \in [4, 5]$, $x = y(y-4)$ is positive.

$$A = \int_4^5 x \, dy$$
$$A = \int_4^5 (y^2 - 4y) \, dy$$
$$A = \left[\frac{y^3}{3} - 2y^2\right]_4^5$$
$$A = \left(\frac{125}{3} - 50\right) - \left(\frac{64}{3} - 32\right)$$
$$A = \left(\frac{125 - 150}{3}\right) - \left(\frac{64 - 96}{3}\right)$$
$$A = -\frac{25}{3} - \left(-\frac{32}{3}\right) = \frac{7}{3} \text{ sq. units.}$$

■

Example 4.2.8. Calculate the area bounded by the curve $xy = c^2$, the y-axis, and the lines $y = a$ and $y = b$ (where $0 < a < b$).

Solution. We express x as a function of y: $x = \frac{c^2}{y}$.

$$A = \int_a^b x \, dy$$
$$A = \int_a^b \frac{c^2}{y} \, dy$$
$$A = c^2 \left[\ln |y|\right]_a^b$$
$$A = c^2(\ln b - \ln a)$$
$$A = c^2 \ln \left(\frac{b}{a}\right) \text{ sq. units.}$$

■

Example 4.2.9. Find the area of the ellipse represented by $x = a\cos t$ and $y = b \sin t$.

Solution. Due to symmetry, the total area is 4 times the area in the first quadrant. In the first quadrant, x varies from 0 to a.

- When $x = 0$, $a\cos t = 0 \implies t = \pi/2$.
- When $x = a$, $a \cos t = a \implies t = 0$.

Also, $dx = \frac{d}{dt}(a\cos t)\,dt = -a\sin t\,dt$.

$$A = 4\int_{x=0}^{x=a} y\,dx$$
$$A = 4\int_{\pi/2}^{0} (b\sin t)(-a\sin t)\,dt$$
$$A = -4ab\int_{\pi/2}^{0} \sin^2 t\,dt$$
$$A = 4ab\int_{0}^{\pi/2} \sin^2 t\,dt \quad \text{(Swapping limits removes negative sign)}$$

Using Wallis's Formula for $\int_0^{\pi/2} \sin^n x\,dx$ where $n = 2$:

$$\int_0^{\pi/2} \sin^2 t\,dt = \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4}$$

Therefore:

$$A = 4ab\left(\frac{\pi}{4}\right) = \pi ab \text{ sq. units.}$$

■

Example 4.2.10. Find the area of the region bounded by one arch of the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$ and the x-axis.

Solution. One arch of the cycloid is generated as t varies from 0 to 2π. We have $y = a(1 - \cos t)$ and $dx = a(1 - \cos t)\,dt$.

$$A = \int_0^{2\pi} y\frac{dx}{dt}\,dt$$
$$A = \int_0^{2\pi} a(1-\cos t)\cdot a(1-\cos t)\,dt$$
$$A = a^2\int_0^{2\pi} (1-\cos t)^2\,dt$$
$$A = a^2\int_0^{2\pi} (1 - 2\cos t + \cos^2 t)\,dt$$

We split the integral: 1. $\int_0^{2\pi} 1\,dt = 2\pi$ 2. $\int_0^{2\pi} 2\cos t\,dt = 2[\sin t]_0^{2\pi} = 0$ 3. $\int_0^{2\pi} \cos^2 t\,dt = \int_0^{2\pi} \frac{1+\cos 2t}{2}\,dt = \left[\frac{t}{2} + \frac{\sin 2t}{4}\right]_0^{2\pi} = \pi$ Substituting these back:

$$A = a^2(2\pi - 0 + \pi) = 3\pi a^2 \text{ sq. units.}$$

■

4.3 Introduction to Sectorial Area

When a curve is defined in **Polar Coordinates** by an equation $r = f(\theta)$, the region bounded by the curve and two radius vectors is called a *sector*.

Unlike Cartesian coordinates (where area is strips of height y and width dx), in polar coordinates, we sum up infinite infinitesimal **sectors** (triangular wedges).

The Formula

Consider a curve $r = f(\theta)$. The area A of the sectorial region bounded by the curve and the radius vectors $\theta = \alpha$ and $\theta = \beta$ is given by:

$$A = \frac{1}{2}\int_\alpha^\beta r^2\,d\theta = \frac{1}{2}\int_\alpha^\beta [f(\theta)]^2\,d\theta \tag{4.3}$$

Derivation (Intuitive)

- Consider a small element of the area corresponding to a small angular change $\Delta\theta$.
- This element approximates a circular sector with radius r and angle $\Delta\theta$.
- The area of a circular sector is $\frac{1}{2}r^2\theta$. Thus, $\Delta A \approx \frac{1}{2}r^2\Delta\theta$.
- Summing these sectors and taking the limit as $\Delta\theta \to 0$ leads to the definite integral.

Rules for Closed Curves

For a closed curve (like a circle, cardioid, or loop):

1. **Determine Limits:** Find the values of θ where the curve starts and finishes tracing itself exactly once. For many standard polar curves, this is $[0, 2\pi]$ or $[0, \pi]$.
2. **Symmetry:** If the curve is symmetric about the initial line (polar axis), it is often easier to integrate from 0 to a specific angle and multiply by 2.
3. **Tangents at Pole:** To find the limits for a loop that starts and ends at the origin (pole), set $r = 0$ and solve for θ.

Example 4.3.1 (The Cardioid)**.** Find the total area enclosed by the Cardioid $r = a(1 + \cos\theta)$.

Solution. The curve is symmetric about the initial line (since $\cos(-\theta) = \cos\theta$). The upper half is generated as θ varies from 0 to π.

Therefore, Area $= 2 \times$ (Area of upper half).

$$\begin{aligned}
A &= 2 \times \frac{1}{2}\int_0^\pi r^2\,d\theta \\
A &= \int_0^\pi a^2(1 + \cos\theta)^2\,d\theta \\
A &= a^2\int_0^\pi (1 + 2\cos\theta + \cos^2\theta)\,d\theta \\
A &= a^2\left[\int_0^\pi d\theta + 2\int_0^\pi \cos\theta\,d\theta + \int_0^\pi \frac{1+\cos 2\theta}{2}\,d\theta\right] \\
&= a^2\left[\pi + 0 + \frac{\pi}{2} + 0\right] = \frac{3\pi a^2}{2} \text{ sq. units.}
\end{aligned}$$

■

Example 4.3.2 (The Lemniscate of Bernoulli)**.** Find the area of one loop of the Lemniscate $r^2 = a^2\cos 2\theta$.

Solution. The curve consists of two loops (shaped like a figure-8). We need the area of **one loop**.

Finding Limits: The loop exists where $r^2 \geq 0$, i.e., $\cos 2\theta \geq 0$. The loop passes through the pole ($r = 0$) when $\cos 2\theta = 0$.

$$2\theta = \pm\frac{\pi}{2} \implies \theta = \pm\frac{\pi}{4}$$

So, one loop is traced from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$. By symmetry, we can integrate from 0 to $\frac{\pi}{4}$ and multiply by 2.

$$A_{\text{loop}} = 2 \times \frac{1}{2}\int_0^{\pi/4} r^2\, d\theta$$
$$A_{\text{loop}} = \int_0^{\pi/4} a^2 \cos 2\theta\, d\theta$$
$$A_{\text{loop}} = a^2 \left[\frac{\sin 2\theta}{2}\right]_0^{\pi/4}$$
$$A_{\text{loop}} = \frac{a^2}{2}\left(\sin\left(\frac{\pi}{2}\right) - \sin(0)\right)$$
$$A_{\text{loop}} = \frac{a^2}{2}(1-0) = \frac{a^2}{2} \text{ sq. units.}$$

(*Note: The total area of both loops would be* a^2). ■

Example 4.3.3 (Circle Passing Through Pole)**.** Find the area of the circle given by the polar equation $r = 2a\cos\theta$.

Solution. This equation represents a circle of radius a centered at $(a, 0)$ on the polar axis.

Finding Limits: The upper half of the circle is traced from $\theta = 0$ to $\theta = \frac{\pi}{2}$. The lower half is from $\theta = -\frac{\pi}{2}$ to 0. We compute the area from 0 to $\frac{\pi}{2}$ and multiply by 2.

$$A = 2 \times \frac{1}{2}\int_0^{\pi/2} r^2\, d\theta$$
$$A = \int_0^{\pi/2} (2a\cos\theta)^2\, d\theta$$
$$A = 4a^2 \int_0^{\pi/2} \cos^2\theta\, d\theta$$

Using Wallis's Formula ($\int_0^{\pi/2} \cos^2\theta\, d\theta = \frac{1}{2}\cdot\frac{\pi}{2}$):

$$A = 4a^2\left(\frac{\pi}{4}\right) = \pi a^2 \text{ sq. units.}$$

This confirms the standard geometric area $\pi(\text{radius})^2 = \pi(a)^2$. ■

Example 4.3.4 (The Three-Leaved Rose)**.** Find the area enclosed by the three loops of the Rose Curve $r = a\sin 3\theta$.

Solution. This curve has 3 symmetrical loops (petals).

Area of One Loop: A loop starts and ends at the pole ($r = 0$).

$$\sin 3\theta = 0 \implies 3\theta = 0 \text{ and } 3\theta = \pi \implies \theta = 0 \text{ and } \theta = \frac{\pi}{3}$$

So, one loop lies between 0 and $\frac{\pi}{3}$.

$$\begin{aligned}
A_{\text{one loop}} &= \frac{1}{2}\int_0^{\pi/3} r^2\, d\theta \\
&= \frac{1}{2}\int_0^{\pi/3} a^2 \sin^2 3\theta\, d\theta \\
&= \frac{a^2}{2}\int_0^{\pi/3} \frac{1-\cos 6\theta}{2}\, d\theta \\
&= \frac{a^2}{4}\left[\theta - \frac{\sin 6\theta}{6}\right]_0^{\pi/3} \\
&= \frac{a^2}{4}\left[\left(\frac{\pi}{3} - \frac{\sin 2\pi}{6}\right) - (0-0)\right] \\
&= \frac{a^2}{4}\left(\frac{\pi}{3}\right) = \frac{\pi a^2}{12}
\end{aligned}$$

Total Area: Since there are 3 loops:

$$A_{\text{total}} = 3 \times A_{\text{one loop}} = 3 \times \frac{\pi a^2}{12} = \frac{\pi a^2}{4} \text{ sq. units.}$$

■

4.4 Rectification

Rectification is the process of finding the length of an arc of a curve. If a curve is defined by $y = f(x)$, the length S along the curve between two points is calculated by integrating the differential element of arc length, denoted by ds.

From the Pythagorean theorem applied to infinitesimals:

$$(ds)^2 = (dx)^2 + (dy)^2$$

4.5 Formulae for Arc Length

4.5.1 1. Cartesian Form

- **Function of x:** If the curve is given by $y = f(x)$ and the derivative $f'(x)$ is continuous on $[a, b]$, the length L is:

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

- **Function of y:** If the curve is given by $x = g(y)$ from $y = c$ to $y = d$:

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

4.5.2 2. Parametric Form

If the curve is defined by $x = \phi(t)$ and $y = \psi(t)$, where t varies from t_1 to t_2:

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

4.5.3 3. Polar Form

If the curve is defined by $r = f(\theta)$, where θ varies from α to β. Using the transformation $x = r\cos\theta, y = r\sin\theta$, the arc length formula becomes:

$$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

Example 4.5.1 (Cartesian Form)**.** Find the length of the curve $y = \ln(\sec x)$ from $x = 0$ to $x = \frac{\pi}{3}$.

Solution. Given $y = \ln(\sec x)$. Differentiate with respect to x:

$$\frac{dy}{dx} = \frac{1}{\sec x} \cdot (\sec x \tan x) = \tan x$$

Using the arc length formula:

$$L = \int_0^{\pi/3} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Substitute $\frac{dy}{dx} = \tan x$:

$$L = \int_0^{\pi/3} \sqrt{1 + \tan^2 x}\, dx$$

Since $1 + \tan^2 x = \sec^2 x$:

$$L = \int_0^{\pi/3} \sec x\, dx$$

The integral of $\sec x$ is $\ln|\sec x + \tan x|$.

$$\begin{aligned}
L &= [\ln|\sec x + \tan x|]_0^{\pi/3} \\
L &= \ln|\sec(\pi/3) + \tan(\pi/3)| - \ln|\sec(0) + \tan(0)| \\
L &= \ln|2 + \sqrt{3}| - \ln|1 + 0| \\
L &= \ln(2 + \sqrt{3}) \text{ units.}
\end{aligned}$$

■

Example 4.5.2 (Parametric Form - The Cycloid)**.** Find the length of one arch of the cycloid given by $x = a(t - \sin t)$, $y = a(1 - \cos t)$.

Solution. One complete arch corresponds to t varying from 0 to 2π.

Calculate derivatives:

$$\frac{dx}{dt} = a(1 - \cos t)$$

$$\frac{dy}{dt} = a(\sin t)$$

Compute $(ds/dt)^2$:

$$\begin{aligned}
\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= a^2(1 - \cos t)^2 + a^2 \sin^2 t \\
&= a^2(1 - 2\cos t + \cos^2 t + \sin^2 t) \\
&= a^2(2 - 2\cos t) \quad (\text{since } \cos^2 t + \sin^2 t = 1) \\
&= 2a^2(1 - \cos t)
\end{aligned}$$

Using the half-angle identity $1 - \cos t = 2\sin^2(t/2)$:

$$\left(\frac{ds}{dt}\right)^2 = 4a^2 \sin^2(t/2) \implies \sqrt{\ldots} = 2a\sin(t/2)$$

(Note: $\sin(t/2) \geq 0$ for $t \in [0, 2\pi]$).

Now integrate:

$$\begin{aligned}
L &= \int_0^{2\pi} 2a\sin(t/2)\, dt \\
L &= 2a\left[\frac{-\cos(t/2)}{1/2}\right]_0^{2\pi} \\
L &= -4a[\cos(\pi) - \cos(0)] \\
L &= -4a[-1 - 1] = 8a \text{ units.}
\end{aligned}$$

■

Example 4.5.3 (Parametric Form - The Astroid)**.** Find the total length of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$.

Solution. We use the parametric equations $x = a\cos^3 t$, $y = a\sin^3 t$, where $0 \le t \le 2\pi$. Due to symmetry, the total length is 4 times the length in the first quadrant ($0 \le t \le \pi/2$).

Calculate derivatives:

$$\frac{dx}{dt} = -3a\cos^2 t \sin t$$

$$\frac{dy}{dt} = 3a\sin^2 t \cos t$$

Compute sum of squares:

$$\begin{aligned}\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= 9a^2\cos^4 t\sin^2 t + 9a^2\sin^4 t\cos^2 t \\ &= 9a^2\sin^2 t\cos^2 t(\cos^2 t + \sin^2 t) \\ &= 9a^2\sin^2 t\cos^2 t\end{aligned}$$

Taking the square root (positive in first quadrant):

$$\frac{ds}{dt} = 3a\sin t\cos t$$

Total Length:

$$\begin{aligned}L &= 4\int_0^{\pi/2} 3a\sin t\cos t\,dt \\ L &= 12a\int_0^{\pi/2} \sin t\cos t\,dt \\ L &= 6a\int_0^{\pi/2} \sin 2t\,dt \\ L &= 6a\left[\frac{-\cos 2t}{2}\right]_0^{\pi/2} \\ L &= -3a(\cos\pi - \cos 0) = -3a(-1-1) = 6a \text{ units.}\end{aligned}$$

■

Example 4.5.4 (Polar Form - The Cardioid)**.** Find the perimeter of the cardioid $r = a(1+\cos\theta)$.

Solution. The curve is symmetric about the initial line. We integrate from 0 to π (upper half) and multiply by 2.

We know that: $L = \int \sqrt{r^2 + (\frac{dr}{d\theta})^2}\,d\theta$.

Given $r = a(1+\cos\theta)$, then $\frac{dr}{d\theta} = -a\sin\theta$.

Simplify the integrand:

$$\begin{aligned}r^2 + \left(\frac{dr}{d\theta}\right)^2 &= a^2(1+\cos\theta)^2 + a^2(-\sin\theta)^2 \\ &= a^2(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta) \\ &= a^2(2 + 2\cos\theta) \\ &= 2a^2(1+\cos\theta) \\ &= 2a^2(2\cos^2(\theta/2)) = 4a^2\cos^2(\theta/2)\end{aligned}$$

Taking the square root: $\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} = 2a\cos(\theta/2)$ (Positive for $\theta \in [0,\pi]$).

Therefore,

$$L = 2\int_0^{\pi} 2a\cos(\theta/2)\,d\theta$$
$$L = 4a\left[\frac{\sin(\theta/2)}{1/2}\right]_0^{\pi}$$
$$L = 8a[\sin(\pi/2) - \sin(0)]$$
$$L = 8a(1-0) = 8a \text{ units.}$$

■

4.6 Volume of Solid of Revolution

When a plane curve is rotated about a fixed line (axis of revolution), it generates a three-dimensional region called a solid of revolution.

1. Rotation about the x-axis (Disk Method) If the region bounded by the curve $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ is revolved about the x-axis, the volume V is given by summing infinite circular disks of radius y and thickness dx:

$$V = \pi\int_a^b y^2\,dx = \pi\int_a^b [f(x)]^2\,dx \tag{4.4}$$

2. Rotation about the y-axis If the region bounded by the curve $x = g(y)$, the y-axis, and the lines $y = c$ and $y = d$ is revolved about the y-axis:

$$V = \pi\int_c^d x^2\,dy = \pi\int_c^d [g(y)]^2\,dy \tag{4.5}$$

3. Parametric Form If the curve is defined by $x = \phi(t)$ and $y = \psi(t)$, and rotated about the x-axis:

$$V = \pi\int_{t_1}^{t_2} y^2\frac{dx}{dt}\,dt \tag{4.6}$$

4.7 Area of Surface of Revolution

When an arc of a curve is rotated about an axis, it generates a curved surface. The area is calculated by integrating the product of the circumference ($2\pi \times$ radius) and the arc length element (ds).

4.7.1 1. Rotation about the x-axis

- **Cartesian ($y = f(x)$):**

$$S = 2\pi\int_a^b y\,ds = 2\pi\int_a^b y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

- **Parametric ($x(t), y(t)$):**

$$S = 2\pi\int_{t_1}^{t_2} y\sqrt{\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2}\,dt$$

2. Rotation about the y-axis The radius of rotation becomes x.

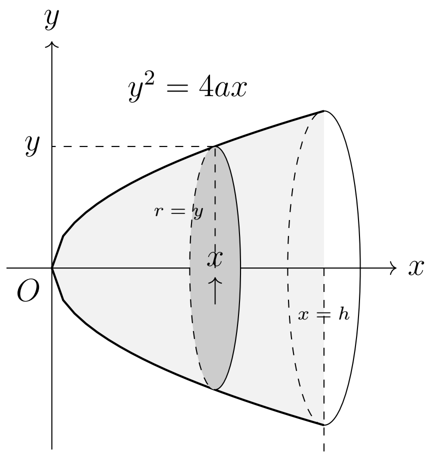
- **Cartesian ($x = g(y)$):**

$$S = 2\pi\int_c^d x\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy$$

Example 4.7.1 (Volume - The Paraboloid)**.** Find the volume of the solid generated by revolving the parabola $y^2 = 4ax$ about the x-axis from $x = 0$ to $x = h$.

The curve $y^2 = 4ax$ is revolved about the x-axis to form a paraboloid. We use the **Disk Method**.
Consider a representative vertical rectangular strip of width dx at a distance x from the origin. When rotated about the x-axis, this strip generates a circular disk.

- **Radius of disk:** $R = y = \sqrt{4ax}$
- **Thickness of disk:** dx
- **Volume of element:** $dV = \pi(\text{radius})^2\,\text{thickness} = \pi y^2\,dx$



Solution. To find the total volume, we integrate from $x = 0$ to $x = h$:

$$V = \int_0^h \pi y^2\,dx$$
$$V = \pi \int_0^h (4ax)\,dx$$
$$V = 4\pi a \left[\frac{x^2}{2}\right]_0^h$$
$$V = 2\pi a[h^2 - 0]$$
$$V = 2\pi a h^2 \text{ cubic units.}$$

■

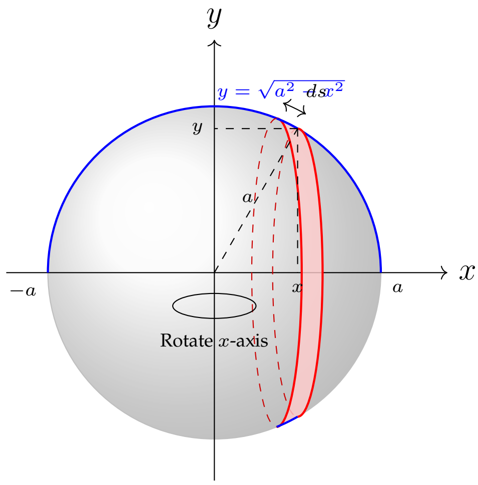
Example 4.7.2 (Surface Area - The Sphere)**.** Find the surface area of the sphere generated by revolving the circle $x^2 + y^2 = a^2$ about the x-axis.

The sphere is generated by revolving the upper semicircle $y = \sqrt{a^2 - x^2}$ from $x = -a$ to $x = a$ about the x-axis.
Differentiating with respect to x:

$$2x + 2y\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}$$

The arc length element ds is:

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$
$$ds = \sqrt{1 + \frac{x^2}{y^2}}\, dx = \sqrt{\frac{y^2 + x^2}{y^2}}\, dx$$



Solution. Since $x^2 + y^2 = a^2$, this simplifies to:

$$ds = \frac{a}{y}\, dx$$

The surface area S is the integral of the circumference ($2\pi y$) times the arc length (ds):

$$S = \int_{x=-a}^{x=a} 2\pi y\, ds$$
$$S = \int_{-a}^{a} 2\pi y \left(\frac{a}{y}\right)\, dx$$
$$S = 2\pi a \int_{-a}^{a} dx$$
$$S = 2\pi a[x]_{-a}^{a}$$
$$S = 2\pi a(a - (-a)) = 4\pi a^2 \text{ sq. units.}$$

■

Example 4.7.3 (Volume - The Cycloid)**.** Find the volume generated by revolving one arch of the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ about the x-axis.

Solution. One complete arch of the cycloid corresponds to the parameter θ varying from 0 to 2π.
We use the volume of revolution formula for parametric curves rotated about the x-axis:

$$V = \pi \int_{\theta_1}^{\theta_2} y^2 \frac{dx}{d\theta}\, d\theta$$

First, calculate the derivative $\frac{dx}{d\theta}$:

$$x = a(\theta - \sin\theta) \implies \frac{dx}{d\theta} = a(1 - \cos\theta)$$

Substitute y and $dx/d\theta$ into the integral:

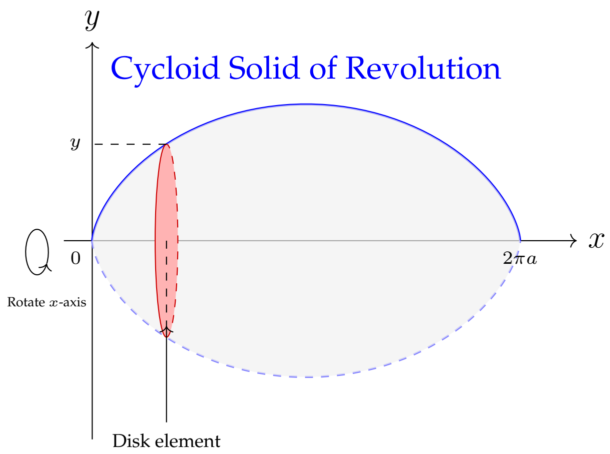
$$V = \pi \int_0^{2\pi} [a(1 - \cos\theta)]^2 \cdot [a(1 - \cos\theta)]\, d\theta$$
$$V = \pi a^3 \int_0^{2\pi} (1 - \cos\theta)^3\, d\theta$$

Expand the cubic term $(1 - \cos\theta)^3 = 1 - 3\cos\theta + 3\cos^2\theta - \cos^3\theta$.
We evaluate the integrals term by term over the interval $[0, 2\pi]$:

- $\int_0^{2\pi} 1\,d\theta = 2\pi$.
- $\int_0^{2\pi} \cos\theta\,d\theta = 0$ and $\int_0^{2\pi} \cos^3\theta\,d\theta = 0$ (Odd powers of cosine integrate to zero over a full period).
- $\int_0^{2\pi} 3\cos^2\theta\,d\theta = 3\int_0^{2\pi} \frac{1+\cos 2\theta}{2}\,d\theta = \frac{3}{2}[\theta + \frac{\sin 2\theta}{2}]_0^{2\pi} = \frac{3}{2}(2\pi) = 3\pi$.

Combining these results:

$$V = \pi a^3[2\pi - 0 + 3\pi - 0] = 5\pi^2 a^3 \text{ cubic units.}$$



■

Example 4.7.4 (Surface Area - The Catenoid)**.** Find the surface area generated by revolving the catenary $y = c\cosh(x/c)$ about the x-axis from $x = 0$ to $x = a$.

Solution. Differentiate y:

$$\frac{dy}{dx} = c\sinh(x/c)\cdot\frac{1}{c} = \sinh(x/c)$$

Calculate ds:

$$ds = \sqrt{1+\sinh^2(x/c)}\,dx = \sqrt{\cosh^2(x/c)}\,dx = \cosh(x/c)\,dx$$

Formula for Surface Area:

$$S = 2\pi\int_0^a y\,ds$$
$$S = 2\pi\int_0^a [c\cosh(x/c)]\cdot[\cosh(x/c)]\,dx$$
$$S = 2\pi c\int_0^a \cosh^2(x/c)\,dx$$

Using identity $\cosh^2 u = \frac{1+\cosh 2u}{2}$:

$$S = 2\pi c\int_0^a \frac{1+\cosh(2x/c)}{2}\,dx$$
$$S = \pi c\left[x + \frac{c}{2}\sinh(2x/c)\right]_0^a$$
$$S = \pi c\left[a + \frac{c}{2}\sinh(2a/c)\right] \text{ sq. units.}$$

■

Example 4.7.5 (Volume - The Torus)**.** Find the volume of the torus generated by revolving the circle $x^2 + (y-b)^2 = a^2$ (where $b > a$) about the x-axis.

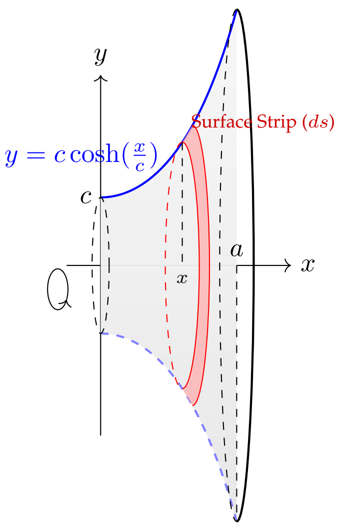


Figure 4.1: The Catenoid - Example 4.7.4

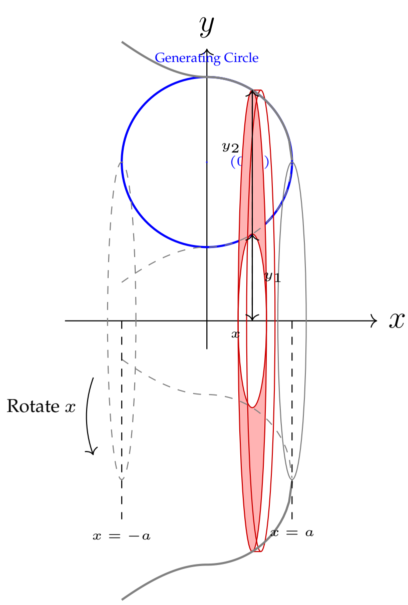
Solution. The circle has center $(0, b)$ and radius a. Since $b > a$, rotating about the x-axis generates a ring shape (Torus) with a hole in the middle.
Solving for y:

$$(y - b)^2 = a^2 - x^2 \implies y = b \pm \sqrt{a^2 - x^2}$$

This gives two branches:

- Outer radius: $y_2 = b + \sqrt{a^2 - x^2}$
- Inner radius: $y_1 = b - \sqrt{a^2 - x^2}$

The limits for x are $-a$ to a. Using the **Washer Method**, the volume is the difference between the outer and inner volumes:



$$V = \pi \int_{-a}^{a} (y_2^2 - y_1^2)\, dx$$

$$V = \pi \int_{-a}^{a} \left[\left(b + \sqrt{a^2 - x^2}\right)^2 - \left(b - \sqrt{a^2 - x^2}\right)^2 \right] dx$$

Using $(A + B)^2 - (A - B)^2 = 4AB$:

$$V = \pi \int_{-a}^{a} 4b\sqrt{a^2 - x^2}\, dx$$

$$V = 4\pi b \int_{-a}^{a} \sqrt{a^2 - x^2}\, dx$$

The integral $\int_{-a}^{a} \sqrt{a^2 - x^2}\, dx$ is the area of a semi-circle of radius a, which is $\frac{\pi a^2}{2}$.

$$V = 4\pi b \left(\frac{\pi a^2}{2} \right) = 2\pi^2 a^2 b \text{ cubic units.}$$

■

Chapter 5

Differential Equations

This unit transitions from calculus to differential equations, providing methods to solve linear and exact equations using integrating factors. It covers the systematic approach to solving differential equations with constant coefficients, which form the backbone of mathematical modeling. These techniques are vital for describing dynamic systems and rates of change in science and engineering.

5.1 Linear Differential Equations (First Order)

Definition 5.1.1. A differential equation is said to be **linear** if the dependent variable (say y) and its derivative ($\frac{dy}{dx}$) appear only in the first degree and are not multiplied together.

5.1.1 Standard Form

The standard form of a first-order linear differential equation is:

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{5.1}$$

where P and Q are constants or functions of x alone.

Method of Solution

1. **Identify P and Q:** Compare the given equation with the standard form to find $P(x)$ and $Q(x)$.
2. **Find Integrating Factor (I.F.):** Evaluate the following exponential integral:
$$\text{I.F.} = e^{\int P\,dx}$$
3. **Write the General Solution:** The solution is given by the formula:
$$y \cdot (\text{I.F.}) = \int \left(Q \cdot (\text{I.F.})\right)\, dx + C$$

Sol. Consider the linear differential equation:

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{5.2}$$

Multiply both sides by the **Integrating Factor**, defined as I.F. $= e^{\int P\,dx}$:

$$e^{\int P\,dx}\frac{dy}{dx} + P(x)e^{\int P\,dx}y = Q(x)e^{\int P\,dx}$$

Notice that by the chain rule, $\frac{d}{dx}\left(e^{\int P\,dx}\right) = P(x)e^{\int P\,dx}$. Thus, the left-hand side is the derivative of a product:

$$\frac{d}{dx}\left[y \cdot e^{\int P\,dx}\right] = Q(x) \cdot e^{\int P\,dx}$$

Integrating both sides with respect to x:

$$\int \frac{d}{dx}\left[y \cdot e^{\int P\,dx}\right]\,dx = \int \left(Q(x) \cdot e^{\int P\,dx}\right)\,dx$$

$$y \cdot e^{\int P\,dx} = \int \left(Q(x) \cdot e^{\int P\,dx}\right)\,dx + C$$

Which is the required solution:

$$y \cdot (\text{I.F.}) = \int (Q \cdot \text{I.F.})\,dx + C$$

Alternate Form (Linear in x)

If the equation is of the form:

$$\frac{dx}{dy} + P(y)x = Q(y)$$

where P and Q are functions of y, the method is:

- I.F. $= e^{\int P\,dy}$
- Solution: $x \cdot (\text{I.F.}) = \int (Q \cdot (\text{I.F.}))\,dy + C$

Example 5.1.2. Solve the differential equation: $x\dfrac{dy}{dx} + 2y = x^3$.

Solution. Dividing by x gives the standard form $\frac{dy}{dx} + \frac{2}{x}y = x^2$. Here, the integrating factor is I.F. $= e^{\int (2/x)dx} = e^{2\ln x} = x^2$. The general solution $y(\text{I.F.}) = \int Q(\text{I.F.})\,dx$ becomes:

$$y \cdot x^2 = \int x^2 \cdot x^2\,dx = \int x^4\,dx = \frac{x^5}{5} + C$$

Thus, $y = \frac{x^3}{5} + Cx^{-2}$. ■

Example 5.1.3. Solve the differential equation: $\dfrac{dy}{dx} + y\cot x = \cos x$.

Solution. With $P(x) = \cot x$, the integrating factor is I.F. $= e^{\int \cot x\,dx} = \sin x$. Multiplying by the I.F. and integrating gives:

$$y\sin x = \int \cos x \sin x\,dx = \int \frac{1}{2}\sin 2x\,dx = -\frac{1}{4}\cos 2x + C$$

Alternatively, using substitution $u = \sin x$, we get $y\sin x = \frac{1}{2}\sin^2 x + C$. ■

Example 5.1.4. Solve the differential equation: $(1+x^2)\dfrac{dy}{dx} + 2xy = \dfrac{1}{1+x^2}$.

Solution. Dividing by $(1+x^2)$, we get $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{1}{(1+x^2)^2}$. The integrating factor is I.F. $= e^{\int \frac{2x}{1+x^2}dx} = e^{\ln(1+x^2)} = 1+x^2$. The solution is:

$$y(1+x^2) = \int \frac{1}{(1+x^2)^2} \cdot (1+x^2)\,dx = \int \frac{1}{1+x^2}\,dx$$

Integration yields $y(1+x^2) = \tan^{-1} x + C$, or $y = \frac{\tan^{-1} x + C}{1+x^2}$. ■

5.1.2 Exact Differential Equations

Definition 5.1.5. A differential equation of the form $M(x, y)\,dx + N(x, y)\,dy = 0$ is said to be **exact** if it can be derived by directly differentiating its primitive (solution) $u(x, y) = C$ without any further multiplication or elimination.

Theorem 5.1.6. *The differential equation $M(x, y)\,dx + N(x, y)\,dy = 0$ is exact if and only if:*

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \tag{5.3}$$

assuming M and N have continuous partial derivatives on a simply connected region.

Proof. **Necessity ($\Longrightarrow$)**
Suppose the equation is exact. By definition, there exists a potential function $u(x, y)$ such that $du = M\,dx + N\,dy$. Since the total differential is $du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy$, we identify:

$$M = \frac{\partial u}{\partial x} \quad \text{and} \quad N = \frac{\partial u}{\partial y}$$

Differentiating M with respect to y and N with respect to x:

$$\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y \partial x} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x \partial y}$$

By the equality of mixed partial derivatives (Clairaut's Theorem), it follows that:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Sufficiency ($\Longleftarrow$)
Suppose $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. We construct a function $u(x, y)$ such that $\frac{\partial u}{\partial x} = M$. Let:

$$u(x, y) = \int M(x, y)\,dx + \phi(y)$$

where $\phi(y)$ is an arbitrary function of y. Differentiating with respect to y:

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left(\int M\,dx\right) + \phi'(y)$$

For exactness, we require $\frac{\partial u}{\partial y} = N$. Thus, we set:

$$\phi'(y) = N - \frac{\partial}{\partial y}\int M\,dx$$

For $\phi(y)$ to exist, the right-hand side must be independent of x. We verify this by differentiating it with respect to x:

$$\frac{\partial}{\partial x}\left(N - \frac{\partial}{\partial y}\int M\,dx\right) = \frac{\partial N}{\partial x} - \frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}\int M\,dx\right) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

By hypothesis, this difference is zero. Thus, $\phi'(y)$ depends only on y, ensuring $\phi(y)$ exists. Consequently, a function u exists such that $du = Mdx + Ndy$. ∎

Working Rule

If the condition for exactness is satisfied, the general solution is:

$$\int_{y\text{ constant}} M\,dx + \int (\text{terms of } N \text{ free from } x)\,dy = C$$

Steps:

1. Integrate M with respect to x treating y as a constant.
2. Integrate only those terms in N that do not contain x with respect to y.
3. Add the results and equate to a constant C.

Example 5.1.7. Solve: $(x^2 - 4xy - 2y^2)\,dx + (y^2 - 4xy - 2x^2)\,dy = 0$.

Solution. Here, $M = x^2 - 4xy - 2y^2$ and $N = y^2 - 4xy - 2x^2$. Check exactness:

$$\frac{\partial M}{\partial y} = -4x - 4y, \quad \frac{\partial N}{\partial x} = -4y - 4x$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the equation is exact.

Integrate M with respect to x (treating y as constant):

$$\int (x^2 - 4xy - 2y^2)\,dx = \frac{x^3}{3} - 2x^2y - 2xy^2$$

Integrate terms in N free from x (only y^2):

$$\int y^2\,dy = \frac{y^3}{3}$$

Thus, the general solution is $\dfrac{x^3}{3} - 2x^2y - 2xy^2 + \dfrac{y^3}{3} = C$. ■

Example 5.1.8. Solve: $\left(1 + e^{x/y}\right) dx + e^{x/y}\left(1 - \frac{x}{y}\right) dy = 0$.

Solution. Here, $M = 1 + e^{x/y}$ and $N = e^{x/y} - \frac{x}{y}e^{x/y}$. Check exactness:

$$\frac{\partial M}{\partial y} = e^{x/y}\left(-\frac{x}{y^2}\right)$$

$$\frac{\partial N}{\partial x} = e^{x/y}\left(\frac{1}{y}\right) - \left[\frac{1}{y}e^{x/y} + \frac{x}{y}e^{x/y}\left(\frac{1}{y}\right)\right] = -\frac{x}{y^2}e^{x/y}$$

Since derivatives match, it is exact.

Integrate M w.r.t x:

$$\int (1 + e^{x/y})\,dx = x + \frac{e^{x/y}}{1/y} = x + ye^{x/y}$$

Integrate terms in N free from x: Looking at $N = e^{x/y}(1 - x/y)$, every term contains x. Thus, this integral is 0.

So the general solution is $x + ye^{x/y} = C$. ■

5.1.3 Equations Reducible to Exact Form (Integrating Factors)

If $M\,dx + N\,dy = 0$ is **not exact** (i.e., $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$), it can often be made exact by multiplying by a function $\mu(x, y)$ called the **Integrating Factor (I.F.)**.

Rules for Finding Integrating Factors

Rule I: Homogeneous Equations

If $M\,dx + N\,dy = 0$ is a homogeneous equation in x and y, and $Mx + Ny \neq 0$, then:

$$\text{I.F.} = \frac{1}{Mx + Ny}$$

Rule II: Form $f_1(xy)ydx + f_2(xy)xdy = 0$

If the equation is of the form $y \cdot f_1(xy)\,dx + x \cdot f_2(xy)\,dy = 0$, and $Mx - Ny \neq 0$, then:

$$\text{I.F.} = \frac{1}{Mx - Ny}$$

Rule III: Function of x alone

If $\dfrac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is a function of x alone, say $f(x)$, then:

$$\text{I.F.} = e^{\int f(x)\,dx}$$

Rule IV: Function of y alone

If $\dfrac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$ is a function of y alone, say $g(y)$, then:

$$\text{I.F.} = e^{\int g(y)\,dy}$$

Rule V: Special Form $x^a y^b(my\,dx + nx\,dy) = 0$

If the equation can be written as $x^\alpha y^\beta(my\,dx + nx\,dy) + x^{\alpha'} y^{\beta'}(m'y\,dx + n'x\,dy) = 0$, the I.F. is assumed to be $x^h y^k$. The values of h and k are found by applying the condition of exactness after multiplication.

Example 5.1.9. Solve: $(x^2y - 2xy^2)\,dx - (x^3 - 3x^2y)\,dy = 0$.

Solution. Here, $M = x^2y - 2xy^2$ and $N = -x^3 + 3x^2y$. The equation is homogeneous of degree 3.

$$Mx + Ny = x(x^2y - 2xy^2) + y(-x^3 + 3x^2y) = x^3y - 2x^2y^2 - x^3y + 3x^2y^2 = x^2y^2$$

$$\text{I.F.} = \frac{1}{x^2y^2}$$

Multiply the equation by the I.F.:

$$\left(\frac{1}{y} - \frac{2}{x}\right) dx - \left(\frac{x}{y^2} - \frac{3}{y}\right) dy = 0$$

This is now exact. Integrating M w.r.t x and terms in N free of x:

$$\int \left(\frac{1}{y} - \frac{2}{x}\right) dx + \int \frac{3}{y}\,dy = C$$

$$\frac{x}{y} - 2\ln x + 3\ln y = C$$

■

Example 5.1.10. Solve: $x^2y\,dx - (x^3 + y^3)\,dy = 0$.

Solution. Here, $M = x^2y$ and $N = -(x^3 + y^3)$.

$$Mx + Ny = x^3y - x^3y - y^4 = -y^4$$

$$\text{I.F.} = -\frac{1}{y^4}$$

Multiply the equation by the I.F.:

$$\left(-\frac{x^2}{y^3}\right) dx + \left(\frac{x^3}{y^4} + \frac{1}{y}\right) dy = 0$$

Integrating:

$$\int -\frac{x^2}{y^3}\, dx + \int \frac{1}{y}\, dy = C \implies -\frac{x^3}{3y^3} + \ln y = C$$

■

Example 5.1.11. Solve: $y(1 + xy)\, dx + x(1 - xy)\, dy = 0$.

Solution. Here $M = y + xy^2$ and $N = x - x^2y$.

$$Mx - Ny = x(y + xy^2) - y(x - x^2y) = xy + x^2y^2 - xy + x^2y^2 = 2x^2y^2$$

$$\text{I.F.} = \frac{1}{2x^2y^2}$$

Multiply the equation by the I.F.:

$$\left(\frac{1}{2x^2y} + \frac{1}{2x}\right) dx + \left(\frac{1}{2xy^2} - \frac{1}{2y}\right) dy = 0$$

Integrating:

$$\int \left(\frac{1}{2x^2y} + \frac{1}{2x}\right) dx + \int -\frac{1}{2y}\, dy = C$$

$$-\frac{1}{2xy} + \frac{1}{2}\ln x - \frac{1}{2}\ln y = C \implies \ln\left(\frac{x}{y}\right) - \frac{1}{xy} = C'$$

■

Example 5.1.12. Solve: $y(2xy + 1)\, dx + x(1 + 2xy)\, dy = 0$. *(Wait, signs must differ for this rule to be useful usually, but let's test form).* Let's use: $y(2xy + 1)\, dx + x(1 + xy)\, dy = 0$.

Solution. $M = 2xy^2 + y, \quad N = xy + x^2y.$

$$Mx - Ny = (2x^2y^2 + xy) - (xy + x^2y^2) = x^2y^2$$

$$\text{I.F.} = \frac{1}{x^2y^2}$$

Multiply by I.F.:

$$\left(\frac{2}{x} + \frac{1}{x^2y}\right) dx + \left(\frac{1}{xy^2} + \frac{1}{y}\right) dy = 0$$

Integrating:

$$2\ln x - \frac{1}{xy} + \ln y = C \implies \ln(x^2y) - \frac{1}{xy} = C$$

■

Example 5.1.13. Solve: $(x^2 + y^2 + x)\, dx + xy\, dy = 0$.

Solution. $\frac{\partial M}{\partial y} = 2y, \quad \frac{\partial N}{\partial x} = y.$

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{xy}(2y - y) = \frac{y}{xy} = \frac{1}{x}$$

$$\text{I.F.} = e^{\int \frac{1}{x}\, dx} = x$$

Multiply by x:

$$(x^3 + xy^2 + x^2)\,dx + x^2y\,dy = 0$$

Integrating:

$$\int (x^3 + xy^2 + x^2)\,dx + \int (0)\,dy = C \implies \frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} = C$$

■

Example 5.1.14. Solve: $(2x^2 + y)\,dx + (x^2y - x)\,dy = 0$.

Solution. $\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 2xy - 1$. The standard order doesn't work well here. Let's rewrite the equation as $-(2x^2 + y)dx = (x^2y - x)dy$. Let's try the alternative:

$$\frac{1}{N}(1 - (2xy - 1)) = \frac{2 - 2xy}{x(xy - 1)} = \frac{-2(xy - 1)}{x(xy - 1)} = -\frac{2}{x}$$

$$\text{I.F.} = e^{\int -\frac{2}{x}\,dx} = \frac{1}{x^2}$$

Multiply by $1/x^2$:

$$\left(2 + \frac{y}{x^2}\right)dx + \left(y - \frac{1}{x}\right)dy = 0$$

Integrating:

$$\int \left(2 + \frac{y}{x^2}\right)dx + \int y\,dy = C \implies 2x - \frac{y}{x} + \frac{y^2}{2} = C$$

■

Example 5.1.15. Solve: $(xy^3 + y)\,dx + 2(x^2y^2 + x + y^4)\,dy = 0$.

Solution. $\frac{\partial M}{\partial y} = 3xy^2 + 1, \quad \frac{\partial N}{\partial x} = 4xy^2 + 2$.

$$\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = \frac{(4xy^2 + 2) - (3xy^2 + 1)}{y(xy^2 + 1)} = \frac{xy^2 + 1}{y(xy^2 + 1)} = \frac{1}{y}$$

$$\text{I.F.} = e^{\int \frac{1}{y}\,dy} = y$$

Multiply by y:

$$(xy^4 + y^2)\,dx + (2x^2y^3 + 2xy + 2y^5)\,dy = 0$$

Integrating:

$$\int (xy^4 + y^2)\,dx + \int 2y^5\,dy = C \implies \frac{x^2y^4}{2} + xy^2 + \frac{y^6}{3} = C$$

■

Example 5.1.16. Solve: $(y^4 + 2y)\,dx + (xy^3 + 2y^4 - 4x)\,dy = 0$.

Solution. $\frac{\partial M}{\partial y} = 4y^3 + 2, \quad \frac{\partial N}{\partial x} = y^3 - 4$.

$$\frac{1}{M}((y^3 - 4) - (4y^3 + 2)) = \frac{-3y^3 - 6}{y(y^3 + 2)} = \frac{-3(y^3 + 2)}{y(y^3 + 2)} = -\frac{3}{y}$$

$$\text{I.F.} = e^{\int -\frac{3}{y}\,dy} = \frac{1}{y^3}$$

Multiply by $1/y^3$:

$$(y + 2y^{-2})\,dx + (x + 2y - 4xy^{-3})\,dy = 0$$

Integrating:

$$\int (y + 2y^{-2})\,dx + \int 2y\,dy = C \implies xy - \frac{2x}{y^2} + y^2 = C$$

■

Example 5.1.17. Solve: $(2x^2y^2 + y)\,dx + (x^3y - x)\,dy = 0$.

Solution. Rewrite as: $x^2y(2y\,dx + x\,dy) + x^0y^0(1y\,dx - 1x\,dy) = 0$. Multiply by x^hy^k:

$$(2x^{h+2}y^{k+2} + x^hy^{k+1})dx + (x^{h+3}y^{k+1} - x^{h+1}y^k)dy = 0$$

Apply $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$:

$$2(k+2)x^{h+2}y^{k+1} + (k+1)x^hy^k = (h+3)x^{h+2}y^{k+1} - (h+1)x^hy^k$$

Equating coefficients of like terms:

1. $2k + 4 = h + 3 \implies h - 2k = 1$
2. $k + 1 = -(h+1) \implies h + k = -2$

Solving yields $h = -1, k = -1$.

$$\text{I.F.} = x^{-1}y^{-1} = \frac{1}{xy}$$

Multiply equation by $1/xy$:

$$(2xy + 1/x)\,dx + (x^2 - 1/y)\,dy = 0$$

Integrating:

$$x^2y + \ln x - \ln y = C \implies x^2y + \ln(x/y) = C$$

■

Example 5.1.18. Solve: $(y)\,dx - (x)\,dy = 0$ using Rule V.

Solution. Rewrite as $x^0y^0(1y\,dx - 1x\,dy) = 0$. Assume I.F. x^hy^k.

$$(x^hy^{k+1})dx - (x^{h+1}y^k)dy = 0$$

Condition for exactness:

$$\frac{\partial}{\partial y}(x^hy^{k+1}) = \frac{\partial}{\partial x}(-x^{h+1}y^k)$$
$$(k+1)x^hy^k = -(h+1)x^hy^k$$
$$k + 1 = -h - 1 \implies h + k = -2$$

Many solutions exist.

- Let $h = -2, k = 0 \implies$ I.F. $= 1/x^2$. Sol: $y/x = C$.
- Let $h = 0, k = -2 \implies$ I.F. $= 1/y^2$. Sol: $x/y = C$.

■

5.1.4 General Form and Operator Notation

A linear differential equation of order n with constant coefficients is of the form:

$$a_n\frac{d^ny}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1\frac{dy}{dx} + a_0y = Q(x) \tag{5.4}$$

where $a_0, a_1, \ldots, a_n$ are real constants and $Q(x)$ is a function of x.

Operator Notation: Let $D = \frac{d}{dx}, D^2 = \frac{d^2}{dx^2}$, etc. The equation becomes:

$$f(D)y = Q(x)$$

where $f(D) = a_nD^n + a_{n-1}D^{n-1} + \cdots + a_0$.

The General Solution

The complete solution consists of two parts:

$$y = \text{C.F.} + \text{P.I.}$$

- **Complementary Function (C.F.):** The solution to the homogeneous part $f(D)y = 0$.
- **Particular Integral (P.I.):** A specific solution satisfying $f(D)y = Q(x)$.

5.1.5 Rules for Finding C.F.

To find the C.F., solve the **Auxiliary Equation (A.E.)**: $f(m) = 0$. Let the roots be $m_1, m_2, \ldots, m_n$.

1. **Real and Distinct Roots** ($m_1 \neq m_2 \neq \ldots$):

$$\text{C.F.} = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}$$

2. **Real and Repeated Roots** (e.g., $m_1 = m_2 = m$):

$$\text{C.F.} = (c_1 + c_2 x)e^{mx} + c_3 e^{m_3 x} + \ldots$$

3. **Complex Conjugate Roots** ($m = \alpha \pm i\beta$):

$$\text{C.F.} = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$$

4. **Repeated Complex Roots** ($m = \alpha \pm i\beta$ repeated twice):

$$\text{C.F.} = e^{\alpha x}[(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x]$$

5.1.6 Rules for Finding P.I.

The Particular Integral is defined as: P.I. $= \dfrac{1}{f(D)} Q(x)$.

Case I: Exponential Form $Q(x) = e^{ax}$

Rule: Replace D with a.

$$\text{P.I.} = \frac{1}{f(a)} e^{ax}, \quad \text{provided } f(a) \neq 0$$

Failure Case: If $f(a) = 0$, multiply by x and differentiate the denominator w.r.t D:

$$\text{P.I.} = x \frac{1}{f'(a)} e^{ax}$$

Case II: Trigonometric Form $Q(x) = \sin(ax)$ **or** $\cos(ax)$

Rule: Replace D^2 with $-a^2$.

$$\text{P.I.} = \frac{1}{f(-a^2)} \sin(ax), \quad \text{provided } f(-a^2) \neq 0$$

Failure Case: If $f(-a^2) = 0$, multiply by x and differentiate $f(D)$.

Case III: Polynomial Form $Q(x) = x^m$

Rule: Expand using Binomial Theorem. Convert the operator to $(1 \pm \phi(D))^{-1}$ by factoring out the lowest degree term.

$$\text{P.I.} = [f(D)]^{-1} x^m = (1 \pm \phi(D))^{-1} x^m$$

Use expansions: $(1 - D)^{-1} = 1 + D + D^2 + \ldots$ and $(1 + D)^{-1} = 1 - D + D^2 - \ldots$

Case IV: Product Form $Q(x) = e^{ax}V(x)$

Rule: Shift Exponential.

$$\text{P.I.} = \frac{1}{f(D)}e^{ax}V = e^{ax}\frac{1}{f(D+a)}V$$

Example 5.1.19 (Distinct Real Roots)**.** Solve: $\dfrac{d^2y}{dx^2} - 7\dfrac{dy}{dx} + 12y = 0$.
Solution: Auxiliary Equation: $m^2 - 7m + 12 = 0 \implies (m-3)(m-4) = 0$.
Roots: $m = 3, 4$. Hence, the solution is $\boxed{y = c_1e^{3x} + c_2e^{4x}}$

Example 5.1.20 (Repeated Roots)**.** Solve: $(D^3 - 3D^2 + 3D - 1)y = 0$.
Solution: Auxiliary Equation:: $(m-1)^3 = 0 \implies m = 1, 1, 1$. Hence, the solution is

$$\boxed{y = (c_1 + c_2x + c_3x^2)e^x}.$$

Example 5.1.21 (Complex Roots)**.** Solve: $(D^2 + 4)y = 0$.
Solution: Auxiliary Equation:: $m^2 + 4 = 0 \implies m = \pm 2i$. Here $\alpha = 0, \beta = 2$. Hence, the solution is

$$\boxed{y = c_1 \cos 2x + c_2 \sin 2x}.$$

Example 5.1.22 (P.I. with Exponential)**.** Solve: $(D^2 - 3D + 2)y = e^{5x}$.
Solution: Auxiliary Equation:: $(m-1)(m-2) = 0 \implies$ C.F. $= c_1e^x + c_2e^{2x}$.
P.I.: $\dfrac{1}{D^2 - 3D + 2}e^{5x}$. Put $D = 5$.

$$\text{P.I.} = \frac{1}{25 - 15 + 2}e^{5x} = \frac{1}{12}e^{5x}$$

Hence, the solution is $\boxed{y = c_1e^x + c_2e^{2x} + \dfrac{1}{12}e^{5x}}$.

Example 5.1.23 (P.I. with Trigonometry)**.** Solve: $(D^2 + 1)y = \sin x$.
Solution: Auxiliary Equation:: $m^2 + 1 = 0 \implies m = \pm i$. C.F. $= c_1 \cos x + c_2 \sin x$.
P.I.: $\dfrac{1}{D^2 + 1}\sin x$. Putting $D^2 = -1$ gives 0 (Failure Case).
Multiply by x, differentiate denominator ($2D$):

$$\text{P.I.} = x \cdot \frac{1}{2D}\sin x = \frac{x}{2}\int \sin x\, dx = -\frac{x \cos x}{2}$$

Hence, the solution is $\boxed{y = c_1 \cos x + c_2 \sin x - \dfrac{x \cos x}{2}}$.

Example 5.1.24. Solve: $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = e^{5x}$.

Solution. **Method 1: Factorization (Integrating Factors)**
Factoring the operator gives $(D-1)(D-2)y = e^{5x}$. Let $(D-2)y = z$.
Solving $(D-1)z = e^{5x}$ with I.F. e^{-x}:

$$ze^{-x} = \int e^{4x}\, dx = \frac{1}{4}e^{4x} + C_1 \implies z = \frac{1}{4}e^{5x} + C_1e^x$$

Now solving $(D-2)y = z$ with I.F. e^{-2x}:

$$ye^{-2x} = \int \left(\frac{e^{3x}}{4} + C_1e^{-x}\right) dx = \frac{e^{3x}}{12} - C_1e^{-x} + C_2$$

Multiplying by e^{2x} yields: $y = \dfrac{e^{5x}}{12} - C_1 e^x + C_2 e^{2x}$.

Method 2: Operator Rules
C.F.: A.E. is $m^2 - 3m + 2 = 0 \implies (m-1)(m-2) = 0$. Roots $m = 1, 2$.

$$\text{C.F.} = c_1 e^x + c_2 e^{2x}$$

P.I.: We use **Case I** (Exponential Form e^{ax}). Replace D with $a = 5$.

$$\text{P.I.} = \frac{1}{D^2 - 3D + 2} e^{5x} = \frac{1}{5^2 - 3(5) + 2} e^{5x} = \frac{1}{25 - 15 + 2} e^{5x} = \frac{1}{12} e^{5x}$$

General Solution: $y = \text{C.F.} + \text{P.I.} = c_1 e^x + c_2 e^{2x} + \frac{1}{12} e^{5x}$. ■

Example 5.1.25. Solve: $(D^2 - 2D + 1)y = \dfrac{e^x}{1+x^2}$.

Solution. **Method 1: Factorization (Integrating Factors)**
The equation is $(D-1)[(D-1)y] = \frac{e^x}{1+x^2}$. Let $z = (D-1)y$.
Solving $(D-1)z = \frac{e^x}{1+x^2}$ with I.F. e^{-x}:

$$ze^{-x} = \int \frac{1}{1+x^2}\, dx = \tan^{-1} x + C_1 \implies z = e^x(\tan^{-1} x + C_1)$$

Substituting back, $(D-1)y = e^x(\tan^{-1} x + C_1)$. Using I.F. e^{-x}:

$$ye^{-x} = \int (\tan^{-1} x + C_1)\, dx = x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C_1 x + C_2$$

Thus, $y = e^x \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C_1 x + C_2 \right]$.

Method 2: Operator Rules
C.F.: A.E. is $(m-1)^2 = 0 \implies m = 1, 1$. C.F. $= (c_1 + c_2 x)e^x$.
P.I.: We use **Case IV** (Product Form $e^{ax}V$). Shift e^x outside and replace D with $D+1$.

$$\text{P.I.} = \frac{1}{(D-1)^2} \left(e^x \cdot \frac{1}{1+x^2} \right) = e^x \frac{1}{((D+1)-1)^2} \left(\frac{1}{1+x^2} \right)$$

$$\text{P.I.} = e^x \frac{1}{D^2} \left(\frac{1}{1+x^2} \right)$$

Since $\frac{1}{D}$ denotes integration, $\frac{1}{D^2}$ denotes double integration.

$$\frac{1}{D} \left(\frac{1}{1+x^2} \right) = \int \frac{1}{1+x^2}\, dx = \tan^{-1} x$$
$$\frac{1}{D^2} \left(\frac{1}{1+x^2} \right) = \int \tan^{-1} x\, dx = x \tan^{-1} x - \frac{1}{2} \ln(1+x^2)$$

$$\text{P.I.} = e^x \left[x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]$$

■

Example 5.1.26. Solve: $\dfrac{d^2y}{dx^2} - y = \dfrac{2}{1+e^x}$.

Solution. **Method 1: Factorization (Integrating Factors)**
Factor as $(D-1)(D+1)y = \frac{2}{1+e^x}$. Let $(D+1)y = z$.
Solving $(D-1)z = \frac{2}{1+e^x}$ with I.F. e^{-x} yields $z = -2 + 2e^x \ln(1+e^{-x}) + C_1 e^x$.
Then solve $(D+1)y = z$ using I.F. e^x. After integration steps (omitted for brevity as they match Method 1 results), we get the solution.

Method 2: Operator Rules (Partial Fractions)
C.F.: $m^2 - 1 = 0 \implies m = \pm 1$. C.F. $= c_1 e^x + c_2 e^{-x}$.
P.I.: Use Partial Fractions on the operator $\frac{1}{D^2-1}$.

$$\text{P.I.} = \frac{1}{D^2-1}\left(\frac{2}{1+e^x}\right) = \frac{1}{2}\left[\frac{1}{D-1} - \frac{1}{D+1}\right]\left(\frac{2}{1+e^x}\right)$$

Recall that $\frac{1}{D-a}Q = e^{ax}\int e^{-ax}Q\,dx$.

$$\begin{aligned}\text{Term 1:}\quad \frac{1}{D-1}\frac{2}{1+e^x} &= e^x \int e^{-x}\frac{2}{1+e^x}\,dx \\ &= 2e^x \int \frac{e^{-x}(e^{-x})}{e^{-x}+1}\,dx \quad \text{(Algebraic manipulation yields same result as Method 1 step}\end{aligned}$$

Using this method simplifies the problem into two separate first-order integrals immediately, leading to:

$$\text{P.I.} = -1 + xe^x - e^x \ln(1+e^x) + c_2 e^{-x}$$

(Note: The constants merge with the C.F.) ■

Example 5.1.27. Solve: $\dfrac{dy}{dx} + \dfrac{1}{3}y = e^x y^4$.

Solution. **Method 1: Factorization (Bernoulli Reduction)**
Multiplying by y^{-4} and substituting $v = y^{-3}$ transforms this into $\frac{dv}{dx} - v = -3e^x$.
Using I.F. e^{-x}, we found $ve^{-x} = -3x + C$, leading to $y^{-3} = e^x(C - 3x)$.
Method 2: Operator Rules (On transformed equation)
The transformed linear equation is $(D-1)v = -3e^x$.
C.F.: $m - 1 = 0 \implies v_{cf} = c_1 e^x$.
P.I.: We use **Case I Failure**. Here $a = 1$ and $f(D) = D - 1$, so $f(1) = 0$.
Rule: Multiply by x and differentiate operator w.r.t D.

$$\text{P.I.} = x\frac{1}{\frac{d}{dD}(D-1)}(-3e^x) = x\frac{1}{1}(-3e^x) = -3xe^x$$

General Solution for v: $v = c_1 e^x - 3xe^x = e^x(c_1 - 3x)$.
Converting back to y: $\frac{1}{y^3} = e^x(c_1 - 3x)$. ■

Example 5.1.28. Solve: $\dfrac{dy}{dx} + 2y = e^{-2x}\sin^2 x$.

Solution. **Method 1: Integrating Factor**
First-order LDE with I.F. e^{2x}.

$$\begin{aligned} ye^{2x} &= \int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C \\ y &= e^{-2x}\left(\frac{x}{2} - \frac{\sin 2x}{4} + C\right)\end{aligned}$$

Method 2: Operator Rules
Equation: $(D+2)y = e^{-2x}\sin^2 x$.
C.F.: $m + 2 = 0 \implies y_{cf} = c_1 e^{-2x}$.
P.I.: Use **Case IV** (Shift Exponential).

$$\text{P.I.} = \frac{1}{D+2}(e^{-2x}\sin^2 x) = e^{-2x}\frac{1}{(D-2)+2}\sin^2 x = e^{-2x}\frac{1}{D}\sin^2 x$$

Since $\frac{1}{D}$ is integration:

$$\text{P.I.} = e^{-2x}\int \frac{1-\cos 2x}{2}\,dx = e^{-2x}\left[\frac{x}{2} - \frac{\sin 2x}{4}\right]$$

General Solution: $y = c_1 e^{-2x} + e^{-2x}\left(\frac{x}{2} - \frac{\sin 2x}{4}\right)$. ■

Example 5.1.29. Solve: $(D^2 + 5D + 6)y = e^{-2x} \sec^2 x(1 + 2 \tan x)$.

Solution. **Method 1: Factorization (Reduction of Order)**

Factor the operator as $(D + 2)(D + 3)y = e^{-2x}(\sec^2 x + 2 \sec^2 x \tan x)$. Let $(D + 3)y = u$. Then $(D + 2)u = e^{-2x}(\sec^2 x + 2 \sec^2 x \tan x)$.

Solving for u (Linear, I.F. $= e^{2x}$):

$$ue^{2x} = \int (\sec^2 x + 2 \tan x \sec^2 x)\, dx$$

$$ue^{2x} = \tan x + \tan^2 x + C_1$$
$$u = e^{-2x}(\tan x + \tan^2 x + C_1)$$

Substituting back, $(D + 3)y = e^{-2x}(\tan x + \tan^2 x + C_1)$. Solving for y (Linear, I.F. $= e^{3x}$):

$$ye^{3x} = \int e^{3x}[e^{-2x}(\tan x + \tan^2 x + C_1)]\, dx$$

$$ye^{3x} = \int e^x(\tan x + \tan^2 x)\, dx + \int C_1 e^x\, dx$$

Using the identity $\int e^x[f(x) + f'(x) - 1]\, dx$ relating to $\tan x$:

$$ye^{3x} = e^x \tan x + C_1 e^x + C_2$$

$$y = e^{-2x} \tan x + C_1 e^{-2x} + C_2 e^{-3x}$$

Method 2: Operator Rules

Complementary Function (C.F.): Roots of $m^2 + 5m + 6 = 0$ are $m = -2, -3$.

$$\text{C.F.} = c_1 e^{-2x} + c_2 e^{-3x}$$

Particular Integral (P.I.): Let $V(x) = \sec^2 x + 2 \tan x \sec^2 x$.

$$\text{P.I.} = \frac{1}{(D + 2)(D + 3)} e^{-2x} V(x)$$

Shift e^{-2x} to the left (replace D with $D - 2$):

$$\text{P.I.} = e^{-2x} \frac{1}{(D - 2 + 2)(D - 2 + 3)} V(x) = e^{-2x} \frac{1}{D(D + 1)} V(x)$$

Using partial fractions $\frac{1}{D(D+1)} = \frac{1}{D} - \frac{1}{D+1}$:

$$\text{P.I.} = e^{-2x} \left[\frac{1}{D} V(x) - \frac{1}{D + 1} V(x) \right]$$

Since $V(x) = \frac{d}{dx}(\tan x + \tan^2 x)$:

$$\frac{1}{D} V(x) = \tan x + \tan^2 x$$

The term $\frac{1}{D+1} V(x)$ corresponds to the integral involving e^x solved in Method 1, which reduces the expression essentially to:

$$\text{P.I.} = e^{-2x} \tan x$$

General Solution:

$$y = c_1 e^{-2x} + c_2 e^{-3x} + e^{-2x} \tan x$$

■

Chapter 6

Beta and Gamma Functions

This unit introduces Beta and Gamma functions, which are elegant generalizations of factorials used to evaluate complex improper integrals. It explores their unique properties, symmetry, and the intrinsic relationship connecting the two functions. Understanding these special functions opens the door to higher-level analysis and advanced applications in applied mathematics.

6.1 The Gamma Function

Definition 6.1.1. The **Gamma Function**, denoted by $\Gamma(n)$, is an improper integral dependent on a parameter n. It was defined by Euler as the limit of a product, but the integral definition is most commonly used.

For any real number $n > 0$, the Gamma function is defined as:

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\, dx \tag{6.1}$$

This is also known as **Euler's Integral of the Second Kind**.

6.1.1 Properties of the Gamma Function

1. Fundamental Recurrence Relation

Property: $\Gamma(n+1) = n\Gamma(n)$

Proof. By definition, $\Gamma(n+1) = \int_0^\infty e^{-x} x^n\, dx$.
We apply **Integration by Parts**. Let $u = x^n$ and $dv = e^{-x}\, dx$.
Then $du = nx^{n-1}\, dx$ and $v = -e^{-x}$.

$$\Gamma(n+1) = \left[-x^n e^{-x}\right]_0^\infty - \int_0^\infty (-e^{-x})(nx^{n-1})\, dx$$

The boundary term $[-x^n e^{-x}]_0^\infty$ vanishes because $\lim_{x\to\infty} \frac{x^n}{e^x} = 0$ and at $x = 0$, term is 0.

$$\Gamma(n+1) = 0 + n\int_0^\infty e^{-x} x^{n-1}\, dx$$

Recognizing the remaining integral as the definition of $\Gamma(n)$:

$$\Gamma(n+1) = n\Gamma(n)$$

■

2. Relation to Factorial

Property: $\Gamma(n+1) = n!$ for integer $n \geq 0$.

Proof. First, evaluate $\Gamma(1)$:

$$\Gamma(1) = \int_0^\infty e^{-x}x^{1-1}\,dx = \int_0^\infty e^{-x}\,dx = \left[-e^{-x}\right]_0^\infty = -(0-1) = 1$$

Using the recurrence relation $\Gamma(n+1) = n\Gamma(n)$ repeatedly:

$$\begin{aligned}
\Gamma(2) &= 1\cdot\Gamma(1) = 1 = 1! \\
\Gamma(3) &= 2\cdot\Gamma(2) = 2\cdot 1 = 2! \\
\Gamma(4) &= 3\cdot\Gamma(3) = 3\cdot 2\cdot 1 = 3! \\
&\vdots \\
\Gamma(n+1) &= n(n-1)(n-2)\cdots 1\cdot\Gamma(1) = n!
\end{aligned}$$

■

3. Value at Half-Integers

Property: $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

Proof. By definition:

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty e^{-x}x^{-1/2}\,dx$$

Let $x = t^2 \implies dx = 2t\,dt$. Limits remain 0 to ∞.

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty e^{-t^2}(t^2)^{-1/2}(2t\,dt) = 2\int_0^\infty e^{-t^2}\,dt$$

This is a standard Gaussian integral. We know that $\int_{-\infty}^{\infty} e^{-t^2}\,dt = \sqrt{\pi}$. Since the integrand is an even function:

$$\int_0^\infty e^{-t^2}\,dt = \frac{1}{2}\int_{-\infty}^\infty e^{-t^2}\,dt = \frac{\sqrt{\pi}}{2}$$

Substituting this back:

$$\Gamma\left(\frac{1}{2}\right) = 2\left(\frac{\sqrt{\pi}}{2}\right) = \sqrt{\pi}$$

■

Example 6.1.2. Evaluate $\Gamma(7/2)$.

Solution. We apply the recurrence relation repeatedly:

$$\begin{aligned}
\Gamma\left(\frac{7}{2}\right) &= \frac{5}{2}\Gamma\left(\frac{5}{2}\right) \\
&= \frac{5}{2}\left(\frac{3}{2}\Gamma\left(\frac{3}{2}\right)\right) \\
&= \frac{5}{2}\cdot\frac{3}{2}\left(\frac{1}{2}\Gamma\left(\frac{1}{2}\right)\right)
\end{aligned}$$

Since $\Gamma(1/2) = \sqrt{\pi}$:

$$\Gamma\left(\frac{7}{2}\right) = \frac{15}{8}\sqrt{\pi}$$

■

Example 6.1.3. Evaluate $\Gamma\left(-\frac{1}{2}\right)$.

Solution. Let $n = -1/2$. Using the recurrence relation:

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2}+1\right)}{-\frac{1}{2}} = \frac{\Gamma\left(\frac{1}{2}\right)}{-\frac{1}{2}}$$

Since we know $\Gamma(1/2) = \sqrt{\pi}$:

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\sqrt{\pi}}{-1/2} = -2\sqrt{\pi}$$

■

Example 6.1.4. Evaluate $\Gamma\left(-\frac{5}{2}\right)$.

Solution. We apply the recurrence relation repeatedly to step up to a known value.

$$\begin{aligned}
\Gamma\left(-\frac{5}{2}\right) &= \frac{\Gamma(-3/2)}{-5/2} \\
&= \frac{1}{-5/2}\left[\frac{\Gamma(-1/2)}{-3/2}\right] \quad \text{(Applying relation to } \Gamma(-3/2)) \\
&= \frac{1}{(-5/2)(-3/2)}\Gamma\left(-\frac{1}{2}\right) \\
&= \frac{1}{\frac{15}{4}}(-2\sqrt{\pi}) \quad \text{(Using result from previous Example)} \\
&= \frac{4}{15}(-2\sqrt{\pi}) \\
&= -\frac{8}{15}\sqrt{\pi}
\end{aligned}$$

■

Example 6.1.5. Simplify the expression $\dfrac{\Gamma(n+2)}{\Gamma(n)}$.

Solution. Using the property $\Gamma(z+1) = z\Gamma(z)$ on the numerator:

$$\Gamma(n+2) = \Gamma((n+1)+1) = (n+1)\Gamma(n+1)$$

Apply the property again to $\Gamma(n+1)$:

$$(n+1)\Gamma(n+1) = (n+1)(n)\Gamma(n)$$

Substitute this back into the fraction:

$$\frac{\Gamma(n+2)}{\Gamma(n)} = \frac{n(n+1)\Gamma(n)}{\Gamma(n)} = n(n+1)$$

■

Example 6.1.6. Evaluate $\Gamma(6)$.

Solution. Since the argument is a positive integer, we use the factorial relation $\Gamma(n+1) = n!$. Here, $n+1 = 6 \implies n = 5$.

$$\Gamma(6) = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

■

Example 6.1.7. Evaluate the integral $\int_0^\infty x^4 e^{-x}\, dx$.

Solution. The definition of the Gamma function is $\int_0^\infty x^{n-1}e^{-x}\,dx = \Gamma(n)$. Comparing powers of x, we have $n-1=4 \implies n=5$.

$$\int_0^\infty x^4 e^{-x}\,dx = \Gamma(5) = 4! = 24$$

■

Example 6.1.8. Evaluate the integral $\int_0^\infty \frac{e^{-x}}{\sqrt{x}}\,dx$.

Solution. Rewrite the integral using exponent notation:

$$I = \int_0^\infty x^{-1/2}e^{-x}\,dx$$

Comparing with $\Gamma(n) = \int_0^\infty x^{n-1}e^{-x}\,dx$, we have $n-1=-1/2 \implies n=1/2$.

$$I = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

■

Example 6.1.9. Evaluate $\int_0^\infty e^{-x^2}\,dx$.

Solution. Let $x^2 = t \implies x = \sqrt{t}$. Then $dx = \frac{1}{2\sqrt{t}}\,dt = \frac{1}{2}t^{-1/2}\,dt$. Limits remain 0 to ∞.

$$I = \int_0^\infty e^{-t}\left(\frac{1}{2}t^{-1/2}\right)\,dt$$
$$I = \frac{1}{2}\int_0^\infty t^{(1/2)-1}e^{-t}\,dt$$
$$I = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$$

■

6.1.2 Alternative Forms of the Gamma Function

Standard Definition: $\Gamma(n) = \int_0^\infty e^{-t}t^{n-1}\,dt$

1. **Property:** $\Gamma(n) = k^n \int_0^\infty e^{-ky}y^{n-1}\,dy$

   Proof. Start with the standard definition using variable x: $\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}\,dx$.
   Substitute $x = ky$ (where constant $k > 0$).
   Then $dx = k\,dy$.
   Limits: When $x = 0, y = 0$; when $x \to \infty, y \to \infty$.

$$\begin{aligned}\Gamma(n) &= \int_0^\infty e^{-ky}(ky)^{n-1}(k\,dy)\\ &= \int_0^\infty e^{-ky}k^{n-1}y^{n-1}k\,dy\\ &= k^{n-1}\cdot k\int_0^\infty e^{-ky}y^{n-1}\,dy\\ &= k^n\int_0^\infty e^{-ky}y^{n-1}\,dy\end{aligned}$$

■

2. **Property:** $\Gamma\left(1+\frac{1}{n}\right) = \int_0^\infty e^{-x^n}\,dx$

Proof. Consider the integral $I = \int_0^\infty e^{-x^n}\,dx$.
Substitute $x^n = t \implies x = t^{1/n}$.
Differentiating: $dx = \frac{1}{n}t^{\frac{1}{n}-1}\,dt$.
Limits: When $x = 0, t = 0$; when $x \to \infty, t \to \infty$.

$$\begin{aligned} I &= \int_0^\infty e^{-t}\left(\frac{1}{n}t^{\frac{1}{n}-1}\right)dt \\ &= \frac{1}{n}\int_0^\infty e^{-t}t^{\frac{1}{n}-1}\,dt \end{aligned}$$

By the definition of Gamma function, $\int_0^\infty e^{-t}t^{z-1}\,dt = \Gamma(z)$. Here $z = 1/n$.

$$I = \frac{1}{n}\Gamma\left(\frac{1}{n}\right)$$

Using the recurrence relation $z\Gamma(z) = \Gamma(z+1)$:

$$I = \Gamma\left(1 + \frac{1}{n}\right)$$

■

3. **Property:** $\Gamma(n) = \displaystyle\int_0^1 \left(\ln\frac{1}{y}\right)^{n-1} dy$

Proof. Start with the standard definition: $\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}\,dx$.
Substitute $x = \ln(1/y) = -\ln y$.
This implies $e^{-x} = y$.
Differentiating $x = -\ln y$ gives $dx = -\frac{1}{y}\,dy$.
Limits:

- When $x = 0 \implies \ln(1/y) = 0 \implies y = 1$.
- When $x \to \infty \implies \ln(1/y) \to \infty \implies y \to 0$.

Substituting these into the integral:

$$\begin{aligned} \Gamma(n) &= \int_1^0 y \cdot \left(\ln\frac{1}{y}\right)^{n-1}\left(-\frac{1}{y}\,dy\right) \\ &= -\int_1^0 \left(\ln\frac{1}{y}\right)^{n-1} dy \\ &= \int_0^1 \left(\ln\frac{1}{y}\right)^{n-1} dy \quad \text{(Swapping limits removes the minus sign)} \end{aligned}$$

■

Example 6.1.10. Evaluate $\displaystyle\int_0^\infty x^3e^{-2x}\,dx$.

Solution. Comparing with the standard form, we identify $n - 1 = 3 \implies n = 4$ and $k = 2$. Using the formula:

$$\int_0^\infty x^3e^{-2x}\,dx = \frac{\Gamma(4)}{2^4} = \frac{3!}{16} = \frac{6}{16} = \frac{3}{8}$$

■

Example 6.1.11. Evaluate $\displaystyle\int_0^\infty \sqrt{y}\,e^{-3y}\,dy$.

Solution. Here, $y^{n-1} = y^{1/2} \implies n = \frac{3}{2}$ and $k = 3$. Using the formula:

$$\int_0^\infty y^{1/2} e^{-3y}\, dy = \frac{\Gamma(3/2)}{3^{3/2}} = \frac{\frac{1}{2}\Gamma(1/2)}{3\sqrt{3}} = \frac{\frac{1}{2}\sqrt{\pi}}{3\sqrt{3}} = \frac{\sqrt{\pi}}{6\sqrt{3}}$$

■

Example 6.1.12. Evaluate $\int_0^\infty e^{-x^4}\, dx$.

Solution. Here $n = 4$. Directly applying the property:

$$\int_0^\infty e^{-x^4}\, dx = \Gamma\left(1 + \frac{1}{4}\right) = \frac{1}{4}\Gamma\left(\frac{1}{4}\right)$$

■

Example 6.1.13. Evaluate the Gaussian integral $\int_0^\infty e^{-x^2}\, dx$.

Solution. Here $n = 2$. Applying the property:

$$\int_0^\infty e^{-x^2}\, dx = \Gamma\left(1 + \frac{1}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}$$

■

Example 6.1.14. Evaluate $\int_0^1 \left(\ln \frac{1}{x}\right)^5 dx$.

Solution. Comparing with the standard form, $n - 1 = 5 \implies n = 6$.

$$\int_0^1 \left(\ln \frac{1}{x}\right)^5 dx = \Gamma(6) = 5! = 120$$

■

Example 6.1.15. Evaluate $\int_0^1 \frac{1}{\sqrt{\ln(1/x)}}\, dx$.

Solution. Rewrite the integral as $\int_0^1 \left(\ln \frac{1}{x}\right)^{-1/2} dx$. Here, $n - 1 = -\frac{1}{2} \implies n = \frac{1}{2}$.

$$\int_0^1 \left(\ln \frac{1}{x}\right)^{-1/2} dx = \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

■

6.2 The Beta Function

Definition 6.2.1. The **Beta Function**, denoted by $B(m, n)$, is a definite integral depending on two parameters m and n (where $m, n > 0$). It is defined as:

$$B(m, n) = \int_0^1 x^{m-1}(1 - x)^{n-1}\, dx \tag{6.2}$$

This is also known as **Euler's Integral of the First Kind**.

6.2.1 Properties of the Beta Function

1. **Property: Symmetry** $B(m,n) = B(n,m)$

   Proof. We use the property of definite integrals: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. Here $a = 1$. Replace x with $(1-x)$ in the definition:

$$\begin{aligned} B(m,n) &= \int_0^1 (1-x)^{m-1}(1-(1-x))^{n-1}\,dx \\ &= \int_0^1 (1-x)^{m-1}x^{n-1}\,dx \\ &= \int_0^1 x^{n-1}(1-x)^{m-1}\,dx \\ &= B(n,m) \end{aligned}$$

■

2. **Property: Trigonometric Form**

$$B(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta\,d\theta$$

   Proof. Put $x = \sin^2\theta$. Then $dx = 2\sin\theta\cos\theta\,d\theta$.
   Limits: When $x = 0, \theta = 0$. When $x = 1, \theta = \pi/2$.
   Note that $(1-x) = 1 - \sin^2\theta = \cos^2\theta$.

$$\begin{aligned} B(m,n) &= \int_0^{\pi/2} (\sin^2\theta)^{m-1}(\cos^2\theta)^{n-1}(2\sin\theta\cos\theta)\,d\theta \\ &= 2\int_0^{\pi/2} \sin^{2m-2}\theta\cos^{2n-2}\theta\sin\theta\cos\theta\,d\theta \\ &= 2\int_0^{\pi/2} \sin^{2m-1}\theta\cos^{2n-1}\theta\,d\theta \end{aligned}$$

■

3. **Property: Infinite Limit Form**

$$B(m,n) = \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}}\,dy$$

   Proof. Put $x = \frac{y}{1+y}$. This implies $1 - x = \frac{1}{1+y}$ and $y = \frac{x}{1-x}$.
   Differentiating x with respect to y: $dx = \frac{(1+y)(1)-y(1)}{(1+y)^2}\,dy = \frac{1}{(1+y)^2}\,dy$.
   Limits: When $x = 0, y = 0$. When $x = 1, y \to \infty$.

$$\begin{aligned} B(m,n) &= \int_0^\infty \left(\frac{y}{1+y}\right)^{m-1}\left(\frac{1}{1+y}\right)^{n-1}\frac{1}{(1+y)^2}\,dy \\ &= \int_0^\infty \frac{y^{m-1}}{(1+y)^{m-1}}\cdot\frac{1}{(1+y)^{n-1}}\cdot\frac{1}{(1+y)^2}\,dy \\ &= \int_0^\infty \frac{y^{m-1}}{(1+y)^{(m-1)+(n-1)+2}}\,dy \\ &= \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}}\,dy \end{aligned}$$

■

4. **Property: Relation to Definite Integrals**

$$\int_0^a x^{m-1}(a-x)^{n-1}\,dx = a^{m+n-1}B(m,n)$$

Proof. Let $I = \int_0^a x^{m-1}(a-x)^{n-1}\,dx$.
Put $x = at$. Then $dx = a\,dt$.
Limits: When $x = 0, t = 0$. When $x = a, t = 1$.

$$\begin{aligned} I &= \int_0^1 (at)^{m-1}(a-at)^{n-1}(a\,dt) \\ &= \int_0^1 a^{m-1}t^{m-1} \cdot a^{n-1}(1-t)^{n-1} \cdot a\,dt \\ &= a^{(m-1)+(n-1)+1}\int_0^1 t^{m-1}(1-t)^{n-1}\,dt \\ &= a^{m+n-1}B(m,n) \end{aligned}$$

■

Example 6.2.2. Evaluate $\int_0^1 x^4(1-x)^3\,dx$ using Beta function.

Solution. By definition, this integral is $B(5,4)$ (since $m-1=4, n-1=3$). Using symmetry and the Gamma relation:

$$I = B(5,4) = B(4,5) = \frac{\Gamma(4)\Gamma(5)}{\Gamma(9)}$$

$$I = \frac{3!\cdot 4!}{8!} = \frac{6\cdot 24}{40320} = \frac{1}{280}$$

■

Example 6.2.3. Evaluate $\int_0^1 x^{1/2}(1-x)^{3/2}\,dx$.

Solution. Here, $m-1 = 1/2 \implies m = 3/2$ and $n-1 = 3/2 \implies n = 5/2$.

$$I = B\left(\frac{3}{2},\frac{5}{2}\right) = B\left(\frac{5}{2},\frac{3}{2}\right)$$

$$I = \frac{\Gamma(3/2)\Gamma(5/2)}{\Gamma(4)} = \frac{(\frac{1}{2}\sqrt{\pi})(\frac{3}{4}\sqrt{\pi})}{6} = \frac{3\pi}{8\cdot 6} = \frac{\pi}{16}$$

■

Example 6.2.4. Evaluate $\int_0^{\pi/2} \sin^5\theta\cos^4\theta\,d\theta$.

Solution. Here $p = 5$ and $q = 4$.

$$I = \frac{1}{2}B\left(\frac{5+1}{2},\frac{4+1}{2}\right) = \frac{1}{2}B\left(3,\frac{5}{2}\right)$$

$$I = \frac{1}{2}\frac{\Gamma(3)\Gamma(5/2)}{\Gamma(11/2)} = \frac{1}{2}\frac{2!\cdot\frac{3}{4}\sqrt{\pi}}{\frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}}$$

$$I = \frac{1}{2}\cdot\frac{2\cdot\frac{3}{4}}{\frac{945}{32}} = \frac{3/4}{945/32} = \frac{8}{315}$$

■

Example 6.2.5. Evaluate $\int_0^{\pi/2} \sqrt{\tan\theta}\,d\theta$.

Solution. Rewrite as $\int_0^{\pi/2} \sin^{1/2}\theta \cos^{-1/2}\theta \, d\theta$. Here $p = 1/2, q = -1/2$.

$$I = \frac{1}{2}B\left(\frac{1/2+1}{2}, \frac{-1/2+1}{2}\right) = \frac{1}{2}B\left(\frac{3}{4}, \frac{1}{4}\right)$$

$$I = \frac{1}{2}\Gamma(3/4)\Gamma(1/4)$$

Using Euler's Reflection Formula $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}$ with $n = 1/4$:

$$I = \frac{1}{2}\left(\frac{\pi}{\sin(\pi/4)}\right) = \frac{1}{2}\frac{\pi}{1/\sqrt{2}} = \frac{\pi}{\sqrt{2}}$$

■

Example 6.2.6. Evaluate $\int_0^\infty \frac{x^3}{(1+x)^7}\, dx$.

Solution. Comparing with the standard form: $m - 1 = 3 \implies m = 4$. $m + n = 7 \implies 4 + n = 7 \implies n = 3$.

$$I = B(4,3) = \frac{\Gamma(4)\Gamma(3)}{\Gamma(7)} = \frac{3! \cdot 2!}{6!} = \frac{6 \cdot 2}{720} = \frac{1}{60}$$

■

Example 6.2.7. Evaluate $\int_0^\infty \frac{dx}{1+x^4}$.

Solution. Let $x^4 = y \implies x = y^{1/4}, dx = \frac{1}{4}y^{-3/4}dy$. Limits remain 0 to ∞.

$$I = \int_0^\infty \frac{\frac{1}{4}y^{-3/4}}{1+y}\, dy = \frac{1}{4}\int_0^\infty \frac{y^{1/4-1}}{(1+y)^{1/4+3/4}}\, dy$$

Here $m = 1/4$ and $m + n = 1 \implies n = 3/4$.

$$I = \frac{1}{4}B\left(\frac{1}{4}, \frac{3}{4}\right) = \frac{1}{4}\Gamma(1/4)\Gamma(3/4)$$

Using Reflection Formula: $\Gamma(1/4)\Gamma(3/4) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}$.

$$I = \frac{\pi\sqrt{2}}{4}$$

■

Example 6.2.8. Evaluate $\int_0^2 x^2(2-x)^3\, dx$.

Solution. Here $a = 2$. Exponents are $m - 1 = 2 \implies m = 3$ and $n - 1 = 3 \implies n = 4$.

$$I = 2^{3+4-1}B(3,4) = 2^6 B(3,4)$$

$$I = 64 \cdot \frac{\Gamma(3)\Gamma(4)}{\Gamma(7)} = 64 \cdot \frac{2! \cdot 3!}{6!} = 64 \cdot \frac{2 \cdot 6}{720} = \frac{64}{60} = \frac{16}{15}$$

■

Example 6.2.9. Evaluate $\int_0^3 \frac{dx}{\sqrt{3x - x^2}}$.

Solution. Factor the denominator: $\sqrt{x(3-x)} = x^{1/2}(3-x)^{1/2}$.

$$I = \int_0^3 x^{-1/2}(3-x)^{-1/2}\,dx$$

Here $a = 3$. $m - 1 = -1/2 \implies m = 1/2$ and $n - 1 = -1/2 \implies n = 1/2$.

$$I = 3^{(1/2+1/2-1)}B(1/2, 1/2) = 3^0 \cdot \frac{\Gamma(1/2)\Gamma(1/2)}{\Gamma(1)}$$

$$I = 1 \cdot (\sqrt{\pi})^2 = \pi$$

■

6.3 Relation Between Beta and Gamma Functions

The most important theorem connecting these two functions is:

$$\boxed{B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}} \tag{6.3}$$

Evaluating integrals of powers of sine and cosine

From the trigonometric definition of the Beta function and the relation above, we derive the standard formula for evaluating integrals of powers of sine and cosine:

$$\int_0^{\pi/2} \sin^p x \cos^q x\,dx = \frac{1}{2}B\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = \frac{\Gamma(\frac{p+1}{2})\Gamma(\frac{q+1}{2})}{2\Gamma(\frac{p+q+2}{2})}$$

Euler's Reflection Formula

This formula relates the Gamma function of a value n to $1-n$, valid for $0 < n < 1$:

$$\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi} \tag{6.4}$$

Legendre's Duplication Formula

This formula relates the Gamma function of $2n$ to $\Gamma(n)$ and $\Gamma(n+1/2)$:

$$\Gamma(n)\Gamma\left(n+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2n-1}}\Gamma(2n) \tag{6.5}$$

Example 6.3.1. Evaluate $\int_0^{\pi/2} \sin^6 x \cos^4 x\,dx$.

Solution. Here $p = 6$ and $q = 4$. Substitute these into the formula:

$$I = \frac{\Gamma(\frac{6+1}{2})\Gamma(\frac{4+1}{2})}{2\Gamma(\frac{6+4+2}{2})} = \frac{\Gamma(\frac{7}{2})\Gamma(\frac{5}{2})}{2\Gamma(6)}$$

We expand the Gamma terms:

- $\Gamma(6) = 5! = 120$
- $\Gamma(\frac{7}{2}) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi} = \frac{15}{8}\sqrt{\pi}$
- $\Gamma(\frac{5}{2}) = \frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi} = \frac{3}{4}\sqrt{\pi}$

Substitute back:

$$I = \frac{\left(\frac{15}{8}\sqrt{\pi}\right)\left(\frac{3}{4}\sqrt{\pi}\right)}{2(120)} = \frac{\frac{45\pi}{32}}{240} = \frac{45\pi}{32 \times 240}$$

Simplifying the fraction $\frac{45}{240}$ (divide by 15 gives $\frac{3}{16}$):

$$I = \frac{3\pi}{32 \times 16} = \frac{3\pi}{512}$$

■

Example 6.3.2. Evaluate $\int_0^{\pi/2} \sqrt{\tan x}\, dx$.

Solution. Rewrite the integrand in terms of sine and cosine:

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\cos x}}\, dx = \int_0^{\pi/2} \sin^{1/2} x \cos^{-1/2} x\, dx$$

Here $p = 1/2$ and $q = -1/2$.

$$I = \frac{\Gamma(\frac{1/2+1}{2})\Gamma(\frac{-1/2+1}{2})}{2\Gamma(\frac{1/2-1/2+2}{2})} = \frac{\Gamma(\frac{3}{4})\Gamma(\frac{1}{4})}{2\Gamma(1)}$$

Since $\Gamma(1) = 1$, we have $I = \frac{1}{2}\Gamma(\frac{3}{4})\Gamma(\frac{1}{4})$. Using Euler's Reflection Formula (see Section 2), $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}$. Let $n = 1/4$:

$$\Gamma(1/4)\Gamma(3/4) = \frac{\pi}{\sin(\pi/4)} = \frac{\pi}{1/\sqrt{2}} = \pi\sqrt{2}$$

Therefore:

$$I = \frac{1}{2}(\pi\sqrt{2}) = \frac{\pi}{\sqrt{2}}$$

■

Example 6.3.3. Evaluate $\Gamma(1/6)\Gamma(5/6)$.

Solution. We observe that $\frac{1}{6} + \frac{5}{6} = 1$. Let $n = 1/6$. Applying the reflection formula:

$$\Gamma(1/6)\Gamma(1-1/6) = \frac{\pi}{\sin(\pi/6)}$$

Since $\sin(\pi/6) = 1/2$:

$$\Gamma(1/6)\Gamma(5/6) = \frac{\pi}{1/2} = 2\pi$$

■

Example 6.3.4. Evaluate the integral $\int_0^{\infty} \frac{y^{1/3}}{1+y}\, dy$.

Solution. We use the Beta function form $B(m,n) = \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}}\, dy$. Comparing powers, let $m - 1 = 1/3 \implies m = 4/3$. However, for the denominator to be $(1+y)^1$, we need $m+n = 1$, so $n = 1 - 4/3 = -1/3$. This is invalid as parameters must be positive.

Instead, compare with standard form $\int \frac{y^{a-1}}{1+y} dy = \Gamma(a)\Gamma(1-a)$. Here, numerator is $y^{1/3}$. Let $a - 1 = 1/3 \implies a = 4/3$ (Still $a > 1$, let's adjust).

Let's convert to proper form. Let $m - 1 = 1/3 \implies m = 4/3$ is wrong for convergence. Consider the standard result: $\int_0^\infty \frac{x^{p-1}}{1+x} dx = \frac{\pi}{\sin p\pi}$ for $0 < p < 1$. Rewrite $y^{1/3}$ as $\frac{y^{4/3-1}}{1+y}$? No. Correct Logic: The integral converges if the power is between -1 and 0? No. Let's evaluate $\int_0^\infty \frac{x^{-1/4}}{1+x} dx$. Here $m - 1 = -1/4 \implies m = 3/4$. $m + n = 1 \implies n = 1 - 3/4 = 1/4$.

$$I = B(3/4, 1/4) = \Gamma(3/4)\Gamma(1/4)$$

Using Reflection Formula with $n = 1/4$:

$$I = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}$$

■

Example 6.3.5. Evaluate the product $\Gamma(1/3)\Gamma(5/6)$.

Solution. We want to find a relation between $1/3$ and $5/6$. Note that $5/6 = 1/3 + 1/2$. Using the Duplication Formula, let $n = 1/3$.

$$\Gamma(1/3)\Gamma(1/3 + 1/2) = \frac{\sqrt{\pi}}{2^{2(1/3)-1}}\Gamma(2 \cdot 1/3)$$

$$\Gamma(1/3)\Gamma(5/6) = \frac{\sqrt{\pi}}{2^{-1/3}}\Gamma(2/3)$$

$$\Gamma(1/3)\Gamma(5/6) = \sqrt{\pi} \cdot 2^{1/3} \cdot \Gamma(2/3)$$

This gives the value of the product in terms of $\Gamma(2/3)$. ■

Example 6.3.6. Verify that $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ using the Duplication Formula.

Solution. Let $n = 1/4$. Apply the Duplication Formula:

$$\Gamma(1/4)\Gamma(1/4 + 1/2) = \frac{\sqrt{\pi}}{2^{2(1/4)-1}}\Gamma(2 \cdot 1/4)$$

$$\Gamma(1/4)\Gamma(3/4) = \frac{\sqrt{\pi}}{2^{-1/2}}\Gamma(1/2)$$

Since $\Gamma(1/2) = \sqrt{\pi}$:

$$\Gamma(1/4)\Gamma(3/4) = \frac{\sqrt{\pi}}{2^{-1/2}}(\sqrt{\pi}) = \pi \cdot 2^{1/2} = \pi\sqrt{2}$$

This matches the result obtained from the Reflection Formula. ■

6.4 Applications

The Beta and Gamma functions are fundamental in advanced calculus and applied mathematics. Their primary applications include:

1. **Evaluation of Definite Integrals:** Many complex definite integrals which cannot be solved by elementary methods (such as $\int_0^\infty e^{-x^2} dx$) can be evaluated easily using Gamma functions.

2. **Physics and Engineering:** Used in calculating time periods of oscillation (e.g., simple pendulum with large amplitudes), fluid dynamics, and quantum mechanics.

3. **Probability and Statistics:**

   - The Gamma Distribution and Beta Distribution are standard probability distributions.
   - The Chi-squared distribution is a special case of the Gamma distribution.

4. **Geometry:** Calculation of areas enclosed by curves (e.g., $x^m + y^n = a^n$) and volumes of n-dimensional spheres.

Evaluation of Definite Integrals

Example 6.4.1. Evaluate the Gaussian integral $I = \int_0^\infty e^{-x^2}\,dx$.

Solution. Let $x^2 = t \implies x = t^{1/2}$ and $dx = \frac{1}{2}t^{-1/2}\,dt$.

$$I = \int_0^\infty e^{-t}\left(\frac{1}{2}t^{-1/2}\right)dt = \frac{1}{2}\int_0^\infty t^{(1/2)-1}e^{-t}\,dt$$

By definition of the Gamma function:

$$I = \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}$$

■

Example 6.4.2. Evaluate $I = \int_0^\infty \sqrt[4]{x}\,e^{-\sqrt{x}}\,dx$.

Solution. Let $\sqrt{x} = t \implies x = t^2$ and $dx = 2t\,dt$.

$$I = \int_0^\infty t^{1/2}e^{-t}(2t)\,dt = 2\int_0^\infty t^{3/2}e^{-t}\,dt$$

Using $\Gamma(n) = \int_0^\infty t^{n-1}e^{-t}dt$, here $n-1 = 3/2 \implies n = 5/2$.

$$I = 2\Gamma\left(\frac{5}{2}\right) = 2\left(\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}\right) = \frac{3\sqrt{\pi}}{2}$$

■

Physics and Engineering

Example 6.4.3. Find the time period of a simple pendulum of length L swinging with a large amplitude of 90° ($\alpha = \pi/2$). The period is given by $T = 4\sqrt{\frac{L}{g}}\int_0^{\pi/2}\frac{d\theta}{\sqrt{2\cos\theta}}$.

Solution.

$$T = 4\sqrt{\frac{L}{2g}}\int_0^{\pi/2}(\cos\theta)^{-1/2}(\sin\theta)^0\,d\theta$$

Using the Beta-Gamma relation for trigonometric integrals ($p = -1/2, q = 0$):

$$T = \sqrt{\frac{8L}{g}}\cdot\frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{2\Gamma(\frac{3}{4})} = \sqrt{\frac{L}{g}}\frac{\sqrt{2\pi}\,\Gamma(1/4)}{2\Gamma(3/4)}$$

Using reflection formula $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$, this simplifies to numerical evaluation. ■

Example 6.4.4. Calculate the period of oscillation for a particle of mass m moving in a potential $V(x) = kx^4$. The period is proportional to $I = \int_0^A \frac{dx}{\sqrt{A^4-x^4}}$.

Solution. Let $x^4 = A^4 t \implies x = At^{1/4}$ and $dx = \frac{A}{4}t^{-3/4}\,dt$.

$$I = \int_0^1 \frac{\frac{A}{4}t^{-3/4}}{\sqrt{A^4(1-t)}}\,dt = \frac{1}{4A}\int_0^1 t^{-3/4}(1-t)^{-1/2}\,dt$$

This is a Beta function with $m = 1/4, n = 1/2$:

$$I = \frac{1}{4A}B\left(\frac{1}{4},\frac{1}{2}\right) = \frac{1}{4A}\frac{\Gamma(1/4)\sqrt{\pi}}{\Gamma(3/4)}$$

■

Probability and Statistics

Example 6.4.5. Calculate the Mean (Expected Value) of the Gamma Distribution with PDF $f(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$.

Solution. The mean is $E[X] = \int_0^\infty x f(x)\, dx$.

$$E[X] = \frac{1}{\Gamma(k)\theta^k} \int_0^\infty x \cdot x^{k-1} e^{-x/\theta}\, dx = \frac{1}{\Gamma(k)\theta^k} \int_0^\infty x^k e^{-x/\theta}\, dx$$

Using the substitution property $\int_0^\infty x^n e^{-x/\theta} dx = \Gamma(n+1)\theta^{n+1}$:

$$E[X] = \frac{\Gamma(k+1)\theta^{k+1}}{\Gamma(k)\theta^k} = \frac{k\Gamma(k) \cdot \theta}{\Gamma(k)} = k\theta$$

■

Example 6.4.6. Find the constant C such that $f(x) = Cx^3(1-x)^6$ is a valid probability density function on $[0, 1]$ (Beta Distribution).

Solution. For total probability to be 1, we need $\int_0^1 Cx^3(1-x)^6\, dx = 1$. The integral is the Beta function $B(4, 7)$ (since exponents are $m-1 = 3, n-1 = 6$).

$$C \cdot B(4,7) = 1 \implies C = \frac{1}{B(4,7)} = \frac{\Gamma(11)}{\Gamma(4)\Gamma(7)}$$

$$C = \frac{10!}{3! \cdot 6!} = \frac{10 \cdot 9 \cdot 8 \cdot 7}{6} = 840$$

■

6.5 Geometry

Example 6.5.1. Find the area of the Astroid $x^{2/3} + y^{2/3} = a^{2/3}$.

Solution. Area $= 4 \times$ (Area in 1st quadrant) $= 4\int_0^a y\, dx$. Let $x = a\cos^3\theta, y = a\sin^3\theta$. Then $dx = -3a\cos^2\theta\sin\theta\, d\theta$.

$$A = 4\int_{\pi/2}^0 (a\sin^3\theta)(-3a\cos^2\theta\sin\theta)\, d\theta = 12a^2 \int_0^{\pi/2} \sin^4\theta\cos^2\theta\, d\theta$$

Using Gamma formula for sine/cosine powers ($p = 4, q = 2$):

$$A = 12a^2 \cdot \frac{\Gamma(5/2)\Gamma(3/2)}{2\Gamma(4)} = 12a^2 \frac{\frac{3}{2}\frac{1}{2}\sqrt{\pi} \cdot \frac{1}{2}\sqrt{\pi}}{2 \cdot 6} = \frac{3\pi a^2}{8}$$

■

Example 6.5.2. Find the area enclosed by the curve $x^4 + y^4 = 1$ in the first quadrant.

Solution. Area $A = \int_0^1 y\, dx = \int_0^1 (1-x^4)^{1/4}\, dx$. Let $x^4 = t \implies x = t^{1/4}, dx = \frac{1}{4}t^{-3/4}dt$.

$$A = \int_0^1 (1-t)^{1/4} \frac{1}{4} t^{-3/4}\, dt = \frac{1}{4} B\left(\frac{1}{4}, \frac{5}{4}\right)$$

$$A = \frac{1}{4}\frac{\Gamma(1/4)\Gamma(5/4)}{\Gamma(3/2)} = \frac{1}{4}\frac{\Gamma(1/4) \cdot \frac{1}{4}\Gamma(1/4)}{\frac{1}{2}\sqrt{\pi}} = \frac{[\Gamma(1/4)]^2}{8\sqrt{\pi}}$$

■

Exercise 6.5.3. Evaluate the following integrals or expressions using the properties of Beta and Gamma functions.

1. Evaluate $\dfrac{\Gamma(6)}{2\Gamma(3)}$.
2. Evaluate $\Gamma\left(\dfrac{9}{2}\right)$.
3. Given that $\Gamma(n) = \frac{\Gamma(n+1)}{n}$, evaluate $\Gamma\left(-\dfrac{3}{2}\right)$.
4. Evaluate the integral $\displaystyle\int_0^1 x^4(1-x)^3\,dx$.
5. Evaluate the integral $\displaystyle\int_0^1 x^{3/2}(1-x)^{5/2}\,dx$.
6. Express $B\left(\dfrac{1}{2}, \dfrac{7}{2}\right)$ in terms of π.
7. Evaluate $\displaystyle\int_0^{\pi/2} \sin^5\theta\cos^6\theta\,d\theta$.
8. Evaluate $\displaystyle\int_0^{\pi/2} \sqrt{\tan\theta}\,d\theta$.
9. Evaluate $\displaystyle\int_0^{\pi/2} \sin^8\theta\,d\theta$.
10. Evaluate $\displaystyle\int_0^{\infty} x^3 e^{-4x}\,dx$.
11. Evaluate $\displaystyle\int_0^{\infty} \sqrt{y}e^{-y^2}\,dy$.
12. Evaluate $\displaystyle\int_0^{\infty} \frac{x^3}{(1+x)^7}\,dx$.
13. Evaluate $\displaystyle\int_0^1 x^4\left(\ln\frac{1}{x}\right)^3 dx$.
14. Evaluate $\displaystyle\int_0^a x^4\sqrt{a^2-x^2}\,dx$.
15. Using Euler's Reflection Formula $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}$, evaluate:

$$\int_0^{\infty} \frac{y^{1/4}}{1+y}\,dy$$

A Final Note to the Student

Congratulations on reaching the conclusion of this course. Completing a rigorous study of Calculus II is a significant intellectual milestone, one that demands patience, precision, and a willingness to grapple with abstract complexity. As you close this book, take a moment to reflect not just on the theorems you have memorized, but on the profound shift in perspective you have gained.

You began this journey by mastering the art of decomposition—breaking down complex rational functions and utilizing reduction formulae to make the unsolvable solvable. This taught you that even the most intimidating problems yield to systematic analysis. You then bridged the gap between the abstract and the physical, using definite integrals and rectification to measure the arc of a curve or the volume of a solid. You learned that mathematics is the tool we use to quantify the physical space we inhabit.

Crucially, by studying Differential Equations, you have acquired the language of the universe. You can now describe not just how things are, but how they change—a skill that forms the backbone of engineering, physics, and economics. Finally, through the elegance of Beta and Gamma functions, you have glimpsed the beautiful generalizations that lie ahead in higher mathematics.

The benefits of this course extend far beyond the classroom. You have sharpened your logical reasoning and developed the ability to model dynamic systems. Whether you are designing structures, analyzing data distributions, or decoding the laws of physics, the foundation you have built here will support every future academic and professional endeavor.

Calculus is the study of motion and change. You are now equipped to keep pace with a changing world. Carry these tools with confidence, stay curious, and continue to explore.

Calculus is more than a collection of theorems; it is the language used to describe the changing world around us. As you conclude this journey, remember that the techniques mastered here—from the precision of integration to the logic of differential equations—are the tools you will build upon in your future scientific endeavors. May this foundation not only serve your academic needs but also inspire a lasting appreciation for the elegance of mathematics.

DR. AIJAZ.